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Lung cancer detection kit or device, and detection method

Abstract

It is intended to provide a kit or a device for the detection of lung cancer and a method for detecting lung cancer. The present invention provides a kit or a device for the detection of lung cancer, comprising a nucleic acid capable of specifically binding to a miRNA in a sample from a subject, and a method for detecting lung cancer, comprising measuring the miRNA in vitro.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2001/0053519	12/2000	Fodor et al.	N/A	N/A
2010/0233704	12/2009	Michot et al.	N/A	N/A
2011/0053158	12/2010	Mambo et al.	N/A	N/A
2011/0117565	12/2010	Zhang et al.	N/A	N/A
2012/0108462	12/2011	Keller et al.	N/A	N/A
2015/0080243	12/2014	Whitney	N/A	N/A
2015/0337332	12/2014	Baker et al.	N/A	N/A
2017/0107581	12/2016	Kawauchi et al.	N/A	N/A
2017/0121778	12/2016	Kondou et al.	N/A	N/A
2017/0130273	12/2016	Sudo et al.	N/A	N/A
2017/0130275	12/2016	Kondou et al.	N/A	N/A
2017/0130278	12/2016	Sudo et al.	N/A	N/A
2017/0166976	12/2016	Kondou et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103173448	12/2012	CN	N/A
106488986	12/2016	CN	N/A
108471132	12/2016	CN	N/A
3156500	12/2016	EP	N/A
2011-505143	12/2010	JP	N/A
2013-502931	12/2012	JP	N/A
WO 2007/081720	12/2006	WO	N/A
WO 2009/070653	12/2008	WO	N/A
WO 2010/139810	12/2009	WO	N/A

WO 2011/025919	12/2010	WO	N/A
WO 2011/076144	12/2010	WO	N/A
WO 2011/146937	12/2010	WO	N/A
WO 2014/013258	12/2013	WO	N/A
WO 2014/192907	12/2013	WO	N/A
WO 2015/012175	12/2014	WO	N/A
WO 2015/115923	12/2014	WO	N/A
WO 2015/190542	12/2014	WO	N/A
WO 2015/190584	12/2014	WO	N/A
WO 2015/194610	12/2014	WO	N/A
WO 2015/194615	12/2014	WO	N/A

OTHER PUBLICATIONS

N.P. Hessvik et al. "Profiling of microRNAs in exosomes released from PC-3 prostate cancer cells" *Biochimica et Biophysica Acta* 1819 (2010) 1154-1163 (Year: 2012). cited by examiner
American Cancer Society, "Lung Cancer (Non-Small Cell)", 2013, total 77 pages, pp. 2-7 and 37-56. cited by applicant

Bai et al., "MIR-296-3p regulates cell growth and multi-drug resistance of human glioblastoma by targeting ether-á-go-go (EAG1)," *European Journal of Cancer*, vol. 49, No. 3, 2013 (available online Sep. 18, 2012), pp. 710-724. cited by applicant

Chen et al., "Identification of ten serum microRNAs from a genome-wide serum microRNA expression profile as novel noninvasive biomarkers for nonsmall cell lung cancer diagnosis", *International Journal Cancer*, vol. 130, May 9, 2011, pp. 1620-1628. cited by applicant

Cheung et al., "Natural variation in human gene expression assessed in lymphoblastoid cells", *Nature Genetics*, vol. 33, Mar. 2003, pp. 422-425. cited by applicant

Chinese Office Action and Search Report dated May 19, 2023 for Application No. 202010875727.X. cited by applicant

Chinese Office Action and Search Report dated Sep. 27, 2023 for Application No. 201880042985.0. cited by applicant

Cobb et al., "Sepsis gene expression profiling: Murine splenic compared with hepatic responses determined by using complementary DNA microarrays", *Critical Care Medicine*, vol. 30, No. 12, 2002, pp. 2711-2721. cited by applicant

Dissertation of Xin Wang, "Micro RNA: Profiling and Functional Implications in Cancer and Metabolism" from University of Houston, Dec. 2012, available online at https://uh-ir.tdl.org/bitstream/handle/10657/540/Diss_XinWang_20121.pdf?sequence=1&isAllowed=y (Year: 2012). cited by applicant

Enard et al., "Intra- and Interspecific Variation in Primate Gene Expression Patterns," *Science*, vol. 296, No. 340, Apr. 12, 2002, pp. 339-343. cited by applicant

Eto et al., "Prospect of microRNA toward laboratory medicine Gastrointestinal Cancer and microRNA", *Clinical Chemistry*, 2014, vol. 43, pp. 99-105. cited by applicant

Foss et al., "miR-1254 and miR-574-5p Serum-Based microRNA Biomarkers for Early-Stage Non-small Cell Lung Cancer", *Journal of Thoracic Oncology*, Mar. 2011, vol. 6, No. 3, pp. 482-488. cited by applicant

GenBank Locus NR_106826, "*Homo sapiens* micro RNA 6768 (MIR6768), micro RNA", (Apr. 3, 2014) from www.ncbi.nlm.nih.gov, printed pp. 1-3. cited by applicant

Hoshikawa et al., "Hypoxia induces different genes in the lungs of rats compared with mice", *Physiological Genomics*, vol. 12, 2003, pp. 209-219. cited by applicant

International Search Report, issued in PCT/JP2015/067533, dated Sep. 15, 2015. cited by applicant

International Search Report, issued in PCT/JP2018/024834, dated Oct. 2, 2018. cited by applicant

Jin et al., "Circulating microRNA: a novel potential biomarker for early diagnosis of Intracranial

Aneurysm Rupture a case control study,” *Journal of Translational Medicine* (2013), vol. 11, No. 296, pp. 1-9. cited by applicant

Keller et al., “Stable serum miRNA profiles as potential tool for non-invasive lung cancer diagnosis”, *RNA Biology*, May 1, 2011, vol. 8, No. 3, pp. 506-516, Supplemental Content. cited by applicant

Kozomara et al., “miRBase: annotating high confidence microR NAs using deep sequencing data”, *Nucleic Acids Research*, Nov. 25, 2013, vol. 42, Database issue, pp. D68-D73. cited by applicant

Leidinger et al., “What makes a blood cell based miRNA expression pattern disease specific?—A miRNome analysis of blood cell subsets in lung cancer patients and healthy controls”, *Oncotarget*, Sep. 19, 2014, vol. 5, No. 19, pp. 9484-9497. cited by applicant

Liu et al., “MicroRNA expression profile of gastric cancer stem cells in the MKN-45 cancer cell line”, *Acta Biochim Biophys Sin*, 2014, vol. 46, pp. 92-99. cited by applicant

Markou et al., “Clinical evaluation of microRNA expression profiling in non small cell lung cancer”, *Lung Cancer*, Sep. 2013, vol. 81, Issue No. 3, pp. 388-396. cited by applicant

MiScript™ miRNA PCR Array (384-well, 384HC) Human miRBase Profiler HC Plate 5, Qiagen, 2012, 10 pages, from <https://b2b.qiagen.com/-/media/genetable/mi/hs/34/mihs-3403z>. cited by applicant

Office Action issued Aug. 23, 2021, in Republic of Korea Patent Application No. 10-2017-7000867. cited by applicant

Office Action issued Jun. 7, 2022, in Japanese Patent Application No. 2019-527064. cited by applicant

Okamura et al., “Diagnostic value of CEA and CYFRA 21-1 tumor markers in primary lung cancer”, *Lung Cancer*, 2013, vol. 80, pp. 45-49. cited by applicant

Ondracek et al., “Global MicroRNA Expression Profiling Identifies Unique MicroRNA Pattern of Radioresistant Glioblastoma Cells”, *Anticancer Research* 37, pp. 1099-1104, 2017. cited by applicant

Partial Supplementary European Search Report, dated Dec. 14, 2017, for European Application No. 15809623.0. cited by applicant

Persson et al., “Identification of New MicroRNAs in Paired Normal and Tumor Breast Tissue Suggests a Dual Role for the ERBB2/Her2 Gene”, *Cancer Research* 71(1), pp. 78-86, 2011. cited by applicant

Oiagen Product Description “mi Script™ miRNA PCR Array (384-well, 384HC) Human miRBase Profiler HC Plate 3” document 1073798, Aug. 2012, from <https://b2b.qiagen.com/-/media/genetable/mi/hs/34/mihs-3403z> (Year: 2012). cited by applicant

Rani et al., “Global analysis of serum microR NAs as potential biomarkers for lung adenocarcinoma”, *Cancer Biology & Therapy*, 2013, vol. 14, Issue 12, pp. 1104-1112. cited by applicant

Roth et al., “Low Levels of Cell-Free Circulating miR-361-3p and miR-625* as Blood-Based Markers for Discriminating Malignant from Benign Lung Tumors”, *PLoS ONE*, Jun. 2012, vol. 7, Issue 6, e38248, pp. 1-10. cited by applicant

Schmidt et al., “Liquid Profiling in Lung Cancer—Quantification of Extracellular miRNAs in Bronchial Lavage”, *Adv Exp Med Biol*. 2016, vol. 924, pp. 33-37. cited by applicant

Shen et al., “Applications of MicroRNAs in the Diagnosis and Prognosis of Lung Cancer,” *Expert Opin. Med. Diagn.* (2012), vol. 6, No. 3, pp. 197-207. cited by applicant

Sobin et al., “TNM Classification of Malignant Tumours, the 7th edition”, *International Union Against Cancer*, 2010, pp. 129-134. cited by applicant

Song et al., “The role of deregulated microRNAs in high metastatic hepatocellular carcinoma”, *Chinese Journal Clinicians (Electronic Edition)*, vol. 7, No. 22, Nov. 15, 2013, pp. 10092-10097 with an English abstract. cited by applicant

Supplementary Partial European Search Report issued in Application No. 18823484.3 dated Mar.

12, 2021. cited by applicant
Tai et al., “Blood-borne miRNA profile-based diagnostic classifier for lung adenocarcinoma”, Scientific Reports, Aug. 10, 2016, 6: 31389, total 9 pages. cited by applicant
Takizawa et al., “miRNA Profiling in Serum Samples Using DNA Chip 3D-Gene®”, BIO Clinica, Jun. 10, 2014, vol. 29, No. 6, pp. 588-589. cited by applicant
U.S. Office Action for U.S. Appl. No. 16/626,781, dated Apr. 15, 2021. cited by applicant
U.S. Office Action for U.S. Appl. No. 16/626,781, dated Feb. 18, 2022. cited by applicant
U.S. Office Action for U.S. Appl. No. 16/626,781, dated Jun. 17, 2022. cited by applicant
U.S. Office Action for U.S. Appl. No. 16/626,781, dated May 25, 2022. cited by applicant
U.S. Office Action for U.S. Appl. No. 16/626,781, dated Sep. 14, 2021. cited by applicant
Volinia et al., “A microRNA expression signature of human solid tumors defines cancer gene targets”, Proc Natl Acad Sci U S A, Feb. 14, 2006, vol. 103, No. 7, pp. 2257-2261. cited by applicant
Written Opinion of the International Searching Authority, issued in PCT/JP2015/067533, dated Sep. 15, 2015. cited by applicant
Written Opinion of the International Searching Authority, issued in PCT/JP2018/024834, dated Oct. 2, 2018. cited by applicant
Yanaihara et al., “Unique microRNA molecular profiles in lung cancer diagnosis and prognosis,” Cancer Cell, vol. 9, No. 3, Mar. 13, 2006, pp. 189-198. cited by applicant
Zhai et al., “Differential expressions of microRNAs in the CD138+ cells of multiple myeloma patients with deletion of Chromosome 13”, Journal of Shandong University (Health Sciences), vol. 51, No. 5, May 2013, pp. 80-84 with an English abstract. cited by applicant
Extended European Search Report for European Application No. 24182689.0, dated Jan. 28, 2025. cited by applicant
Korean Office Action for Korean Application No. 10-2019-7033142, dated Jan. 22, 2024. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Divisional of application Ser. No. 17/977,377, filed on Oct. 31, 2022, which is a Divisional of application Ser. No. 16/800,755 filed Feb. 25, 2020 (now U.S. Pat. No. 11,519,927), which is a Divisional of application Ser. No. 15/319,695, filed on Dec. 16, 2016 (now U.S. Pat. No. 10,620,228), which is the National Phase under 35 U.S.C. § 371 of International Application No. PCT/JP2015/067533, filed on Jun. 18, 2015, and claims the benefit under 35 U.S.C. § 119(a) to Patent Application No. 2014-125561, filed in Japan on Jun. 18, 2014, all of which are hereby expressly incorporated by reference into the present application.

REFERENCE TO ELECTRONIC SEQUENCE LISTING

(1) The application contains a Sequence Listing which has been submitted electronically in .XML format and is hereby incorporated by reference in its entirety. Said .XML copy, created on Feb. 1, 2023, is named “PH-6239-PCT.xml” and is 557,029 bytes in size. The sequence listing contained in this .XML file is part of the specification and is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

(2) The present invention relates to a kit or a device for the detection of lung cancer, comprising a nucleic acid capable of specifically binding to a particular miRNA, which is used for examining the presence or absence of lung cancer in a subject, and a method for detecting lung cancer, comprising measuring an expression level of the miRNA using the nucleic acid.

BACKGROUND ART

(3) The lungs have important functions of supplying oxygen into the body through respiration and eliminating carbon dioxide. Air taken up from the mouth or the nose passes through the trachea and the bronchus, then separately enters the left lung and the right lung, and spreads throughout the lung through the thinner bronchial tubes. Eventually, oxygen is taken up into blood in the alveoli while carbon dioxide is eliminated (Non Patent Literature 1).

(4) According to the 2012 cancer type-specific statistics in Japan disclosed by the Center for Cancer Control and Information Services, National Cancer Center, the number of individuals affected by lung cancer was 107,241 people. Namely, it is estimated that one out of 10 males and one out of 22 females experience lung cancer. The number of incidences of this cancer among other cancer types takes the 3rd in place. Men are twice as likely as women to develop lung cancer. The number of lung cancer deaths in men and women together climbs to 71,518 people and takes the 1st in place among other cancer types. The estimated number of American individuals affected by lung cancer climbed to 224,210 people in 2014, among which approximately 159,260 people reportedly died (Non Patent Literature 1).

(5) Lung cancer has multiple histological types. Small-cell lung cancer occupies approximately 15%, while the remaining histological types are called non-small cell lung cancer. The non-small cell lung cancer is further broadly classified into three subtypes; adenocarcinoma, squamous cell carcinoma, and large-cell carcinoma. These histological types differ largely in the site of origin, the manner and rate of progression, symptoms, etc., and therefore differ in treatment methods.

(6) The stages of lung cancer progression are classified into stages 0 to 4 according to the degrees of tumor spread (T0, Tis, and T1 to T4), lymph node metastasis (N0 to N3), and distant metastasis (M0 and M1). Particularly, as for the tumor spread, T1 denotes tumor of 3 cm or less in greatest diameter; T2 denotes tumor of more than 3 cm but 7 cm or less across; T3 denotes tumor of more than 7 cm across or found to have invaded adjacent sites; and T4 denotes tumor that has invaded adjacent sites more widely regardless of its size.

(7) The survival rate of lung cancer differs depending on the stages of progression. According to the report of Non Patent Literature 1, the 5-year relative survival rate of non-small cell lung cancer is 45 to 49% for stage 1, 30 to 31% for stage 2, 5 to 14% for stage 3, and 1% for stage 4. Thus, the detection and treatment of lung cancer at an early stage makes a significant contribution to improvement in the survival rate.

(8) The treatment of lung cancer is mainly performed by surgical resection, radiotherapy, and anticancer drug treatment. Particularly, in early lung cancer, surgery is applicable and the cancer is likely to be completely cured (Non Patent Literature 1). For early lung cancer, there are some therapeutic options, and for example, treatment that places less burden on patients, such as thoracoscopic surgery, stereotactic body radiotherapy (SBRT), photo dynamic therapy, laser treatment, and brachytherapy, which delivers radiation from within the body, can also be applied to such lung cancer (Non Patent Literature 1).

(9) As described in Non Patent Literature 1, diagnostic tests of lung cancer are medical history check and physical examination as well as chest X-ray examination which is most commonly conducted. When there are findings that suspects lung cancer by the chest X-ray examination, more precise diagnostic imaging such as CT, MRI, or PET is carried out. Alternatively, as tests using samples, sputum cytology, pleural fluid analysis, or pathological examination which involves inserting a needle into a lesion and collecting cells or tissues, which are then examined under a microscope is carried out. Furthermore, CEA and CYFRA21-1 are known as tumor markers for the detection of lung cancer.

(10) As shown in Patent Literatures 1 and 2, there are reports, albeit at a research stage, on the detection of lung cancer using the expression levels of microRNAs (miRNAs) or combinations of the expression levels of miRNAs and the expression levels of additional protein markers in biological samples including blood.

(11) Patent Literature 1 discloses a method for detecting lung cancer or other lung diseases using miR-19b (miR-19b-3p) and the like in serum.

(12) Patent Literature 2 discloses a method for detecting lung cancer using miR-1268 and miR-1228 in serum or plasma.

(13) Patent Literature 3 discloses a method for detecting lung cancer using miR-1307 and the like in blood cells.

CITATION LIST

Patent Literature

(14) Patent Literature 1: JP Patent Publication (Kohyo) No. 2013-502931 A (2013) Patent

Literature 2: International Publication No. WO 2011/146937 Patent Literature 3: U.S. patent application Ser. No. 13/376,281

Non Patent Literature

(15) Non Patent Literature 1: American Cancer Society, "Lung Cancer (Non-Small Cell)", 2013, p.

2 to 7 and 37 to 56 Non Patent Literature 2: Sobin, L. et al., "TNM Classification of Malignant

Tumours, the 7th edition", 2010, p. 129-134 Non Patent Literature 3: Okamura, K. et al, Lung Cancer, 2013, Vol. 80 (1), p. 45-9

SUMMARY OF INVENTION

Technical Problem

(16) An object of the present invention is to find a novel tumor marker for lung cancer and to provide a method that can effectively detect lung cancer using a nucleic acid capable of specifically binding to the marker. Chest X-ray examination is being commonly practiced as a test of lung cancer. Nonetheless, the number of lung cancer deaths is increasing yearly and takes the first place by cancer type. For these reasons, it is not always true that the X-ray examination works as a deterrent for lung cancer. Although CT and MRI are capable of detecting lung cancer with high performance, these tests are not suitable for widespread use as 1st tests because of the necessity of their special apparatuses and expensive examination cost.

(17) For example, CEA and CYFRA21-1 are known as tumor markers in blood for the detection of lung cancer (Non Patent Literature 3). The usefulness thereof, however, has not yet been established. The lung cancer guidebook provided by the American Cancer Society makes no mention about these markers (Non Patent Literature 1). According to the report of Non Patent Literature 3, these tumor markers in blood have general lung cancer detection sensitivity of 69% (CEA) and 43% (CYFRA21-1). The tumor markers such as CEA and CYFRA21-1 may elevate for reasons other than lung cancer and therefore allegedly fail to determine the presence or absence of lung cancer. The false diagnosis of other cancers as lung cancer wastes appropriate therapeutic opportunity or places unnecessary economical and physical burdens on patients due to the application of wrong medicine.

(18) As described below, there are reports, albeit at a research stage, on the determination of lung cancer using the expression levels of microRNAs (miRNAs) in biological samples including blood, none of which, however, have yet been brought into practical use.

(19) Patent Literature 1 discloses a method for detecting lung cancer or other lung diseases using miR-19b (miR-19b-3p) and the like in serum. However, the number of samples from healthy subjects used as negative controls was as small as a dozen. Therefore, the universality of the marker for the difference among subjects is not insured. Thus, this method has low reliability as a method for detecting lung cancer.

(20) Patent Literature 2 discloses a method for detecting lung cancer using miR-1268 and miR-1228 in serum or plasma. These markers, however, were validated in only 3 mesothelioma cases as

a cancer other than lung cancer. Thus, the possibility that these markers have a high rate of false positives and detect cancers other than lung cancer cannot be excluded.

(21) Patent Literature 3 discloses a method for detecting lung cancer using miR-1307 and the like in blood cells. However, a marker obtained using one case group was not validated in another independent case group. Thus, this method has low reliability as a method for testing lung cancer.

(22) As mentioned above, the existing tumor markers exhibit low performance in the detection of lung cancer, or neither performance nor detection methods are specifically shown as to the markers at a research stage. Therefore, use of these markers might lead to carrying out needless extra examination due to the false detection of healthy subjects as being lung cancer patients, or might waste therapeutic opportunity because of overlooking lung cancer patients. In addition, the measurement of several dozens to several hundreds of miRNAs increases examination cost and is therefore difficult to use in large-scale screening such as medical checkup. Furthermore, the collection of lung tissues for measuring the tumor markers is highly invasive to patients and is not favorable. Hence, there is a demand for a highly accurate lung cancer marker that is detectable from blood, which can be collected with limited invasiveness, and is capable of correctly determining a lung cancer patient as a lung cancer patient and a healthy subject as a healthy subject. Particularly, the early detection of lung cancer can increase the applicability of surgery and drastically improve the survival rates. For early lung cancer, there are multiple therapeutic options. There is a possibility that treatment that places less burden on patients, such as thoracoscopic surgery or stereotactic body radiotherapy, can also be applied to such lung cancer. Therefore, a highly sensitive lung cancer marker that can detect lung cancer even at an early stage of progression is desired.

Solution to Problem

(23) The present inventors have conducted diligent studies to attain the object and consequently completed the present invention by finding multiple genes usable as markers for the detection of lung cancer from blood, which can be collected with limited invasiveness, and finding that lung cancer can be significantly detected by using a nucleic acid capable of specifically binding to any of these markers.

SUMMARY OF INVENTION

(24) Specifically, the present invention has the following features:

(25) (1) A kit for the detection of lung cancer, comprising a nucleic acid capable of specifically binding to at least one or more polynucleotide(s) selected from the group consisting of lung cancer markers miR-6768-5p, miR-6836-3p, miR-6782-5p, miR-3663-3p, miR-1908-3p, miR-6726-5p, miR-4258, miR-1343-3p, miR-4516, miR-6875-5p, miR-4651, miR-6825-5p, miR-6840-3p, miR-6780b-5p, miR-6749-5p, miR-8063, miR-6784-5p, miR-3679-5p, miR-3184-5p, miR-663b, miR-6880-5p, miR-1908-5p, miR-92a-2-5p, miR-7975, miR-7110-5p, miR-6842-5p, miR-6857-5p, miR-5572, miR-3197, miR-6131, miR-6889-5p, miR-4454, miR-1199-5p, miR-1247-3p, miR-6800-5p, miR-6872-3p, miR-4649-5p, miR-6791-5p, miR-4433b-3p, miR-3135b, miR-128-2-5p, miR-4675, miR-4472, miR-6785-5p, miR-6741-5p, miR-7977, miR-3665, miR-128-1-5p, miR-4286, miR-6765-3p, miR-4632-5p, miR-365a-5p, miR-6088, miR-6816-5p, miR-6885-5p, miR-711, miR-6765-5p, miR-3180, miR-4442, miR-4792, miR-6721-5p, miR-6798-5p, miR-3162-5p, miR-6126, miR-4758-5p, miR-2392, miR-486-3p, miR-6727-5p, miR-4728-5p, miR-6746-5p, miR-4270, miR-3940-5p, miR-4725-3p, miR-7108-5p, miR-3656, miR-6879-5p, miR-6738-5p, miR-1260a, miR-4446-3p, miR-3131, miR-4463, miR-3185, miR-6870-5p, miR-6779-5p, miR-1273g-3p, miR-8059, miR-4697-5p, miR-4674, miR-4433-3p, miR-4257, miR-1915-5p, miR-4417, miR-1343-5p, miR-6781-5p, miR-4695-5p, miR-1237-5p, miR-6775-5p, miR-7845-5p, miR-4746-3p, miR-7641, miR-7847-3p, miR-6806-5p, miR-4467, miR-4726-5p, miR-4648, miR-6089, miR-1260b, miR-4532, miR-5195-3p, miR-3188, miR-6848-5p, miR-1233-5p, miR-6717-5p, miR-3195, miR-6757-5p, miR-8072, miR-4745-5p, miR-6511a-5p, miR-6776-5p, miR-371a-5p, miR-1227-5p, miR-7150, miR-1915-3p, miR-187-5p, miR-614, miR-1225-5p, miR-451a, miR-939-5p,

miR-223-3p, miR-125a-3p, miR-92b-5p, miR-22-3p, miR-6073, miR-6845-5p, miR-6769b-5p, miR-4665-3p, miR-1913, miR-1228-3p, miR-940, miR-296-3p, miR-4690-5p, miR-548q, miR-663a, miR-1249, miR-1202, miR-7113-3p, miR-1225-3p, miR-4783-3p, miR-4448 and miR-4534.

(26) (2) The kit according to (1), wherein miR-6768-5p is hsa-miR-6768-5p, miR-6836-3p is hsa-miR-6836-3p, miR-6782-5p is hsa-miR-6782-5p, miR-3663-3p is hsa-miR-3663-3p, miR-1908-3p is hsa-miR-1908-3p, miR-6726-5p is hsa-miR-6726-5p, miR-4258 is hsa-miR-4258, miR-1343-3p is hsa-miR-1343-3p, miR-4516 is hsa-miR-4516, miR-6875-5p is hsa-miR-6875-5p, miR-4651 is hsa-miR-4651, miR-6825-5p is hsa-miR-6825-5p, miR-6840-3p is hsa-miR-6840-3p, miR-6780b-5p is hsa-miR-6780b-5p, miR-6749-5p is hsa-miR-6749-5p, miR-8063 is hsa-miR-8063, miR-6784-5p is hsa-miR-6784-5p, miR-3679-5p is hsa-miR-3679-5p, miR-3184-5p is hsa-miR-3184-5p, miR-663b is hsa-miR-663b, miR-6880-5p is hsa-miR-6880-5p, miR-1908-5p is hsa-miR-1908-5p, miR-92a-2-5p is hsa-miR-92a-2-5p, miR-7975 is hsa-miR-7975, miR-7110-5p is hsa-miR-7110-5p, miR-6842-5p is hsa-miR-6842-5p, miR-6857-5p is hsa-miR-6857-5p, miR-5572 is hsa-miR-5572, miR-3197 is hsa-miR-3197, miR-6131 is hsa-miR-6131, miR-6889-5p is hsa-miR-6889-5p, miR-4454 is hsa-miR-4454, miR-1199-5p is hsa-miR-1199-5p, miR-1247-3p is hsa-miR-1247-3p, miR-6800-5p is hsa-miR-6800-5p, miR-6872-3p is hsa-miR-6872-3p, miR-4649-5p is hsa-miR-4649-5p, miR-6791-5p is hsa-miR-6791-5p, miR-4433b-3p is hsa-miR-4433b-3p, miR-3135b is hsa-miR-3135b, miR-128-2-5p is hsa-miR-128-2-5p, miR-4675 is hsa-miR-4675, miR-4472 is hsa-miR-4472, miR-6785-5p is hsa-miR-6785-5p, miR-6741-5p is hsa-miR-6741-5p, miR-7977 is hsa-miR-7977, miR-3665 is hsa-miR-3665, miR-128-1-5p is hsa-miR-128-1-5p, miR-4286 is hsa-miR-4286, miR-6765-3p is hsa-miR-6765-3p, miR-4632-5p is hsa-miR-4632-5p, miR-365a-5p is hsa-miR-365a-5p, miR-6088 is hsa-miR-6088, miR-6816-5p is hsa-miR-6816-5p, miR-6885-5p is hsa-miR-6885-5p, miR-711 is hsa-miR-711, miR-6765-5p is hsa-miR-6765-5p, miR-3180 is hsa-miR-3180, miR-4442 is hsa-miR-4442, miR-4792 is hsa-miR-4792, miR-6721-5p is hsa-miR-6721-5p, miR-6798-5p is hsa-miR-6798-5p, miR-3162-5p is hsa-miR-3162-5p, miR-6126 is hsa-miR-6126, miR-4758-5p is hsa-miR-4758-5p, miR-2392 is hsa-miR-2392, miR-486-3p is hsa-miR-486-3p, miR-6727-5p is hsa-miR-6727-5p, miR-4728-5p is hsa-miR-4728-5p, miR-6746-5p is hsa-miR-6746-5p, miR-4270 is hsa-miR-4270, miR-3940-5p is hsa-miR-3940-5p, miR-4725-3p is hsa-miR-4725-3p, miR-7108-5p is hsa-miR-7108-5p, miR-3656 is hsa-miR-3656, miR-6879-5p is hsa-miR-6879-5p, miR-6738-5p is hsa-miR-6738-5p, miR-1260a is hsa-miR-1260a, miR-4446-3p is hsa-miR-4446-3p, miR-3131 is hsa-miR-3131, miR-4463 is hsa-miR-4463, miR-3185 is hsa-miR-3185, miR-6870-5p is hsa-miR-6870-5p, miR-6779-5p is hsa-miR-6779-5p, miR-1273g-3p is hsa-miR-1273g-3p, miR-8059 is hsa-miR-8059, miR-4697-5p is hsa-miR-4697-5p, miR-4674 is hsa-miR-4674, miR-4433-3p is hsa-miR-4433-3p, miR-4257 is hsa-miR-4257, miR-1915-5p is hsa-miR-1915-5p, miR-4417 is hsa-miR-4417, miR-1343-5p is hsa-miR-1343-5p, miR-6781-5p is hsa-miR-6781-5p, miR-4695-5p is hsa-miR-4695-5p, miR-1237-5p is hsa-miR-1237-5p, miR-6775-5p is hsa-miR-6775-5p, miR-7845-5p is hsa-miR-7845-5p, miR-4746-3p is hsa-miR-4746-3p, miR-7641 is hsa-miR-7641, miR-7847-3p is hsa-miR-7847-3p, miR-6806-5p is hsa-miR-6806-5p, miR-4467 is hsa-miR-4467, miR-4726-5p is hsa-miR-4726-5p, miR-4648 is hsa-miR-4648, miR-6089 is hsa-miR-6089, miR-1260b is hsa-miR-1260b, miR-4532 is hsa-miR-4532, miR-5195-3p is hsa-miR-5195-3p, miR-3188 is hsa-miR-3188, miR-6848-5p is hsa-miR-6848-5p, miR-1233-5p is hsa-miR-1233-5p, miR-6717-5p is hsa-miR-6717-5p, miR-3195 is hsa-miR-3195, miR-6757-5p is hsa-miR-6757-5p, miR-8072 is hsa-miR-8072, miR-4745-5p is hsa-miR-4745-5p, miR-6511a-5p is hsa-miR-6511a-5p, miR-6776-5p is hsa-miR-6776-5p, miR-371a-5p is hsa-miR-371a-5p, miR-1227-5p is hsa-miR-1227-5p, miR-7150 is hsa-miR-7150, miR-1915-3p is hsa-miR-1915-3p, miR-187-5p is hsa-miR-187-5p, miR-614 is hsa-miR-614, miR-1225-5p is hsa-miR-1225-5p, miR-451a is hsa-miR-451a, miR-939-5p is hsa-miR-939-5p, miR-223-3p is hsa-miR-223-3p, miR-125a-3p is hsa-miR-125a-3p, miR-92b-5p is hsa-miR-92b-5p, miR-22-3p is hsa-miR-22-3p, miR-6073 is hsa-miR-6073, miR-6845-5p is hsa-miR-6845-5p, miR-6769b-5p is hsa-miR-6769b-5p, miR-4665-3p is hsa-miR-4665-3p, miR-1913 is hsa-miR-1913, miR-1228-3p is hsa-miR-1228-3p, miR-940 is

hsa-miR-940, miR-296-3p is hsa-miR-296-3p, miR-4690-5p is hsa-miR-4690-5p, miR-548q is hsa-miR-548q, miR-663a is hsa-miR-663a, miR-1249 is hsa-miR-1249, miR-1202 is hsa-miR-1202, miR-7113-3p is hsa-miR-7113-3p, miR-1225-3p is hsa-miR-1225-3p, miR-4783-3p is hsa-miR-4783-3p, miR-4448 is hsa-miR-4448, and miR-4534 is hsa-miR-4534.

(27) (3) The kit according to (1) or (2), wherein the nucleic acid is a polynucleotide selected from the group consisting of the following polynucleotides (a) to (e): (a) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (b) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578, (c) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (d) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and (e) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (a) to (d).

(28) (4) The kit according to any of (1) to (3), wherein the kit further comprises a nucleic acid capable of specifically binding to at least one or more polynucleotide(s) selected from the group consisting of other lung cancer markers miR-19b-3p, miR-1228-5p, and miR-1307-3p.

(29) (5) The kit according to (4), wherein miR-19b-3p is hsa-miR-19b-3p, miR-1228-5p is hsa-miR-1228-5p, and miR-1307-3p is hsa-miR-1307-3p.

(30) (6) The kit according to (4) or (5), wherein the nucleic acid is a polynucleotide selected from the group consisting of the following polynucleotides (f) to (j): (f) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (g) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579, (h) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (i) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and (j) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (f) to (i).

(31) (7) The kit according to any of (1) to (6), wherein the kit further comprises a nucleic acid capable of specifically binding to at least one or more polynucleotide(s) selected from the group consisting of other lung cancer markers miR-4271, miR-642b-3p, miR-6075, miR-6125, miR-887-3p, miR-6851-5p, miR-6763-5p, miR-3928-3p, miR-4443, miR-3648, miR-149-3p, miR-4689, miR-4763-3p, miR-6729-5p, miR-3196, miR-8069, miR-1268a, miR-4739, miR-1268b, miR-5698, miR-6752-5p, miR-4507, miR-564, miR-4497, miR-6877-5p, miR-6087, miR-4731-5p, miR-615-5p, miR-760, miR-6891-5p, miR-6887-5p, miR-4525, miR-1914-3p, miR-619-5p, miR-5001-5p, miR-6722-3p, miR-3621, miR-4298, miR-675-5p and miR-4655-5p.

(32) (8) The kit according to (7), wherein miR-4271 is hsa-miR-4271, miR-642b-3p is hsa-miR-642b-3p, miR-6075 is hsa-miR-6075, miR-6125 is hsa-miR-6125, miR-887-3p is hsa-miR-887-3p, miR-6851-5p is hsa-miR-6851-5p, miR-6763-5p is hsa-miR-6763-5p, miR-3928-3p is hsa-miR-3928-3p, miR-4443 is hsa-miR-4443, miR-3648 is hsa-miR-3648, miR-149-3p is hsa-miR-149-3p,

miR-4689 is hsa-miR-4689, miR-4763-3p is hsa-miR-4763-3p, miR-6729-5p is hsa-miR-6729-5p, miR-3196 is hsa-miR-3196, miR-8069 is hsa-miR-8069, miR-1268a is hsa-miR-1268a, miR-4739 is hsa-miR-4739, miR-1268b is hsa-miR-1268b, miR-5698 is hsa-miR-5698, miR-6752-5p is hsa-miR-6752-5p, miR-4507 is hsa-miR-4507, miR-564 is hsa-miR-564, miR-4497 is hsa-miR-4497, miR-6877-5p is hsa-miR-6877-5p, miR-6087 is hsa-miR-6087, miR-4731-5p is hsa-miR-4731-5p, miR-615-5p is hsa-miR-615-5p, miR-760 is hsa-miR-760, miR-6891-5p is hsa-miR-6891-5p, miR-6887-5p is hsa-miR-6887-5p, miR-4525 is hsa-miR-4525, miR-1914-3p is hsa-miR-1914-3p, miR-619-5p is hsa-miR-619-5p, miR-5001-5p is hsa-miR-5001-5p, miR-6722-3p is hsa-miR-6722-3p, miR-3621 is hsa-miR-3621, miR-4298 is hsa-miR-4298, miR-675-5p is hsa-miR-675-5p, and miR-4655-5p is hsa-miR-4655-5p.

(33) (9) The kit according to (7) or (8), wherein the nucleic acid is a polynucleotide selected from the group consisting of the following polynucleotides (k) to (o): (k) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (l) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174, (m) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (n) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and (o) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (k) to (n).

(34) (10) The kit according to any one of (1) to (9), wherein the kit comprises at least two or more nucleic acids capable of specifically binding to at least two or more polynucleotides, respectively, selected from the group consisting of all of the lung cancer markers according to (1) or (2).

(35) (11) A device for the detection of lung cancer, comprising a nucleic acid capable of specifically binding to at least one or more polynucleotide(s) selected from the group consisting of lung cancer markers miR-6768-5p, miR-6836-3p, miR-6782-5p, miR-3663-3p, miR-1908-3p, miR-6726-5p, miR-4258, miR-1343-3p, miR-4516, miR-6875-5p, miR-4651, miR-6825-5p, miR-6840-3p, miR-6780b-5p, miR-6749-5p, miR-8063, miR-6784-5p, miR-3679-5p, miR-3184-5p, miR-663b, miR-6880-5p, miR-1908-5p, miR-92a-2-5p, miR-7975, miR-7110-5p, miR-6842-5p, miR-6857-5p, miR-5572, miR-3197, miR-6131, miR-6889-5p, miR-4454, miR-1199-5p, miR-1247-3p, miR-6800-5p, miR-6872-3p, miR-4649-5p, miR-6791-5p, miR-4433b-3p, miR-3135b, miR-128-2-5p, miR-4675, miR-4472, miR-6785-5p, miR-6741-5p, miR-7977, miR-3665, miR-128-1-5p, miR-4286, miR-6765-3p, miR-4632-5p, miR-365a-5p, miR-6088, miR-6816-5p, miR-6885-5p, miR-711, miR-6765-5p, miR-3180, miR-4442, miR-4792, miR-6721-5p, miR-6798-5p, miR-3162-5p, miR-6126, miR-4758-5p, miR-2392, miR-486-3p, miR-6727-5p, miR-4728-5p, miR-6746-5p, miR-4270, miR-3940-5p, miR-4725-3p, miR-7108-5p, miR-3656, miR-6879-5p, miR-6738-5p, miR-1260a, miR-4446-3p, miR-3131, miR-4463, miR-3185, miR-6870-5p, miR-6779-5p, miR-1273g-3p, miR-8059, miR-4697-5p, miR-4674, miR-4433-3p, miR-4257, miR-1915-5p, miR-4417, miR-1343-5p, miR-6781-5p, miR-4695-5p, miR-1237-5p, miR-6775-5p, miR-7845-5p, miR-4746-3p, miR-7641, miR-7847-3p, miR-6806-5p, miR-4467, miR-4726-5p, miR-4648, miR-6089, miR-1260b, miR-4532, miR-5195-3p, miR-3188, miR-6848-5p, miR-1233-5p, miR-6717-5p, miR-3195, miR-6757-5p, miR-8072, miR-4745-5p, miR-6511a-5p, miR-6776-5p, miR-371a-5p, miR-1227-5p, miR-7150, miR-1915-3p, miR-187-5p, miR-614, miR-1225-5p, miR-451a, miR-939-5p, miR-223-3p, miR-125a-3p, miR-92b-5p, miR-22-3p, miR-6073, miR-6845-5p, miR-6769b-5p, miR-4665-3p, miR-1913, miR-1228-3p, miR-940, miR-296-3p, miR-4690-5p, miR-548q, miR-663a, miR-1249, miR-1202, miR-7113-3p, miR-1225-3p, miR-4783-3p, miR-4448 and miR-4534.

(36) (12) The device according to (11), wherein miR-6768-5p is hsa-miR-6768-5p, miR-6836-3p is hsa-miR-6836-3p, miR-6782-5p is hsa-miR-6782-5p, miR-3663-3p is hsa-miR-3663-3p, miR-1908-3p is hsa-miR-1908-3p, miR-6726-5p is hsa-miR-6726-5p, miR-4258 is hsa-miR-4258, miR-1343-3p is hsa-miR-1343-3p, miR-4516 is hsa-miR-4516, miR-6875-5p is hsa-miR-6875-5p, miR-4651 is hsa-miR-4651, miR-6825-5p is hsa-miR-6825-5p, miR-6840-3p is hsa-miR-6840-3p, miR-6780b-5p is hsa-miR-6780b-5p, miR-6749-5p is hsa-miR-6749-5p, miR-8063 is hsa-miR-8063, miR-6784-5p is hsa-miR-6784-5p, miR-3679-5p is hsa-miR-3679-5p, miR-3184-5p is hsa-miR-3184-5p, miR-663b is hsa-miR-663b, miR-6880-5p is hsa-miR-6880-5p, miR-1908-5p is hsa-miR-1908-5p, miR-92a-2-5p is hsa-miR-92a-2-5p, miR-7975 is hsa-miR-7975, miR-7110-5p is hsa-miR-7110-5p, miR-6842-5p is hsa-miR-6842-5p, miR-6857-5p is hsa-miR-6857-5p, miR-5572 is hsa-miR-5572, miR-3197 is hsa-miR-3197, miR-6131 is hsa-miR-6131, miR-6889-5p is hsa-miR-6889-5p, miR-4454 is hsa-miR-4454, miR-1199-5p is hsa-miR-1199-5p, miR-1247-3p is hsa-miR-1247-3p, miR-6800-5p is hsa-miR-6800-5p, miR-6872-3p is hsa-miR-6872-3p, miR-4649-5p is hsa-miR-4649-5p, miR-6791-5p is hsa-miR-6791-5p, miR-4433b-3p is hsa-miR-4433b-3p, miR-3135b is hsa-miR-3135b, miR-128-2-5p is hsa-miR-128-2-5p, miR-4675 is hsa-miR-4675, miR-4472 is hsa-miR-4472, miR-6785-5p is hsa-miR-6785-5p, miR-6741-5p is hsa-miR-6741-5p, miR-7977 is hsa-miR-7977, miR-3665 is hsa-miR-3665, miR-128-1-5p is hsa-miR-128-1-5p, miR-4286 is hsa-miR-4286, miR-6765-3p is hsa-miR-6765-3p, miR-4632-5p is hsa-miR-4632-5p, miR-365a-5p is hsa-miR-365a-5p, miR-6088 is hsa-miR-6088, miR-6816-5p is hsa-miR-6816-5p, miR-6885-5p is hsa-miR-6885-5p, miR-711 is hsa-miR-711, miR-6765-5p is hsa-miR-6765-5p, miR-3180 is hsa-miR-3180, miR-4442 is hsa-miR-4442, miR-4792 is hsa-miR-4792, miR-6721-5p is hsa-miR-6721-5p, miR-6798-5p is hsa-miR-6798-5p, miR-3162-5p is hsa-miR-3162-5p, miR-6126 is hsa-miR-6126, miR-4758-5p is hsa-miR-4758-5p, miR-2392 is hsa-miR-2392, miR-486-3p is hsa-miR-486-3p, miR-6727-5p is hsa-miR-6727-5p, miR-4728-5p is hsa-miR-4728-5p, miR-6746-5p is hsa-miR-6746-5p, miR-4270 is hsa-miR-4270, miR-3940-5p is hsa-miR-3940-5p, miR-4725-3p is hsa-miR-4725-3p, miR-7108-5p is hsa-miR-7108-5p, miR-3656 is hsa-miR-3656, miR-6879-5p is hsa-miR-6879-5p, miR-6738-5p is hsa-miR-6738-5p, miR-1260a is hsa-miR-1260a, miR-4446-3p is hsa-miR-4446-3p, miR-3131 is hsa-miR-3131, miR-4463 is hsa-miR-4463, miR-3185 is hsa-miR-3185, miR-6870-5p is hsa-miR-6870-5p, miR-6779-5p is hsa-miR-6779-5p, miR-1273g-3p is hsa-miR-1273g-3p, miR-8059 is hsa-miR-8059, miR-4697-5p is hsa-miR-4697-5p, miR-4674 is hsa-miR-4674, miR-4433-3p is hsa-miR-4433-3p, miR-4257 is hsa-miR-4257, miR-1915-5p is hsa-miR-1915-5p, miR-4417 is hsa-miR-4417, miR-1343-5p is hsa-miR-1343-5p, miR-6781-5p is hsa-miR-6781-5p, miR-4695-5p is hsa-miR-4695-5p, miR-1237-5p is hsa-miR-1237-5p, miR-6775-5p is hsa-miR-6775-5p, miR-7845-5p is hsa-miR-7845-5p, miR-4746-3p is hsa-miR-4746-3p, miR-7641 is hsa-miR-7641, miR-7847-3p is hsa-miR-7847-3p, miR-6806-5p is hsa-miR-6806-5p, miR-4467 is hsa-miR-4467, miR-4726-5p is hsa-miR-4726-5p, miR-4648 is hsa-miR-4648, miR-6089 is hsa-miR-6089, miR-1260b is hsa-miR-1260b, miR-4532 is hsa-miR-4532, miR-5195-3p is hsa-miR-5195-3p, miR-3188 is hsa-miR-3188, miR-6848-5p is hsa-miR-6848-5p, miR-1233-5p is hsa-miR-1233-5p, miR-6717-5p is hsa-miR-6717-5p, miR-3195 is hsa-miR-3195, miR-6757-5p is hsa-miR-6757-5p, miR-8072 is hsa-miR-8072, miR-4745-5p is hsa-miR-4745-5p, miR-6511a-5p is hsa-miR-6511a-5p, miR-6776-5p is hsa-miR-6776-5p, miR-371a-5p is hsa-miR-371a-5p, miR-1227-5p is hsa-miR-1227-5p, miR-7150 is hsa-miR-7150, miR-1915-3p is hsa-miR-1915-3p, miR-187-5p is hsa-miR-187-5p, miR-614 is hsa-miR-614, miR-1225-5p is hsa-miR-1225-5p, miR-451a is hsa-miR-451a, miR-939-5p is hsa-miR-939-5p, miR-223-3p is hsa-miR-223-3p, miR-125a-3p is hsa-miR-125a-3p, miR-92b-5p is hsa-miR-92b-5p, miR-22-3p is hsa-miR-22-3p, miR-6073 is hsa-miR-6073, miR-6845-5p is hsa-miR-6845-5p, miR-6769b-5p is hsa-miR-6769b-5p, miR-4665-3p is hsa-miR-4665-3p, miR-1913 is hsa-miR-1913, miR-1228-3p is hsa-miR-1228-3p, miR-940 is hsa-miR-940, miR-296-3p is hsa-miR-296-3p, miR-4690-5p is hsa-miR-4690-5p, miR-548q is hsa-miR-548q, miR-663a is hsa-miR-663a, miR-1249 is hsa-miR-1249, miR-1202 is hsa-miR-1202, miR-7113-3p is hsa-miR-7113-3p, miR-1225-3p is hsa-miR-1225-3p, miR-4783-3p is hsa-miR-

4783-3p, miR-4448 is hsa-miR-4448, and miR-4534 is hsa-miR-4534.

(37) (13) The device according to (11) or (12), wherein the nucleic acid is a polynucleotide selected from the group consisting of the following polynucleotides (a) to (e): (a) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (b) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578, (c) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (d) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and (e) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (a) to (d).

(38) (14) The device according to any of (11) to (13), wherein the device further comprises a nucleic acid capable of specifically binding to at least one or more polynucleotide(s) selected from the group consisting of other lung cancer markers miR-19b-3p, miR-1228-5p, and miR-1307-3p.

(39) (15) The device according to (14), wherein miR-19b-3p is hsa-miR-19b-3p, miR-1228-5p is hsa-miR-1228-5p, and miR-1307-3p is hsa-miR-1307-3p.

(40) (16) The device according to (14) or (15), wherein the nucleic acid is a polynucleotide selected from the group consisting of the following polynucleotides (f) to (j): (f) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (g) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579, (h) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (i) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and (j) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (f) to (i).

(41) (17) The device according to any of (11) to (16), wherein the device further comprises a nucleic acid capable of specifically binding to at least one or more polynucleotide(s) selected from the group consisting of other lung cancer markers miR-4271, miR-642b-3p, miR-6075, miR-6125, miR-887-3p, miR-6851-5p, miR-6763-5p, miR-3928-3p, miR-4443, miR-3648, miR-149-3p, miR-4689, miR-4763-3p, miR-6729-5p, miR-3196, miR-8069, miR-1268a, miR-4739, miR-1268b, miR-5698, miR-6752-5p, miR-4507, miR-564, miR-4497, miR-6877-5p, miR-6087, miR-4731-5p, miR-615-5p, miR-760, miR-6891-5p, miR-6887-5p, miR-4525, miR-1914-3p, miR-619-5p, miR-5001-5p, miR-6722-3p, miR-3621, miR-4298, miR-675-5p and miR-4655-5p.

(42) (18) The device according to (17), wherein miR-4271 is hsa-miR-4271, miR-642b-3p is hsa-miR-642b-3p, miR-6075 is hsa-miR-6075, miR-6125 is hsa-miR-6125, miR-887-3p is hsa-miR-887-3p, miR-6851-5p is hsa-miR-6851-5p, miR-6763-5p is hsa-miR-6763-5p, miR-3928-3p is hsa-miR-3928-3p, miR-4443 is hsa-miR-4443, miR-3648 is hsa-miR-3648, miR-149-3p is hsa-miR-149-3p, miR-4689 is hsa-miR-4689, miR-4763-3p is hsa-miR-4763-3p, miR-6729-5p is hsa-miR-6729-5p, miR-3196 is hsa-miR-3196, miR-8069 is hsa-miR-8069, miR-1268a is hsa-miR-1268a, miR-4739 is hsa-miR-4739, miR-1268b is hsa-miR-1268b, miR-5698 is hsa-miR-5698, miR-6752-

5p is hsa-miR-6752-5p, miR-4507 is hsa-miR-4507, miR-564 is hsa-miR-564, miR-4497 is hsa-miR-4497, miR-6877-5p is hsa-miR-6877-5p, miR-6087 is hsa-miR-6087, miR-4731-5p is hsa-miR-4731-5p, miR-615-5p is hsa-miR-615-5p, miR-760 is hsa-miR-760, miR-6891-5p is hsa-miR-6891-5p, miR-6887-5p is hsa-miR-6887-5p, miR-4525 is hsa-miR-4525, miR-1914-3p is hsa-miR-1914-3p, miR-619-5p is hsa-miR-619-5p, miR-5001-5p is hsa-miR-5001-5p, miR-6722-3p is hsa-miR-6722-3p, miR-3621 is hsa-miR-3621, miR-4298 is hsa-miR-4298, miR-675-5p is hsa-miR-675-5p, and miR-4655-5p is hsa-miR-4655-5p.

(43) (19) The device according to (17) or (18), wherein the nucleic acid is a polynucleotide selected from the group consisting of the following polynucleotides (k) to (o): (k) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (l) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174, (m) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (n) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and (o) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (k) to (n).

(44) (20) The device according to any one of (11) to (19), wherein the device is a device for measurement by a hybridization technique.

(45) (21) The device according to (20), wherein the hybridization technique is a nucleic acid array technique.

(46) (22) The device according to any one of (11) to (21), wherein the device comprises at least two or more nucleic acids capable of specifically binding to at least two or more polynucleotides, respectively, selected from all of the lung cancer markers according to (11) or (12).

(47) (23) A method for detecting lung cancer, comprising measuring an expression level of a target nucleic acid in a sample from a subject using a kit according to any one of (1) to (10) or a device according to any one of (11) to (22), and evaluating in vitro whether or not the subject has lung cancer using both of the measured expression level and a control expression level in a sample from a healthy subject measured in the same way.

(48) (24) The method according to (23), wherein the subject is a human.

(49) (25) The method according to (23) or (24), wherein the sample is blood, serum, or plasma.

Definition of Terms

(50) The terms used herein are defined as follows.

(51) Abbreviations or terms such as nucleotide, polynucleotide, DNA, and RNA abide by “Guidelines for the preparation of specification which contain nucleotide and/or amino acid sequences” (edited by Japan Patent Office) and common use in the art.

(52) The term “polynucleotide” used herein is used for a nucleic acid including any of RNA, DNA, and RNA/DNA (chimera). The DNA includes any of cDNA, genomic DNA, and synthetic DNA. The RNA includes any of total RNA, mRNA, rRNA, miRNA, siRNA, snoRNA, snRNA, non-coding RNA and synthetic RNA. Here the “synthetic DNA” and the “synthetic RNA” refer to DNA and RNA artificially prepared using, for example, an automatic nucleic acid synthesizer, on the basis of predetermined nucleotide sequences (which may be any of natural and non-natural sequences). The “non-natural sequence” is intended to be used in a broad sense and includes, for example, a sequence comprising substitution, deletion, insertion, and/or addition of one or more nucleotide(s) (i.e., a variant sequence) and a sequence comprising one or more modified nucleotide(s) (i.e., a modified sequence), which are different from the natural sequence. Herein, the

term “polynucleotide” is used interchangeably with the term “nucleic acid”.

(53) The term “fragment” used herein is a polynucleotide having a nucleotide sequence that consists of a consecutive portion of a polynucleotide and desirably has a length of 15 or more nucleotides, preferably 17 or more nucleotides, more preferably 19 or more nucleotides.

(54) The term “gene” used herein is intended to include not only RNA and double-stranded DNA but each single-stranded DNA such as a plus strand (or a sense strand) or a complementary strand (or an antisense strand) constituting the duplex. The gene is not particularly limited by its length.

(55) Thus, the “gene” used herein includes any of double-stranded DNA including human genomic DNA, single-stranded DNA (plus strand), single-stranded DNA having a sequence complementary to the plus strand (complementary strand) including cDNA, microRNA (miRNA), and their fragments, and transcripts, unless otherwise specified. The “gene” includes not only a “gene” represented by a particular nucleotide sequence (or SEQ ID NO) but “nucleic acids” encoding RNAs having biological functions equivalent to an RNA encoded by the gene, for example, a congener (i.e., a homolog or an ortholog), a variant (e.g., a genetic polymorph), and a derivative. Specific examples of such a “nucleic acid” encoding a congener, a variant, or a derivative can include a “nucleic acid” having a nucleotide sequence hybridizing under stringent conditions described later to a complementary sequence of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 618, or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t. The “gene” is not particularly limited by its functional region and can contain, for example, an expression regulatory region, a coding region, an exon, or an intron. The “gene” may be contained in a cell or may exist alone after being released into the outside of a cell. Alternatively, the “gene” may be in a state enclosed in a vesicle called exosome.

(56) The term “exosome” used herein is a vesicle that is encapsulated by a lipid bilayer and secreted from a cell. The exosome is derived from a multivesicular endosome and may incorporate biomaterials such as a “gene” (e.g., RNA or DNA) or a protein when released into an extracellular environment. The exosome is known to be contained in a body fluid such as blood, serum, plasma, or lymph.

(57) The term “transcript” used herein refers to an RNA synthesized with the DNA sequence of a gene as a template. RNA polymerase binds to a site called a promoter located upstream of the gene and adds ribonucleotides complementary to the nucleotide sequence of the DNA to the 3' end to synthesize RNA. This RNA contains not only the gene itself but also the whole sequence from a transcription initiation site to the end of a polyA sequence, including an expression regulatory region, a coding region, an exon, or an intron.

(58) The term “microRNA (miRNA)” used herein is intended to mean a 15- to 25-nucleotide non-coding RNA that is involved in the suppression of translation of mRNA, and that transcribed as an RNA precursor having a hairpin-like structure, cleaved by a dsRNA-cleaving enzyme which has RNase III cleavage activity, and integrated into a protein complex called RISC, unless otherwise specified. The term “miRNA” used herein includes not only a “miRNA” represented by a particular nucleotide sequence (or SEQ ID NO) but a precursor of the “miRNA” (pre-miRNA or pri-miRNA), and miRNAs having biological functions equivalent thereto, for example, a congener (i.e., a homolog or an ortholog), a variant (e.g., a genetic polymorph), and a derivative. Such a precursor, a congener, a variant, or a derivative can be specifically identified using miRBase Release 20 (<http://www.mirbase.org/>), and examples thereof can include a “miRNA” having a nucleotide sequence hybridizing under stringent conditions described later to a complementary sequence of any particular nucleotide sequence represented by any of SEQ ID NOs: 1 to 618. The term “miRNA” used herein may be a gene product of a miR gene. Such a gene product includes a mature miRNA (e.g., a 15- to 25-nucleotide or 19- to 25-nucleotide non-coding RNA involved in the suppression of translation of mRNA as described above) or a miRNA precursor (e.g., pre-miRNA or pri-miRNA as described above).

(59) The term “probe” used herein includes a polynucleotide that is used for specifically detecting

RNA resulting from the expression of a gene or a polynucleotide derived from the RNA, and/or a polynucleotide complementary thereto.

(60) The term “primer” used herein includes a polynucleotide that specifically recognizes and amplifies RNA resulting from the expression of a gene or a polynucleotide derived from the RNA, and/or a polynucleotide complementary thereto.

(61) In this context, the complementary polynucleotide (complementary strand or reverse strand) means a polynucleotide in a complementary base relationship of A:T (U) and G:C base pairs with the full-length sequence of a polynucleotide consisting of a nucleotide sequence defined by any of SEQ ID NOs: 1 to 618 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, or a partial sequence thereof (here, this full-length or partial sequence is referred to as a plus strand for the sake of convenience). However, such a complementary strand is not limited to a sequence completely complementary to the nucleotide sequence of the target plus strand and may have a complementary relationship to an extent that permits hybridization under stringent conditions to the target plus strand.

(62) The term “stringent conditions” used herein refers to conditions under which a nucleic acid probe hybridizes to its target sequence to a larger extent (e.g., a measurement value equal to or larger than a mean of background measurement values+ a standard deviation of the background measurement values $\times$ 2) than that for other sequences. The stringent conditions are dependent on a sequence and differ depending on an environment where hybridization is performed. A target sequence that is 100% complementary to the nucleic acid probe can be identified by controlling the stringency of hybridization and/or washing conditions. Specific examples of the “stringent conditions” will be mentioned later.

(63) The term “T_m value” used herein means a temperature at which the double-stranded moiety of a polynucleotide is denatured into single strands so that the double strands and the single strands exist at a ratio of 1:1.

(64) The term “variant” used herein means, in the case of a nucleic acid, a natural variant attributed to polymorphism, mutation, or the like; a variant containing the deletion, substitution, addition, or insertion of 1, 2, or 3 or more nucleotides in a nucleotide sequence represented by any of SEQ ID NOs: 1 to 618 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, or a partial sequence thereof; a variant containing the deletion, substitution, addition, or insertion of 1 or 2 or more nucleotides in a nucleotide sequence of a premature miRNA of a sequence represented by any of SEQ ID NOs: 1 to 618 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, or a partial sequence thereof; a variant that exhibits identity of approximately 90% or higher, approximately 95% or higher, approximately 97% or higher, approximately 98% or higher, approximately 99% or higher to each of these nucleotide sequences or the partial sequence thereof; or a nucleic acid hybridizing under the stringent conditions defined above to a polynucleotide or an oligonucleotide comprising each of these nucleotide sequences or the partial sequence thereof.

(65) The term “several” used herein means an integer of approximately 10, 9, 8, 7, 6, 5, 4, 3, or 2.

(66) The variant used herein can be prepared by use of a well-known technique such as site-directed mutagenesis or PCR-based mutagenesis.

(67) The term “identity” used herein can be determined with or without an introduced gap, using a protein or gene search system based on BLAST or FASTA described above (Zheng Zhang et al., 2000, J. Comput. Biol., Vol. 7, p. 203-214; Altschul, S. F. et al., 1990, Journal of Molecular Biology, Vol. 215, p. 403-410; and Pearson, W. R. et al., 1988, Proc. Natl. Acad. Sci. U.S.A., Vol. 85, p. 2444-2448).

(68) The term “derivative” used herein is meant to include a modified nucleic acid, for example, a derivative labeled with a fluorophore or the like, a derivative containing a modified nucleotide (e.g., a nucleotide containing a group such as halogen, alkyl such as methyl, alkoxy such as methoxy, thio, or carboxymethyl, and a nucleotide that has undergone base rearrangement, double

bond saturation, deamination, replacement of an oxygen molecule with a sulfur atom, etc.), PNA (peptide nucleic acid; Nielsen, P. E. et al., 1991, Science, Vol. 254, p. 1497-500), and LNA (locked nucleic acid; Obika, S. et al., 1998, Tetrahedron Lett., Vol. 39, p. 5401-5404) without any limitation.

(69) As used herein, the “nucleic acid” capable of specifically binding to a polynucleotide selected from the lung cancer marker miRNAs described above is a synthesized or prepared nucleic acid and specifically includes a “nucleic acid probe” or a “primer”. The “nucleic acid” is utilized directly or indirectly for detecting the presence or absence of lung cancer in a subject, for diagnosing the presence or absence of lung cancer, the severity of lung cancer, the presence or absence of amelioration or the degree of amelioration of lung cancer, or the sensitivity of lung cancer for treatment, or for screening for a candidate substance useful in the prevention, amelioration, or treatment of lung cancer. The “nucleic acid” includes a nucleotide, an oligonucleotide, and a polynucleotide capable of specifically recognizing and binding to a transcript represented by any of SEQ ID NOs: 1 to 618 or a synthetic cDNA nucleic acid thereof in vivo, particularly, in a sample such as a body fluid (e.g., blood or urine), in relation to the development of lung cancer. The nucleotide, the oligonucleotide, and the polynucleotide can be effectively used as probes for detecting the aforementioned gene expressed in vivo, in tissues, in cells, or the like on the basis of the properties described above, or as primers for amplifying the aforementioned gene expressed in vivo.

(70) The term “detection” used herein is interchangeable with the term “examination”, “measurement”, “detection”, or “decision support”. As used herein, the term “evaluation” is meant to include diagnosing or evaluation-supporting on the basis of examination results or measurement results.

(71) The term “subject” used herein means a mammal such as a primate including a human and a chimpanzee, a pet animal including a dog and a cat, a livestock animal including cattle, a horse, sheep, and a goat, and a rodent including a mouse and a rat. The term “healthy subject” also means such a mammal without the cancer to be detected.

(72) The term “P” or “P value” used herein refers to a probability at which a more extreme statistic than that actually calculated from data under null hypothesis is observed in a statistical test. Thus, smaller “P” or “P value” means more significant difference between subjects to be compared.

(73) The term “sensitivity” used herein means a value of $(\text{the number of true positives})/(\text{the number of true positives} + \text{the number of false negatives})$. High sensitivity allows lung cancer to be detected early, leading to the complete resection of cancer sites and reduction in the rate of recurrence.

(74) The term “specificity” used herein means a value of $(\text{the number of true negatives})/(\text{the number of true negatives} + \text{the number of false positives})$. High specificity prevents needless extra examination for healthy subjects misjudged as being lung cancer patients, leading to reduction in burden on patients and reduction in medical expense.

(75) The term “accuracy” used herein means a value of $(\text{the number of true positives} + \text{the number of true negatives})/(\text{the total number of cases})$. The accuracy indicates the ratio of samples that correctly identified in the discriminant results to all samples, and serves as a primary index for evaluating detection performance.

(76) As used herein, the “sample” that is subject to determination, detection, or diagnosis refers to a tissue and a biological material in which the expression of the gene of the present invention varies as lung cancer develops, lung cancer progresses, and therapeutic effects on lung cancer are exerted. Specifically, the “sample” refers to a lung tissue, a peripulmonary vascular channel, lymph node, and organ, an organ suspected of having metastasis, the skin, a body fluid such as blood, urine, saliva, sweat, or tissue exudates, serum or plasma prepared from blood, feces, hair, and the like. The “sample” further refers to a biological sample extracted therefrom, specifically, a gene such as RNA or miRNA.

(77) The term “hsa-miR-6768-5p gene” or “hsa-miR-6768-5p” used herein includes the hsa-miR-

6768-5p gene (miRBase Accession No. MIMAT0027436) described in SEQ ID NO: 1, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6768-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6768” (miRBase Accession No. MI0022613, SEQ ID NO: 175) having a hairpin-like structure is known as a precursor of “hsa-miR-6768-5p”.

(78) The term “hsa-miR-6836-3p gene” or “hsa-miR-6836-3p” used herein includes the hsa-miR-6836-3p gene (miRBase Accession No. MIMAT0027575) described in SEQ ID NO: 2, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6836-3p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6836” (miRBase Accession No. MI0022682, SEQ ID NO: 176) having a hairpin-like structure is known as a precursor of “hsa-miR-6836-3p”.

(79) The term “hsa-miR-6782-5p gene” or “hsa-miR-6782-5p” used herein includes the hsa-miR-6782-5p gene (miRBase Accession No. MIMAT0027464) described in SEQ ID NO: 3, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6782-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6782” (miRBase Accession No. MI0022627, SEQ ID NO: 177) having a hairpin-like structure is known as a precursor of “hsa-miR-6782-5p”.

(80) The term “hsa-miR-3663-3p gene” or “hsa-miR-3663-3p” used herein includes the hsa-miR-3663-3p gene (miRBase Accession No. MIMAT0018085) described in SEQ ID NO: 4, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3663-3p gene can be obtained by a method described in Liao J Y et al., 2010, *PLoS One*, Vol. 5, e10563. Also, “hsa-mir-3663” (miRBase Accession No. MI0016064, SEQ ID NO: 178) having a hairpin-like structure is known as a precursor of “hsa-miR-3663-3p”.

(81) The term “hsa-miR-1908-3p gene” or “hsa-miR-1908-3p” used herein includes the hsa-miR-1908-3p gene (miRBase Accession No. MIMAT0026916) described in SEQ ID NO: 5, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1908-3p gene can be obtained by a method described in Bar M et al., 2008, *Stem Cells*, Vol. 26, p. 2496-2505. Also, “hsa-mir-1908” (miRBase Accession No. MI0008329, SEQ ID NO: 179) having a hairpin-like structure is known as a precursor of “hsa-miR-1908-3p”.

(82) The term “hsa-miR-6726-5p gene” or “hsa-miR-6726-5p” used herein includes the hsa-miR-6726-5p gene (miRBase Accession No. MIMAT0027353) described in SEQ ID NO: 6, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6726-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6726” (miRBase Accession No. MI0022571, SEQ ID NO: 180) having a hairpin-like structure is known as a precursor of “hsa-miR-6726-5p”.

(83) The term “hsa-miR-4258 gene” or “hsa-miR-4258” used herein includes the hsa-miR-4258 gene (miRBase Accession No. MIMAT0016879) described in SEQ ID NO: 7, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4258 gene can be obtained by a method described in Goff L A et al., 2009, *PLoS One*, Vol. 4, e7192. Also, “hsa-mir-4258” (miRBase Accession No. M10015857, SEQ ID NO: 181) having a hairpin-like structure is known as a precursor of “hsa-miR-4258”.

(84) The term “hsa-miR-1343-3p gene” or “hsa-miR-1343-3p” used herein includes the hsa-miR-1343-3p gene (miRBase Accession No. MIMAT0019776) described in SEQ ID NO: 8, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1343-3p gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, “hsa-mir-1343” (miRBase Accession No. M10017320, SEQ ID NO: 182) having a hairpin-like structure is known as a precursor of “hsa-miR-1343-3p”.

(85) The term “hsa-miR-4516 gene” or “hsa-miR-4516” used herein includes the hsa-miR-4516 gene (miRBase Accession No. MIMAT0019053) described in SEQ ID NO: 9, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4516 gene can be obtained by a

method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4516” (miRBase Accession No. M10016882, SEQ ID NO: 183) having a hairpin-like structure is known as a precursor of “hsa-miR-4516”.

(86) The term “hsa-miR-6875-5p gene” or “hsa-miR-6875-5p” used herein includes the hsa-miR-6875-5p gene (miRBase Accession No. MIMAT0027650) described in SEQ ID NO: 10, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6875-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6875” (miRBase Accession No. M10022722, SEQ ID NO: 184) having a hairpin-like structure is known as a precursor of “hsa-miR-6875-5p”.

(87) The term “hsa-miR-4651 gene” or “hsa-miR-4651” used herein includes the hsa-miR-4651 gene (miRBase Accession No. MIMAT0019715) described in SEQ ID NO: 11, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4651 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4651” (miRBase Accession No. MI0017279, SEQ ID NO: 185) having a hairpin-like structure is known as a precursor of “hsa-miR-4651”.

(88) The term “hsa-miR-6825-5p gene” or “hsa-miR-6825-5p” used herein includes the hsa-miR-6825-5p gene (miRBase Accession No. MIMAT0027550) described in SEQ ID NO: 12, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6825-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6825” (miRBase Accession No. MI0022670, SEQ ID NO: 186) having a hairpin-like structure is known as a precursor of “hsa-miR-6825-5p”.

(89) The term “hsa-miR-6840-3p gene” or “hsa-miR-6840-3p” used herein includes the hsa-miR-6840-3p gene (miRBase Accession No. MIMAT0027583) described in SEQ ID NO: 13, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6840-3p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6840” (miRBase Accession No. M10022686, SEQ ID NO: 187) having a hairpin-like structure is known as a precursor of “hsa-miR-6840-3p”.

(90) The term “hsa-miR-6780b-5p gene” or “hsa-miR-6780b-5p” used herein includes the hsa-miR-6780b-5p gene (miRBase Accession No. MIMAT0027572) described in SEQ ID NO: 14, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6780b-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6780b” (miRBase Accession No. M10022681, SEQ ID NO: 188) having a hairpin-like structure is known as a precursor of “hsa-miR-6780b-5p”.

(91) The term “hsa-miR-6749-5p gene” or “hsa-miR-6749-5p” used herein includes the hsa-miR-6749-5p gene (miRBase Accession No. MIMAT0027398) described in SEQ ID NO: 15, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6749-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6749” (miRBase Accession No. MI0022594, SEQ ID NO: 189) having a hairpin-like structure is known as a precursor of “hsa-miR-6749-5p”.

(92) The term “hsa-miR-8063 gene” or “hsa-miR-8063” used herein includes the hsa-miR-8063 gene (miRBase Accession No. MIMAT0030990) described in SEQ ID NO: 16, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-8063 gene can be obtained by a method described in Wang H J et al., 2013, Shock, Vol. 39, p. 480-487. Also, “hsa-mir-8063” (miRBase Accession No. MI0025899, SEQ ID NO: 190) having a hairpin-like structure is known as a precursor of “hsa-miR-8063”.

(93) The term “hsa-miR-6784-5p gene” or “hsa-miR-6784-5p” used herein includes the hsa-miR-6784-5p gene (miRBase Accession No. MIMAT0027468) described in SEQ ID NO: 17, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6784-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6784” (miRBase Accession No. MI0022629, SEQ ID NO: 191) having a hairpin-

like structure is known as a precursor of “hsa-miR-6784-5p”.

(94) The term “hsa-miR-3679-5p gene” or “hsa-miR-3679-5p” used herein includes the hsa-miR-3679-5p gene (miRBase Accession No. MIMAT0018104) described in SEQ ID NO: 18, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3679-5p gene can be obtained by a method described in Creighton C J et al., 2010, PLoS One, Vol. 5, e9637. Also, “hsa-mir-3679” (miRBase Accession No. MI0016080, SEQ ID NO: 192) having a hairpin-like structure is known as a precursor of “hsa-miR-3679-5p”.

(95) The term “hsa-miR-3184-5p gene” or “hsa-miR-3184-5p” used herein includes the hsa-miR-3184-5p gene (miRBase Accession No. MIMAT0015064) described in SEQ ID NO: 19, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3184-5p gene can be obtained by a method described in Stark M S et al., 2010, PLoS One, Vol. 5, e9685. Also, “hsa-mir-3184” (miRBase Accession No. MI0014226, SEQ ID NO: 193) having a hairpin-like structure is known as a precursor of “hsa-miR-3184-5p”.

(96) The term “hsa-miR-663b gene” or “hsa-miR-663b” used herein includes the hsa-miR-663b gene (miRBase Accession No. MIMAT0005867) described in SEQ ID NO: 20, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-663b gene can be obtained by a method described in Takada S et al., 2008, Leukemia, Vol. 22, p. 1274-1278. Also, “hsa-mir-663b” (miRBase Accession No. MI0006336, SEQ ID NO: 194) having a hairpin-like structure is known as a precursor of “hsa-miR-663b”.

(97) The term “hsa-miR-6880-5p gene” or “hsa-miR-6880-5p” used herein includes the hsa-miR-6880-5p gene (miRBase Accession No. MIMAT0027660) described in SEQ ID NO: 21, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6880-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6880” (miRBase Accession No. MI0022727, SEQ ID NO: 195) having a hairpin-like structure is known as a precursor of “hsa-miR-6880-5p”.

(98) The term “hsa-miR-1908-5p gene” or “hsa-miR-1908-5p” used herein includes the hsa-miR-1908-5p gene (miRBase Accession No. MIMAT0007881) described in SEQ ID NO: 22, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1908-5p gene can be obtained by a method described in Bar M et al., 2008, Stem Cells, Vol. 26, p. 2496-2505. Also, “hsa-mir-1908” (miRBase Accession No. MI0008329, SEQ ID NO: 179) having a hairpin-like structure is known as a precursor of “hsa-miR-1908-5p”.

(99) The term “hsa-miR-92a-2-5p gene” or “hsa-miR-92a-2-5p” used herein includes the hsa-miR-92a-2-5p gene (miRBase Accession No. MIMAT0004508) described in SEQ ID NO: 23, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-92a-2-5p gene can be obtained by a method described in Mourelatos Z et al., 2002, Genes Dev, Vol. 16, p. 720-728. Also, “hsa-mir-92a-2” (miRBase Accession No. M10000094, SEQ ID NO: 196) having a hairpin-like structure is known as a precursor of “hsa-miR-92a-2-5p”.

(100) The term “hsa-miR-7975 gene” or “hsa-miR-7975” used herein includes the hsa-miR-7975 gene (miRBase Accession No. MIMAT0031178) described in SEQ ID NO: 24, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7975 gene can be obtained by a method described in Velthut-Meikas A et al., 2013, Mol Endocrinol, online. Also, “hsa-mir-7975” (miRBase Accession No. M10025751, SEQ ID NO: 197) having a hairpin-like structure is known as a precursor of “hsa-miR-7975”.

(101) The term “hsa-miR-7110-5p gene” or “hsa-miR-7110-5p” used herein includes the hsa-miR-7110-5p gene (miRBase Accession No. MIMAT0028117) described in SEQ ID NO: 25, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7110-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-7110” (miRBase Accession No. M10022961, SEQ ID NO: 198) having a hairpin-like structure is known as a precursor of “hsa-miR-7110-5p”.

(102) The term “hsa-miR-6842-5p gene” or “hsa-miR-6842-5p” used herein includes the hsa-miR-

6842-5p gene (miRBase Accession No. MIMAT0027586) described in SEQ ID NO: 26, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6842-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6842” (miRBase Accession No. MI0022688, SEQ ID NO: 199) having a hairpin-like structure is known as a precursor of “hsa-miR-6842-5p”.

(103) The term “hsa-miR-6857-5p gene” or “hsa-miR-6857-5p” used herein includes the hsa-miR-6857-5p gene (miRBase Accession No. MIMAT0027614) described in SEQ ID NO: 27, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6857-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6857” (miRBase Accession No. MI0022703, SEQ ID NO: 200) having a hairpin-like structure is known as a precursor of “hsa-miR-6857-5p”.

(104) The term “hsa-miR-5572 gene” or “hsa-miR-5572” used herein includes the hsa-miR-5572 gene (miRBase Accession No. MIMAT0022260) described in SEQ ID NO: 28, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-5572 gene can be obtained by a method described in Tandon M et al., 2012, *Oral Dis*, Vol. 18, p. 127-131. Also, “hsa-mir-5572” (miRBase Accession No. MI0019117, SEQ ID NO: 201) having a hairpin-like structure is known as a precursor of “hsa-miR-5572”.

(105) The term “hsa-miR-3197 gene” or “hsa-miR-3197” used herein includes the hsa-miR-3197 gene (miRBase Accession No. MIMAT0015082) described in SEQ ID NO: 29, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3197 gene can be obtained by a method described in Stark M S et al., 2010, *PLoS One*, Vol. 5, e9685. Also, “hsa-mir-3197” (miRBase Accession No. MI0014245, SEQ ID NO: 202) having a hairpin-like structure is known as a precursor of “hsa-miR-3197”.

(106) The term “hsa-miR-6131 gene” or “hsa-miR-6131” used herein includes the hsa-miR-6131 gene (miRBase Accession No. MIMAT0024615) described in SEQ ID NO: 30, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6131 gene can be obtained by a method described in Dannemann M et al., 2012, *Genome Biol Evol*, Vol. 4, p. 552-564. Also, “hsa-mir-6131” (miRBase Accession No. MI0021276, SEQ ID NO: 203) having a hairpin-like structure is known as a precursor of “hsa-miR-6131”.

(107) The term “hsa-miR-6889-5p gene” or “hsa-miR-6889-5p” used herein includes the hsa-miR-6889-5p gene (miRBase Accession No. MIMAT0027678) described in SEQ ID NO: 31, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6889-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6889” (miRBase Accession No. MI0022736, SEQ ID NO: 204) having a hairpin-like structure is known as a precursor of “hsa-miR-6889-5p”.

(108) The term “hsa-miR-4454 gene” or “hsa-miR-4454” used herein includes the hsa-miR-4454 gene (miRBase Accession No. MIMAT0018976) described in SEQ ID NO: 32, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4454 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4454” (miRBase Accession No. M10016800, SEQ ID NO: 205) having a hairpin-like structure is known as a precursor of “hsa-miR-4454”.

(109) The term “hsa-miR-1199-5p gene” or “hsa-miR-1199-5p” used herein includes the hsa-miR-1199-5p gene (miRBase Accession No. MIMAT0031119) described in SEQ ID NO: 33, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1199-5p gene can be obtained by a method described in Salvi A et al., 2013, *Int J Oncol*, Vol. 42, p. 391-402. Also, “hsa-mir-1199” (miRBase Accession No. MI0020340, SEQ ID NO: 206) having a hairpin-like structure is known as a precursor of “hsa-miR-1199-5p”.

(110) The term “hsa-miR-1247-3p gene” or “hsa-miR-1247-3p” used herein includes the hsa-miR-1247-3p gene (miRBase Accession No. MIMAT0022721) described in SEQ ID NO: 34, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1247-3p gene can be

obtained by a method described in Morin R D et al., 2008, Genome Res, Vol. 18, p. 610-621. Also, “hsa-mir-1247” (miRBase Accession No. MI0006382, SEQ ID NO: 207) having a hairpin-like structure is known as a precursor of “hsa-miR-1247-3p”.

(111) The term “hsa-miR-6800-5p gene” or “hsa-miR-6800-5p” used herein includes the hsa-miR-6800-5p gene (miRBase Accession No. MIMAT0027500) described in SEQ ID NO: 35, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6800-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6800” (miRBase Accession No. M10022645, SEQ ID NO: 208) having a hairpin-like structure is known as a precursor of “hsa-miR-6800-5p”.

(112) The term “hsa-miR-6872-3p gene” or “hsa-miR-6872-3p” used herein includes the hsa-miR-6872-3p gene (miRBase Accession No. MIMAT0027645) described in SEQ ID NO: 36, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6872-3p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6872” (miRBase Accession No. M10022719, SEQ ID NO: 209) having a hairpin-like structure is known as a precursor of “hsa-miR-6872-3p”.

(113) The term “hsa-miR-4649-5p gene” or “hsa-miR-4649-5p” used herein includes the hsa-miR-4649-5p gene (miRBase Accession No. MIMAT0019711) described in SEQ ID NO: 37, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4649-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4649” (miRBase Accession No. MI0017276, SEQ ID NO: 210) having a hairpin-like structure is known as a precursor of “hsa-miR-4649-5p”.

(114) The term “hsa-miR-6791-5p gene” or “hsa-miR-6791-5p” used herein includes the hsa-miR-6791-5p gene (miRBase Accession No. MIMAT0027482) described in SEQ ID NO: 38, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6791-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6791” (miRBase Accession No. MI0022636, SEQ ID NO: 211) having a hairpin-like structure is known as a precursor of “hsa-miR-6791-5p”.

(115) The term “hsa-miR-4433b-3p gene” or “hsa-miR-4433b-3p” used herein includes the hsa-miR-4433b-3p gene (miRBase Accession No. MIMAT0030414) described in SEQ ID NO: 39, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4433b-3p gene can be obtained by a method described in Ple H et al., 2012, PLoS One, Vol. 7, e50746. Also, “hsa-mir-4433b” (miRBase Accession No. MI0025511, SEQ ID NO: 212) having a hairpin-like structure is known as a precursor of “hsa-miR-4433b-3p”.

(116) The term “hsa-miR-3135b gene” or “hsa-miR-3135b” used herein includes the hsa-miR-3135b gene (miRBase Accession No. MIMAT0018985) described in SEQ ID NO: 40, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3135b gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-3135b” (miRBase Accession No. MI0016809, SEQ ID NO: 213) having a hairpin-like structure is known as a precursor of “hsa-miR-3135b”.

(117) The term “hsa-miR-128-2-5p gene” or “hsa-miR-128-2-5p” used herein includes the hsa-miR-128-2-5p gene (miRBase Accession No. MIMAT0031095) described in SEQ ID NO: 41, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-128-2-5p gene can be obtained by a method described in Lagos-Quintana M et al., 2002, Curr Biol, Vol. 12, p. 735-739. Also, “hsa-mir-128-2” (miRBase Accession No. MI0000727, SEQ ID NO: 214) having a hairpin-like structure is known as a precursor of “hsa-miR-128-2-5p”.

(118) The term “hsa-miR-4675 gene” or “hsa-miR-4675” used herein includes the hsa-miR-4675 gene (miRBase Accession No. MIMAT0019757) described in SEQ ID NO: 42, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4675 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4675” (miRBase Accession No. MI0017306, SEQ ID NO: 215) having a hairpin-like structure is known

as a precursor of “hsa-miR-4675”.

(119) The term “hsa-miR-4472 gene” or “hsa-miR-4472” used herein includes the hsa-miR-4472 gene (miRBase Accession No. MIMAT0018999) described in SEQ ID NO: 43, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4472 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4472-1 and hsa-mir-4472-2” (miRBase Accession Nos. M10016823 and M10016824, SEQ ID NOs: 216 and 217) having a hairpin-like structure are known as precursors of “hsa-miR-4472”.

(120) The term “hsa-miR-6785-5p gene” or “hsa-miR-6785-5p” used herein includes the hsa-miR-6785-5p gene (miRBase Accession No. MIMAT0027470) described in SEQ ID NO: 44, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6785-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6785” (miRBase Accession No. MI0022630, SEQ ID NO: 218) having a hairpin-like structure is known as a precursor of “hsa-miR-6785-5p”.

(121) The term “hsa-miR-6741-5p gene” or “hsa-miR-6741-5p” used herein includes the hsa-miR-6741-5p gene (miRBase Accession No. MIMAT0027383) described in SEQ ID NO: 45, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6741-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6741” (miRBase Accession No. M10022586, SEQ ID NO: 219) having a hairpin-like structure is known as a precursor of “hsa-miR-6741-5p”.

(122) The term “hsa-miR-7977 gene” or “hsa-miR-7977” used herein includes the hsa-miR-7977 gene (miRBase Accession No. MIMAT0031180) described in SEQ ID NO: 46, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7977 gene can be obtained by a method described in Velthut-Meikas A et al., 2013, Mol Endocrinol, online. Also, “hsa-mir-7977” (miRBase Accession No. MI0025753, SEQ ID NO: 220) having a hairpin-like structure is known as a precursor of “hsa-miR-7977”.

(123) The term “hsa-miR-3665 gene” or “hsa-miR-3665” used herein includes the hsa-miR-3665 gene (miRBase Accession No. MIMAT0018087) described in SEQ ID NO: 47, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3665 gene can be obtained by a method described in Xie X et al., 2005, Nature, Vol. 434, p. 338-345. Also, “hsa-mir-3665” (miRBase Accession No. MI0016066, SEQ ID NO: 221) having a hairpin-like structure is known as a precursor of “hsa-miR-3665”.

(124) The term “hsa-miR-128-1-5p gene” or “hsa-miR-128-1-5p” used herein includes the hsa-miR-128-1-5p gene (miRBase Accession No. MIMAT0026477) described in SEQ ID NO: 48, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-128-1-5p gene can be obtained by a method described in Lagos-Quintana M et al., 2002, Curr Biol, Vol. 12, p. 735-739. Also, “hsa-mir-128-1” (miRBase Accession No. MI0000447, SEQ ID NO: 222) having a hairpin-like structure is known as a precursor of “hsa-miR-128-1-5p”.

(125) The term “hsa-miR-4286 gene” or “hsa-miR-4286” used herein includes the hsa-miR-4286 gene (miRBase Accession No. MIMAT0016916) described in SEQ ID NO: 49, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4286 gene can be obtained by a method described in Goff L A et al., 2009, PLoS One, Vol. 4, e7192. Also, “hsa-mir-4286” (miRBase Accession No. MI0015894, SEQ ID NO: 223) having a hairpin-like structure is known as a precursor of “hsa-miR-4286”.

(126) The term “hsa-miR-6765-3p gene” or “hsa-miR-6765-3p” used herein includes the hsa-miR-6765-3p gene (miRBase Accession No. MIMAT0027431) described in SEQ ID NO: 50, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6765-3p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6765” (miRBase Accession No. M10022610, SEQ ID NO: 224) having a hairpin-like structure is known as a precursor of “hsa-miR-6765-3p”.

(127) The term “hsa-miR-4632-5p gene” or “hsa-miR-4632-5p” used herein includes the hsa-miR-

4632-5p gene (miRBase Accession No. MIMAT0022977) described in SEQ ID NO: 51, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4632-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4632” (miRBase Accession No. MI0017259, SEQ ID NO: 225) having a hairpin-like structure is known as a precursor of “hsa-miR-4632-5p”.

(128) The term “hsa-miR-365a-5p gene” or “hsa-miR-365a-5p” used herein includes the hsa-miR-365a-5p gene (miRBase Accession No. MIMAT0009199) described in SEQ ID NO: 52, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-365a-5p gene can be obtained by a method described in Xie X et al., 2005, Nature, Vol. 434, p. 338-345. Also, “hsa-mir-365a” (miRBase Accession No. MI0000767, SEQ ID NO: 226) having a hairpin-like structure is known as a precursor of “hsa-miR-365a-5p”.

(129) The term “hsa-miR-6088 gene” or “hsa-miR-6088” used herein includes the hsa-miR-6088 gene (miRBase Accession No. MIMAT0023713) described in SEQ ID NO: 53, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6088 gene can be obtained by a method described in Yoo J K et al., 2012, Stem Cells Dev, Vol. 21, p. 2049-2057. Also, “hsa-mir-6088” (miRBase Accession No. MI0020365, SEQ ID NO: 227) having a hairpin-like structure is known as a precursor of “hsa-miR-6088”.

(130) The term “hsa-miR-6816-5p gene” or “hsa-miR-6816-5p” used herein includes the hsa-miR-6816-5p gene (miRBase Accession No. MIMAT0027532) described in SEQ ID NO: 54, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6816-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6816” (miRBase Accession No. MI0022661, SEQ ID NO: 228) having a hairpin-like structure is known as a precursor of “hsa-miR-6816-5p”.

(131) The term “hsa-miR-6885-5p gene” or “hsa-miR-6885-5p” used herein includes the hsa-miR-6885-5p gene (miRBase Accession No. MIMAT0027670) described in SEQ ID NO: 55, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6885-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6885” (miRBase Accession No. M10022732, SEQ ID NO: 229) having a hairpin-like structure is known as a precursor of “hsa-miR-6885-5p”.

(132) The term “hsa-miR-711 gene” or “hsa-miR-711” used herein includes the hsa-miR-711 gene (miRBase Accession No. MIMAT0012734) described in SEQ ID NO: 56, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-711 gene can be obtained by a method described in Artzi S et al., 2008, BMC Bioinformatics, Vol. 9, p. 39. Also, “hsa-mir-711” (miRBase Accession No. M10012488, SEQ ID NO: 230) having a hairpin-like structure is known as a precursor of “hsa-miR-711”.

(133) The term “hsa-miR-6765-5p gene” or “hsa-miR-6765-5p” used herein includes the hsa-miR-6765-5p gene (miRBase Accession No. MIMAT0027430) described in SEQ ID NO: 57, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6765-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6765” (miRBase Accession No. MI0022610, SEQ ID NO: 224) having a hairpin-like structure is known as a precursor of “hsa-miR-6765-5p”.

(134) The term “hsa-miR-3180 gene” or “hsa-miR-3180” used herein includes the hsa-miR-3180 gene (miRBase Accession No. MIMAT0018178) described in SEQ ID NO: 58, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3180 gene can be obtained by a method described in Creighton C J et al., 2010, PLoS One, Vol. 5, e9637. Also, “hsa-mir-3180-4 and hsa-mir-3180-5” (miRBase Accession Nos. MI0016408 and MI0016409, SEQ ID NOs: 231 and 232) having a hairpin-like structure are known as precursors of “hsa-miR-3180”.

(135) The term “hsa-miR-4442 gene” or “hsa-miR-4442” used herein includes the hsa-miR-4442 gene (miRBase Accession No. MIMAT0018960) described in SEQ ID NO: 59, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4442 gene can be obtained by a

method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4442” (miRBase Accession No. M10016785, SEQ ID NO: 233) having a hairpin-like structure is known as a precursor of “hsa-miR-4442”.

(136) The term “hsa-miR-4792 gene” or “hsa-miR-4792” used herein includes the hsa-miR-4792 gene (miRBase Accession No. MIMAT0019964) described in SEQ ID NO: 60, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4792 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4792” (miRBase Accession No. M10017439, SEQ ID NO: 234) having a hairpin-like structure is known as a precursor of “hsa-miR-4792”.

(137) The term “hsa-miR-6721-5p gene” or “hsa-miR-6721-5p” used herein includes the hsa-miR-6721-5p gene (miRBase Accession No. MIMAT0025852) described in SEQ ID NO: 61, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6721-5p gene can be obtained by a method described in Li Y et al., 2012, Gene, Vol. 497, p. 330-335. Also, “hsa-mir-6721” (miRBase Accession No. M10022556, SEQ ID NO: 235) having a hairpin-like structure is known as a precursor of “hsa-miR-6721-5p”.

(138) The term “hsa-miR-6798-5p gene” or “hsa-miR-6798-5p” used herein includes the hsa-miR-6798-5p gene (miRBase Accession No. MIMAT0027496) described in SEQ ID NO: 62, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6798-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6798” (miRBase Accession No. M10022643, SEQ ID NO: 236) having a hairpin-like structure is known as a precursor of “hsa-miR-6798-5p”.

(139) The term “hsa-miR-3162-5p gene” or “hsa-miR-3162-5p” used herein includes the hsa-miR-3162-5p gene (miRBase Accession No. MIMAT0015036) described in SEQ ID NO: 63, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3162-5p gene can be obtained by a method described in Stark M S et al., 2010, PLoS One, Vol. 5, e9685. Also, “hsa-mir-3162” (miRBase Accession No. MI0014192, SEQ ID NO: 237) having a hairpin-like structure is known as a precursor of “hsa-miR-3162-5p”.

(140) The term “hsa-miR-6126 gene” or “hsa-miR-6126” used herein includes the hsa-miR-6126 gene (miRBase Accession No. MIMAT0024599) described in SEQ ID NO: 64, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6126 gene can be obtained by a method described in Smith J L et al., 2012, J Virol, Vol. 86, p. 5278-5287. Also, “hsa-mir-6126” (miRBase Accession No. MI0021260, SEQ ID NO: 238) having a hairpin-like structure is known as a precursor of “hsa-miR-6126”.

(141) The term “hsa-miR-4758-5p gene” or “hsa-miR-4758-5p” used herein includes the hsa-miR-4758-5p gene (miRBase Accession No. MIMAT0019903) described in SEQ ID NO: 65, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4758-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4758” (miRBase Accession No. MI0017399, SEQ ID NO: 239) having a hairpin-like structure is known as a precursor of “hsa-miR-4758-5p”.

(142) The term “hsa-miR-2392 gene” or “hsa-miR-2392” used herein includes the hsa-miR-2392 gene (miRBase Accession No. MIMAT0019043) described in SEQ ID NO: 66, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-2392 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-2392” (miRBase Accession No. MI0016870, SEQ ID NO: 240) having a hairpin-like structure is known as a precursor of “hsa-miR-2392”.

(143) The term “hsa-miR-486-3p gene” or “hsa-miR-486-3p” used herein includes the hsa-miR-486-3p gene (miRBase Accession No. MIMAT0004762) described in SEQ ID NO: 67, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-486-3p gene can be obtained by a method described in Fu H et al., 2005, FEBS Lett, Vol. 579, p. 3849-3854. Also, “hsa-mir-486 and hsa-mir-486-2” (miRBase Accession Nos. M10002470 and M10023622, SEQ ID

NOs: 241 and 242) having a hairpin-like structure are known as precursors of “hsa-miR-486-3p”.

(144) The term “hsa-miR-6727-5p gene” or “hsa-miR-6727-5p” used herein includes the hsa-miR-6727-5p gene (miRBase Accession No. MIMAT0027355) described in SEQ ID NO: 68, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6727-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6727” (miRBase Accession No. MI0022572, SEQ ID NO: 243) having a hairpin-like structure is known as a precursor of “hsa-miR-6727-5p”.

(145) The term “hsa-miR-4728-5p gene” or “hsa-miR-4728-5p” used herein includes the hsa-miR-4728-5p gene (miRBase Accession No. MIMAT0019849) described in SEQ ID NO: 69, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4728-5p gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, “hsa-mir-4728” (miRBase Accession No. MI0017365, SEQ ID NO: 244) having a hairpin-like structure is known as a precursor of “hsa-miR-4728-5p”.

(146) The term “hsa-miR-6746-5p gene” or “hsa-miR-6746-5p” used herein includes the hsa-miR-6746-5p gene (miRBase Accession No. MIMAT0027392) described in SEQ ID NO: 70, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6746-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6746” (miRBase Accession No. MI0022591, SEQ ID NO: 245) having a hairpin-like structure is known as a precursor of “hsa-miR-6746-5p”.

(147) The term “hsa-miR-4270 gene” or “hsa-miR-4270” used herein includes the hsa-miR-4270 gene (miRBase Accession No. MIMAT0016900) described in SEQ ID NO: 71, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4270 gene can be obtained by a method described in Goff L A et al., 2009, *PLoS One*, Vol. 4, e7192. Also, “hsa-mir-4270” (miRBase Accession No. MI0015878, SEQ ID NO: 246) having a hairpin-like structure is known as a precursor of “hsa-miR-4270”.

(148) The term “hsa-miR-3940-5p gene” or “hsa-miR-3940-5p” used herein includes the hsa-miR-3940-5p gene (miRBase Accession No. MIMAT0019229) described in SEQ ID NO: 72, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3940-5p gene can be obtained by a method described in Liao J Y et al., 2010, *PLoS One*, Vol. 5, e10563. Also, “hsa-mir-3940” (miRBase Accession No. MI0016597, SEQ ID NO: 247) having a hairpin-like structure is known as a precursor of “hsa-miR-3940-5p”.

(149) The term “hsa-miR-4725-3p gene” or “hsa-miR-4725-3p” used herein includes the hsa-miR-4725-3p gene (miRBase Accession No. MIMAT0019844) described in SEQ ID NO: 73, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4725-3p gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, “hsa-mir-4725” (miRBase Accession No. MI0017362, SEQ ID NO: 248) having a hairpin-like structure is known as a precursor of “hsa-miR-4725-3p”.

(150) The term “hsa-miR-7108-5p gene” or “hsa-miR-7108-5p” used herein includes the hsa-miR-7108-5p gene (miRBase Accession No. MIMAT0028113) described in SEQ ID NO: 74, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7108-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-7108” (miRBase Accession No. MI0022959, SEQ ID NO: 249) having a hairpin-like structure is known as a precursor of “hsa-miR-7108-5p”.

(151) The term “hsa-miR-3656 gene” or “hsa-miR-3656” used herein includes the hsa-miR-3656 gene (miRBase Accession No. MIMAT0018076) described in SEQ ID NO: 75, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3656 gene can be obtained by a method described in Meiri E et al., 2010, *Nucleic Acids Res*, Vol. 38, p. 6234-6246. Also, “hsa-mir-3656” (miRBase Accession No. MI0016056, SEQ ID NO: 250) having a hairpin-like structure is known as a precursor of “hsa-miR-3656”.

(152) The term “hsa-miR-6879-5p gene” or “hsa-miR-6879-5p” used herein includes the hsa-miR-

6879-5p gene (miRBase Accession No. MIMAT0027658) described in SEQ ID NO: 76, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6879-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6879” (miRBase Accession No. MI0022726, SEQ ID NO: 251) having a hairpin-like structure is known as a precursor of “hsa-miR-6879-5p”.

(153) The term “hsa-miR-6738-5p gene” or “hsa-miR-6738-5p” used herein includes the hsa-miR-6738-5p gene (miRBase Accession No. MIMAT0027377) described in SEQ ID NO: 77, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6738-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6738” (miRBase Accession No. MI0022583, SEQ ID NO: 252) having a hairpin-like structure is known as a precursor of “hsa-miR-6738-5p”.

(154) The term “hsa-miR-1260a gene” or “hsa-miR-1260a” used herein includes the hsa-miR-1260a gene (miRBase Accession No. MIMAT0005911) described in SEQ ID NO: 78, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1260a gene can be obtained by a method described in Morin R D et al., 2008, Genome Res, Vol. 18, p. 610-621. Also, “hsa-mir-1260a” (miRBase Accession No. MI0006394, SEQ ID NO: 253) having a hairpin-like structure is known as a precursor of “hsa-miR-1260a”.

(155) The term “hsa-miR-4446-3p gene” or “hsa-miR-4446-3p” used herein includes the hsa-miR-4446-3p gene (miRBase Accession No. MIMAT0018965) described in SEQ ID NO: 79, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4446-3p gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4446” (miRBase Accession No. M10016789, SEQ ID NO: 254) having a hairpin-like structure is known as a precursor of “hsa-miR-4446-3p”.

(156) The term “hsa-miR-3131 gene” or “hsa-miR-3131” used herein includes the hsa-miR-3131 gene (miRBase Accession No. MIMAT0014996) described in SEQ ID NO: 80, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3131 gene can be obtained by a method described in Stark M S et al., 2010, PLoS One, Vol. 5, e9685. Also, “hsa-mir-3131” (miRBase Accession No. MI0014151, SEQ ID NO: 255) having a hairpin-like structure is known as a precursor of “hsa-miR-3131”.

(157) The term “hsa-miR-4463 gene” or “hsa-miR-4463” used herein includes the hsa-miR-4463 gene (miRBase Accession No. MIMAT0018987) described in SEQ ID NO: 81, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4463 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4463” (miRBase Accession No. M10016811, SEQ ID NO: 256) having a hairpin-like structure is known as a precursor of “hsa-miR-4463”.

(158) The term “hsa-miR-3185 gene” or “hsa-miR-3185” used herein includes the hsa-miR-3185 gene (miRBase Accession No. MIMAT0015065) described in SEQ ID NO: 82, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3185 gene can be obtained by a method described in Stark M S et al., 2010, PLoS One, Vol. 5, e9685. Also, “hsa-mir-3185” (miRBase Accession No. MI0014227, SEQ ID NO: 257) having a hairpin-like structure is known as a precursor of “hsa-miR-3185”.

(159) The term “hsa-miR-6870-5p gene” or “hsa-miR-6870-5p” used herein includes the hsa-miR-6870-5p gene (miRBase Accession No. MIMAT0027640) described in SEQ ID NO: 83, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6870-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6870” (miRBase Accession No. M10022717, SEQ ID NO: 258) having a hairpin-like structure is known as a precursor of “hsa-miR-6870-5p”.

(160) The term “hsa-miR-6779-5p gene” or “hsa-miR-6779-5p” used herein includes the hsa-miR-6779-5p gene (miRBase Accession No. MIMAT0027458) described in SEQ ID NO: 84, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6779-5p gene can be

obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6779” (miRBase Accession No. M10022624, SEQ ID NO: 259) having a hairpin-like structure is known as a precursor of “hsa-miR-6779-5p”.

(161) The term “hsa-miR-1273g-3p gene” or “hsa-miR-1273g-3p” used herein includes the hsa-miR-1273g-3p gene (miRBase Accession No. MIMAT0022742) described in SEQ ID NO: 85, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1273g-3p gene can be obtained by a method described in Reshmi G et al., 2011, Genomics, Vol. 97, p. 333-340. Also, “hsa-mir-1273g” (miRBase Accession No. MI0018003, SEQ ID NO: 260) having a hairpin-like structure is known as a precursor of “hsa-miR-1273g-3p”.

(162) The term “hsa-miR-8059 gene” or “hsa-miR-8059” used herein includes the hsa-miR-8059 gene (miRBase Accession No. MIMAT0030986) described in SEQ ID NO: 86, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-8059 gene can be obtained by a method described in Wang H J et al., 2013, Shock, Vol. 39, p. 480-487. Also, “hsa-mir-8059” (miRBase Accession No. MI0025895, SEQ ID NO: 261) having a hairpin-like structure is known as a precursor of “hsa-miR-8059”.

(163) The term “hsa-miR-4697-5p gene” or “hsa-miR-4697-5p” used herein includes the hsa-miR-4697-5p gene (miRBase Accession No. MIMAT0019791) described in SEQ ID NO: 87, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4697-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4697” (miRBase Accession No. MI0017330, SEQ ID NO: 262) having a hairpin-like structure is known as a precursor of “hsa-miR-4697-5p”.

(164) The term “hsa-miR-4674 gene” or “hsa-miR-4674” used herein includes the hsa-miR-4674 gene (miRBase Accession No. MIMAT0019756) described in SEQ ID NO: 88, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4674 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4674” (miRBase Accession No. MI0017305, SEQ ID NO: 263) having a hairpin-like structure is known as a precursor of “hsa-miR-4674”.

(165) The term “hsa-miR-4433-3p gene” or “hsa-miR-4433-3p” used herein includes the hsa-miR-4433-3p gene (miRBase Accession No. MIMAT0018949) described in SEQ ID NO: 89, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4433-3p gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4433” (miRBase Accession No. MI0016773, SEQ ID NO: 264) having a hairpin-like structure is known as a precursor of “hsa-miR-4433-3p”.

(166) The term “hsa-miR-4257 gene” or “hsa-miR-4257” used herein includes the hsa-miR-4257 gene (miRBase Accession No. MIMAT0016878) described in SEQ ID NO: 90, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4257 gene can be obtained by a method described in Goff L A et al., 2009, PLoS One, Vol. 4, e7192. Also, “hsa-mir-4257” (miRBase Accession No. MI0015856, SEQ ID NO: 265) having a hairpin-like structure is known as a precursor of “hsa-miR-4257”.

(167) The term “hsa-miR-1915-5p gene” or “hsa-miR-1915-5p” used herein includes the hsa-miR-1915-5p gene (miRBase Accession No. MIMAT0007891) described in SEQ ID NO: 91, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1915-5p gene can be obtained by a method described in Bar M et al., 2008, Stem Cells, Vol. 26, p. 2496-2505. Also, “hsa-mir-1915” (miRBase Accession No. MI0008336, SEQ ID NO: 266) having a hairpin-like structure is known as a precursor of “hsa-miR-1915-5p”.

(168) The term “hsa-miR-4417 gene” or “hsa-miR-4417” used herein includes the hsa-miR-4417 gene (miRBase Accession No. MIMAT0018929) described in SEQ ID NO: 92, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4417 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4417” (miRBase Accession No. MI0016753, SEQ ID NO: 267) having a hairpin-like structure is known

as a precursor of “hsa-miR-4417”.

(169) The term “hsa-miR-1343-5p gene” or “hsa-miR-1343-5p” used herein includes the hsa-miR-1343-5p gene (miRBase Accession No. MIMAT0027038) described in SEQ ID NO: 93, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1343-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-1343” (miRBase Accession No. MI0017320, SEQ ID NO: 182) having a hairpin-like structure is known as a precursor of “hsa-miR-1343-5p”.

(170) The term “hsa-miR-6781-5p gene” or “hsa-miR-6781-5p” used herein includes the hsa-miR-6781-5p gene (miRBase Accession No. MIMAT0027462) described in SEQ ID NO: 94, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6781-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6781” (miRBase Accession No. MI0022626, SEQ ID NO: 268) having a hairpin-like structure is known as a precursor of “hsa-miR-6781-5p”.

(171) The term “hsa-miR-4695-5p gene” or “hsa-miR-4695-5p” used herein includes the hsa-miR-4695-5p gene (miRBase Accession No. MIMAT0019788) described in SEQ ID NO: 95, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4695-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4695” (miRBase Accession No. MI0017328, SEQ ID NO: 269) having a hairpin-like structure is known as a precursor of “hsa-miR-4695-5p”.

(172) The term “hsa-miR-1237-5p gene” or “hsa-miR-1237-5p” used herein includes the hsa-miR-1237-5p gene (miRBase Accession No. MIMAT0022946) described in SEQ ID NO: 96, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1237-5p gene can be obtained by a method described in Berezikov E et al., 2007, Mol Cell, Vol. 28, p. 328-336. Also, “hsa-mir-1237” (miRBase Accession No. MI0006327, SEQ ID NO: 270) having a hairpin-like structure is known as a precursor of “hsa-miR-1237-5p”.

(173) The term “hsa-miR-6775-5p gene” or “hsa-miR-6775-5p” used herein includes the hsa-miR-6775-5p gene (miRBase Accession No. MIMAT0027450) described in SEQ ID NO: 97, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6775-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6775” (miRBase Accession No. MI0022620, SEQ ID NO: 271) having a hairpin-like structure is known as a precursor of “hsa-miR-6775-5p”.

(174) The term “hsa-miR-7845-5p gene” or “hsa-miR-7845-5p” used herein includes the hsa-miR-7845-5p gene (miRBase Accession No. MIMAT0030420) described in SEQ ID NO: 98, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7845-5p gene can be obtained by a method described in Ple H et al., 2012, PLoS One, Vol. 7, e50746. Also, “hsa-mir-7845” (miRBase Accession No. M10025515, SEQ ID NO: 272) having a hairpin-like structure is known as a precursor of “hsa-miR-7845-5p”.

(175) The term “hsa-miR-4746-3p gene” or “hsa-miR-4746-3p” used herein includes the hsa-miR-4746-3p gene (miRBase Accession No. MIMAT0019881) described in SEQ ID NO: 99, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4746-3p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4746” (miRBase Accession No. MI0017385, SEQ ID NO: 273) having a hairpin-like structure is known as a precursor of “hsa-miR-4746-3p”.

(176) The term “hsa-miR-7641 gene” or “hsa-miR-7641” used herein includes the hsa-miR-7641 gene (miRBase Accession No. MIMAT0029782) described in SEQ ID NO: 100, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7641 gene can be obtained by a method described in Yoo J K et al., 2013, Arch Pharm Res, Vol. 36, p. 353-358. Also, “hsa-mir-7641-1 and hsa-mir-7641-2” (miRBase Accession Nos. M10024975 and M10024976, SEQ ID NOs: 274 and 275) having a hairpin-like structure are known as precursors of “hsa-miR-7641”.

(177) The term “hsa-miR-7847-3p gene” or “hsa-miR-7847-3p” used herein includes the hsa-miR-

7847-3p gene (miRBase Accession No. MIMAT0030422) described in SEQ ID NO: 101, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7847-3p gene can be obtained by a method described in Ple H et al., 2012, PLoS One, Vol. 7, e50746. Also, “hsa-mir-7847” (miRBase Accession No. MI0025517, SEQ ID NO: 276) having a hairpin-like structure is known as a precursor of “hsa-miR-7847-3p”.

(178) The term “hsa-miR-6806-5p gene” or “hsa-miR-6806-5p” used herein includes the hsa-miR-6806-5p gene (miRBase Accession No. MIMAT0027512) described in SEQ ID NO: 102, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6806-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6806” (miRBase Accession No. MI0022651, SEQ ID NO: 277) having a hairpin-like structure is known as a precursor of “hsa-miR-6806-5p”.

(179) The term “hsa-miR-4467 gene” or “hsa-miR-4467” used herein includes the hsa-miR-4467 gene (miRBase Accession No. MIMAT0018994) described in SEQ ID NO: 103, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4467 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4467” (miRBase Accession No. MI0016818, SEQ ID NO: 278) having a hairpin-like structure is known as a precursor of “hsa-miR-4467”.

(180) The term “hsa-miR-4726-5p gene” or “hsa-miR-4726-5p” used herein includes the hsa-miR-4726-5p gene (miRBase Accession No. MIMAT0019845) described in SEQ ID NO: 104, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4726-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4726” (miRBase Accession No. M10017363, SEQ ID NO: 279) having a hairpin-like structure is known as a precursor of “hsa-miR-4726-5p”.

(181) The term “hsa-miR-4648 gene” or “hsa-miR-4648” used herein includes the hsa-miR-4648 gene (miRBase Accession No. MIMAT0019710) described in SEQ ID NO: 105, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4648 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4648” (miRBase Accession No. MI0017275, SEQ ID NO: 280) having a hairpin-like structure is known as a precursor of “hsa-miR-4648”.

(182) The term “hsa-miR-6089 gene” or “hsa-miR-6089” used herein includes the hsa-miR-6089 gene (miRBase Accession No. MIMAT0023714) described in SEQ ID NO: 106, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6089 gene can be obtained by a method described in Yoo J K et al., 2012, Stem Cells Dev, Vol. 21, p. 2049-2057. Also, “hsa-mir-6089-1 and hsa-mir-6089-2” (miRBase Accession Nos. M10020366 and M10023563, SEQ ID NOs: 281 and 282) having a hairpin-like structure are known as precursors of “hsa-miR-6089”.

(183) The term “hsa-miR-1260b gene” or “hsa-miR-1260b” used herein includes the hsa-miR-1260b gene (miRBase Accession No. MIMAT0015041) described in SEQ ID NO: 107, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1260b gene can be obtained by a method described in Stark M S et al., 2010, PLoS One, Vol. 5, e9685. Also, “hsa-mir-1260b” (miRBase Accession No. MI0014197, SEQ ID NO: 283) having a hairpin-like structure is known as a precursor of “hsa-miR-1260b”.

(184) The term “hsa-miR-4532 gene” or “hsa-miR-4532” used herein includes the hsa-miR-4532 gene (miRBase Accession No. MIMAT0019071) described in SEQ ID NO: 108, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4532 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4532” (miRBase Accession No. MI0016899, SEQ ID NO: 284) having a hairpin-like structure is known as a precursor of “hsa-miR-4532”.

(185) The term “hsa-miR-5195-3p gene” or “hsa-miR-5195-3p” used herein includes the hsa-miR-5195-3p gene (miRBase Accession No. MIMAT0021127) described in SEQ ID NO: 109, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-5195-3p gene

can be obtained by a method described in Schotte D et al., 2011, *Leukemia*, Vol. 25, p. 1389-1399. Also, “hsa-mir-5195” (miRBase Accession No. MI0018174, SEQ ID NO: 285) having a hairpin-like structure is known as a precursor of “hsa-miR-5195-3p”.

(186) The term “hsa-miR-3188 gene” or “hsa-miR-3188” used herein includes the hsa-miR-3188 gene (miRBase Accession No. MIMAT0015070) described in SEQ ID NO: 110, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3188 gene can be obtained by a method described in Stark M S et al., 2010, *PLoS One*, Vol. 5, e9685. Also, “hsa-mir-3188” (miRBase Accession No. MI0014232, SEQ ID NO: 286) having a hairpin-like structure is known as a precursor of “hsa-miR-3188”.

(187) The term “hsa-miR-6848-5p gene” or “hsa-miR-6848-5p” used herein includes the hsa-miR-6848-5p gene (miRBase Accession No. MIMAT0027596) described in SEQ ID NO: 111, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6848-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6848” (miRBase Accession No. MI0022694, SEQ ID NO: 287) having a hairpin-like structure is known as a precursor of “hsa-miR-6848-5p”.

(188) The term “hsa-miR-1233-5p gene” or “hsa-miR-1233-5p” used herein includes the hsa-miR-1233-5p gene (miRBase Accession No. MIMAT0022943) described in SEQ ID NO: 112, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1233-5p gene can be obtained by a method described in Berezikov E et al., 2007, *Mol Cell*, Vol. 28, p. 328-336. Also, “hsa-mir-1233-1 and hsa-mir-1233-2” (miRBase Accession Nos. MI0006323 and M10015973, SEQ ID NOs: 288 and 289) having a hairpin-like structure are known as precursors of “hsa-miR-1233-5p”.

(189) The term “hsa-miR-6717-5p gene” or “hsa-miR-6717-5p” used herein includes the hsa-miR-6717-5p gene (miRBase Accession No. MIMAT0025846) described in SEQ ID NO: 113, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6717-5p gene can be obtained by a method described in Li Y et al., 2012, *Gene*, Vol. 497, p. 330-335. Also, “hsa-mir-6717” (miRBase Accession No. MI0022551, SEQ ID NO: 290) having a hairpin-like structure is known as a precursor of “hsa-miR-6717-5p”.

(190) The term “hsa-miR-3195 gene” or “hsa-miR-3195” used herein includes the hsa-miR-3195 gene (miRBase Accession No. MIMAT0015079) described in SEQ ID NO: 114, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3195 gene can be obtained by a method described in Stark M S et al., 2010, *PLoS One*, Vol. 5, e9685. Also, “hsa-mir-3195” (miRBase Accession No. MI0014240, SEQ ID NO: 291) having a hairpin-like structure is known as a precursor of “hsa-miR-3195”.

(191) The term “hsa-miR-6757-5p gene” or “hsa-miR-6757-5p” used herein includes the hsa-miR-6757-5p gene (miRBase Accession No. MIMAT0027414) described in SEQ ID NO: 115, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6757-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6757” (miRBase Accession No. MI0022602, SEQ ID NO: 292) having a hairpin-like structure is known as a precursor of “hsa-miR-6757-5p”.

(192) The term “hsa-miR-8072 gene” or “hsa-miR-8072” used herein includes the hsa-miR-8072 gene (miRBase Accession No. MIMAT0030999) described in SEQ ID NO: 116, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-8072 gene can be obtained by a method described in Wang H J et al., 2013, *Shock*, Vol. 39, p. 480-487. Also, “hsa-mir-8072” (miRBase Accession No. MI0025908, SEQ ID NO: 293) having a hairpin-like structure is known as a precursor of “hsa-miR-8072”.

(193) The term “hsa-miR-4745-5p gene” or “hsa-miR-4745-5p” used herein includes the hsa-miR-4745-5p gene (miRBase Accession No. MIMAT0019878) described in SEQ ID NO: 117, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4745-5p gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86.

Also, “hsa-mir-4745” (miRBase Accession No. M10017384, SEQ ID NO: 294) having a hairpin-like structure is known as a precursor of “hsa-miR-4745-5p”.

(194) The term “hsa-miR-6511a-5p gene” or “hsa-miR-6511a-5p” used herein includes the hsa-miR-6511a-5p gene (miRBase Accession No. MIMAT0025478) described in SEQ ID NO: 118, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6511a-5p gene can be obtained by a method described in Joyce C E et al., 2011, Hum Mol Genet, Vol. 20, p. 4025-4040. Also, “hsa-mir-6511a-1, hsa-mir-6511a-2, hsa-mir-6511a-3, and hsa-mir-6511a-4” (miRBase Accession Nos. MI0022223, MI0023564, MI0023565, and MI0023566, SEQ ID NOs: 295, 296, 297, and 298) having a hairpin-like structure are known as precursors of “hsa-miR-6511a-5p”.

(195) The term “hsa-miR-6776-5p gene” or “hsa-miR-6776-5p” used herein includes the hsa-miR-6776-5p gene (miRBase Accession No. MIMAT0027452) described in SEQ ID NO: 119, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6776-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6776” (miRBase Accession No. MI0022621, SEQ ID NO: 299) having a hairpin-like structure is known as a precursor of “hsa-miR-6776-5p”.

(196) The term “hsa-miR-371a-5p gene” or “hsa-miR-371a-5p” used herein includes the hsa-miR-371a-5p gene (miRBase Accession No. MIMAT0004687) described in SEQ ID NO: 120, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-371a-5p gene can be obtained by a method described in Suh M R et al., 2004, Dev Biol, Vol. 270, p. 488-498. Also, “hsa-mir-371a” (miRBase Accession No. MI0000779, SEQ ID NO: 300) having a hairpin-like structure is known as a precursor of “hsa-miR-371a-5p”.

(197) The term “hsa-miR-1227-5p gene” or “hsa-miR-1227-5p” used herein includes the hsa-miR-1227-5p gene (miRBase Accession No. MIMAT0022941) described in SEQ ID NO: 121, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1227-5p gene can be obtained by a method described in Berezikov E et al., 2007, Mol Cell, Vol. 28, p. 328-336. Also, “hsa-mir-1227” (miRBase Accession No. MI0006316, SEQ ID NO: 301) having a hairpin-like structure is known as a precursor of “hsa-miR-1227-5p”.

(198) The term “hsa-miR-7150 gene” or “hsa-miR-7150” used herein includes the hsa-miR-7150 gene (miRBase Accession No. MIMAT0028211) described in SEQ ID NO: 122, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7150 gene can be obtained by a method described in Oulas A et al., 2009, Nucleic Acids Res, Vol. 37, p. 3276-3287. Also, “hsa-mir-7150” (miRBase Accession No. MI0023610, SEQ ID NO: 302) having a hairpin-like structure is known as a precursor of “hsa-miR-7150”.

(199) The term “hsa-miR-1915-3p gene” or “hsa-miR-1915-3p” used herein includes the hsa-miR-1915-3p gene (miRBase Accession No. MIMAT0007892) described in SEQ ID NO: 123, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1915-3p gene can be obtained by a method described in Bar M et al., 2008, Stem Cells, Vol. 26, p. 2496-2505. Also, “hsa-mir-1915” (miRBase Accession No. MI0008336, SEQ ID NO: 266) having a hairpin-like structure is known as a precursor of “hsa-miR-1915-3p”.

(200) The term “hsa-miR-187-5p gene” or “hsa-miR-187-5p” used herein includes the hsa-miR-187-5p gene (miRBase Accession No. MIMAT0004561) described in SEQ ID NO: 124, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-187-5p gene can be obtained by a method described in Lim L P et al., 2003, Science, Vol. 299, p. 1540. Also, “hsa-mir-187” (miRBase Accession No. MI0000274, SEQ ID NO: 303) having a hairpin-like structure is known as a precursor of “hsa-miR-187-5p”.

(201) The term “hsa-miR-614 gene” or “hsa-miR-614” used herein includes the hsa-miR-614 gene (miRBase Accession No. MIMAT0003282) described in SEQ ID NO: 125, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-614 gene can be obtained by a method described in Cummins J M et al., 2006, Proc Natl Acad Sci USA, Vol. 103, p. 3687-3692. Also, “hsa-mir-614” (miRBase Accession No. MI0003627, SEQ ID NO: 304) having a hairpin-like

structure is known as a precursor of “hsa-miR-614”.

(202) The term “hsa-miR-19b-3p gene” or “hsa-miR-19b-3p” used herein includes the hsa-miR-19b-3p gene (miRBase Accession No. MIMAT0000074) described in SEQ ID NO: 126, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-19b-3p gene can be obtained by a method described in Lagos-Quintana M et al., 2001, Science, Vol. 294, p. 853-858. Also, “hsa-mir-19b-1 and hsa-mir-19b-2” (miRBase Accession Nos. M10000074 and M10000075, SEQ ID NOs: 305 and 306) having a hairpin-like structure are known as precursors of “hsa-miR-19b-3p”.

(203) The term “hsa-miR-1225-5p gene” or “hsa-miR-1225-5p” used herein includes the hsa-miR-1225-5p gene (miRBase Accession No. MIMAT0005572) described in SEQ ID NO: 127, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1225-5p gene can be obtained by a method described in Berezikov E et al., 2007, Mol Cell, Vol. 28, p. 328-336. Also, “hsa-mir-1225” (miRBase Accession No. M10006311, SEQ ID NO: 307) having a hairpin-like structure is known as a precursor of “hsa-miR-1225-5p”.

(204) The term “hsa-miR-451a gene” or “hsa-miR-451a” used herein includes the hsa-miR-451a gene (miRBase Accession No. MIMAT0001631) described in SEQ ID NO: 128, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-451a gene can be obtained by a method described in Altuvia Y et al., 2005, Nucleic Acids Res, Vol. 33, p. 2697-2706. Also, “hsa-mir-451a” (miRBase Accession No. MI0001729, SEQ ID NO: 308) having a hairpin-like structure is known as a precursor of “hsa-miR-451a”.

(205) The term “hsa-miR-939-5p gene” or “hsa-miR-939-5p” used herein includes the hsa-miR-939-5p gene (miRBase Accession No. MIMAT0004982) described in SEQ ID NO: 129, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-939-5p gene can be obtained by a method described in Lui W O et al., 2007, Cancer Res, Vol. 67, p. 6031-6043. Also, “hsa-mir-939” (miRBase Accession No. MI0005761, SEQ ID NO: 309) having a hairpin-like structure is known as a precursor of “hsa-miR-939-5p”.

(206) The term “hsa-miR-223-3p gene” or “hsa-miR-223-3p” used herein includes the hsa-miR-223-3p gene (miRBase Accession No. MIMAT0000280) described in SEQ ID NO: 130, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-223-3p gene can be obtained by a method described in Lim L P et al., 2003, Science, Vol. 299, p. 1540. Also, “hsa-mir-223” (miRBase Accession No. MI0000300, SEQ ID NO: 310) having a hairpin-like structure is known as a precursor of “hsa-miR-223-3p”.

(207) The term “hsa-miR-1228-5p gene” or “hsa-miR-1228-5p” used herein includes the hsa-miR-1228-5p gene (miRBase Accession No. MIMAT0005582) described in SEQ ID NO: 131, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1228-5p gene can be obtained by a method described in Berezikov E et al., 2007, Mol Cell, Vol. 28, p. 328-336. Also, “hsa-mir-1228” (miRBase Accession No. MI0006318, SEQ ID NO: 311) having a hairpin-like structure is known as a precursor of “hsa-miR-1228-5p”.

(208) The term “hsa-miR-125a-3p gene” or “hsa-miR-125a-3p” used herein includes the hsa-miR-125a-3p gene (miRBase Accession No. MIMAT0004602) described in SEQ ID NO: 132, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-125a-3p gene can be obtained by a method described in Lagos-Quintana M et al., 2002, Curr Biol, Vol. 12, p. 735-739. Also, “hsa-mir-125a” (miRBase Accession No. MI0000469, SEQ ID NO: 312) having a hairpin-like structure is known as a precursor of “hsa-miR-125a-3p”.

(209) The term “hsa-miR-92b-5p gene” or “hsa-miR-92b-5p” used herein includes the hsa-miR-92b-5p gene (miRBase Accession No. MIMAT0004792) described in SEQ ID NO: 133, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-92b-5p gene can be obtained by a method described in Cummins J M et al., 2006, Proc Natl Acad Sci USA, Vol. 103, p. 3687-3692. Also, “hsa-mir-92b” (miRBase Accession No. MI0003560, SEQ ID NO: 313) having a hairpin-like structure is known as a precursor of “hsa-miR-92b-5p”.

(210) The term “hsa-miR-22-3p gene” or “hsa-miR-22-3p” used herein includes the hsa-miR-22-3p gene (miRBase Accession No. MIMAT0000077) described in SEQ ID NO: 134, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-22-3p gene can be obtained by a method described in Lagos-Quintana M et al., 2001, Science, Vol. 294, p. 853-858. Also, “hsa-mir-22” (miRBase Accession No. MI0000078, SEQ ID NO: 314) having a hairpin-like structure is known as a precursor of “hsa-miR-22-3p”.

(211) The term “hsa-miR-4271 gene” or “hsa-miR-4271” used herein includes the hsa-miR-4271 gene (miRBase Accession No. MIMAT0016901) described in SEQ ID NO: 135, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4271 gene can be obtained by a method described in Goff L A et al., 2009, PLoS One, Vol. 4, e7192. Also, “hsa-mir-4271” (miRBase Accession No. MI0015879, SEQ ID NO: 315) having a hairpin-like structure is known as a precursor of “hsa-miR-4271”.

(212) The term “hsa-miR-642b-3p gene” or “hsa-miR-642b-3p” used herein includes the hsa-miR-642b-3p gene (miRBase Accession No. MIMAT0018444) described in SEQ ID NO: 136, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-642b-3p gene can be obtained by a method described in Witten D et al., 2010, BMC Biol, Vol. 8, p. 58. Also, “hsa-mir-642b” (miRBase Accession No. MI0016685, SEQ ID NO: 316) having a hairpin-like structure is known as a precursor of “hsa-miR-642b-3p”.

(213) The term “hsa-miR-6075 gene” or “hsa-miR-6075” used herein includes the hsa-miR-6075 gene (miRBase Accession No. MIMAT0023700) described in SEQ ID NO: 137, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6075 gene can be obtained by a method described in Voellenkle C et al., 2012, RNA, Vol. 18, p. 472-484. Also, “hsa-mir-6075” (miRBase Accession No. MI0020352, SEQ ID NO: 317) having a hairpin-like structure is known as a precursor of “hsa-miR-6075”.

(214) The term “hsa-miR-6125 gene” or “hsa-miR-6125” used herein includes the hsa-miR-6125 gene (miRBase Accession No. MIMAT0024598) described in SEQ ID NO: 138, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6125 gene can be obtained by a method described in Smith J L et al., 2012, J Virol, Vol. 86, p. 5278-5287. Also, “hsa-mir-6125” (miRBase Accession No. MI0021259, SEQ ID NO: 318) having a hairpin-like structure is known as a precursor of “hsa-miR-6125”.

(215) The term “hsa-miR-887-3p gene” or “hsa-miR-887-3p” used herein includes the hsa-miR-887-3p gene (miRBase Accession No. MIMAT0004951) described in SEQ ID NO: 139, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-887-3p gene can be obtained by a method described in Berezikov E et al., 2006, Genome Res, Vol. 16, p. 1289-1298. Also, “hsa-mir-887” (miRBase Accession No. MI0005562, SEQ ID NO: 319) having a hairpin-like structure is known as a precursor of “hsa-miR-887-3p”.

(216) The term “hsa-miR-6851-5p gene” or “hsa-miR-6851-5p” used herein includes the hsa-miR-6851-5p gene (miRBase Accession No. MIMAT0027602) described in SEQ ID NO: 140, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6851-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6851” (miRBase Accession No. MI0022697, SEQ ID NO: 320) having a hairpin-like structure is known as a precursor of “hsa-miR-6851-5p”.

(217) The term “hsa-miR-6763-5p gene” or “hsa-miR-6763-5p” used herein includes the hsa-miR-6763-5p gene (miRBase Accession No. MIMAT0027426) described in SEQ ID NO: 141, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6763-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6763” (miRBase Accession No. MI0022608, SEQ ID NO: 321) having a hairpin-like structure is known as a precursor of “hsa-miR-6763-5p”.

(218) The term “hsa-miR-3928-3p gene” or “hsa-miR-3928-3p” used herein includes the hsa-miR-3928-3p gene (miRBase Accession No. MIMAT0018205) described in SEQ ID NO: 142, a

homolog or an ortholog of a different organism species, and the like. The hsa-miR-3928-3p gene can be obtained by a method described in Creighton C J et al., 2010, PLoS One, Vol. 5, e9637. Also, “hsa-mir-3928” (miRBase Accession No. MI0016438, SEQ ID NO: 322) having a hairpin-like structure is known as a precursor of “hsa-miR-3928-3p”.

(219) The term “hsa-miR-4443 gene” or “hsa-miR-4443” used herein includes the hsa-miR-4443 gene (miRBase Accession No. MIMAT0018961) described in SEQ ID NO: 143, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4443 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4443” (miRBase Accession No. MI0016786, SEQ ID NO: 323) having a hairpin-like structure is known as a precursor of “hsa-miR-4443”.

(220) The term “hsa-miR-3648 gene” or “hsa-miR-3648” used herein includes the hsa-miR-3648 gene (miRBase Accession No. MIMAT0018068) described in SEQ ID NO: 144, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3648 gene can be obtained by a method described in Meiri E et al., 2010, Nucleic Acids Res, Vol. 38, p. 6234-6246. Also, “hsa-mir-3648” (miRBase Accession No. MI0016048, SEQ ID NO: 324) having a hairpin-like structure is known as a precursor of “hsa-miR-3648”.

(221) The term “hsa-miR-149-3p gene” or “hsa-miR-149-3p” used herein includes the hsa-miR-149-3p gene (miRBase Accession No. MIMAT0004609) described in SEQ ID NO: 145, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-149-3p gene can be obtained by a method described in Lagos-Quintana M et al., 2002, Curr Biol, Vol. 12, p. 735-739. Also, “hsa-mir-149” (miRBase Accession No. MI0000478, SEQ ID NO: 325) having a hairpin-like structure is known as a precursor of “hsa-miR-149-3p”.

(222) The term “hsa-miR-4689 gene” or “hsa-miR-4689” used herein includes the hsa-miR-4689 gene (miRBase Accession No. MIMAT0019778) described in SEQ ID NO: 146, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4689 gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4689” (miRBase Accession No. MI0017322, SEQ ID NO: 326) having a hairpin-like structure is known as a precursor of “hsa-miR-4689”.

(223) The term “hsa-miR-4763-3p gene” or “hsa-miR-4763-3p” used herein includes the hsa-miR-4763-3p gene (miRBase Accession No. MIMAT0019913) described in SEQ ID NO: 147, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4763-3p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4763” (miRBase Accession No. MI0017404, SEQ ID NO: 327) having a hairpin-like structure is known as a precursor of “hsa-miR-4763-3p”.

(224) The term “hsa-miR-6729-5p gene” or “hsa-miR-6729-5p” used herein includes the hsa-miR-6729-5p gene (miRBase Accession No. MIMAT0027359) described in SEQ ID NO: 148, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6729-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6729” (miRBase Accession No. MI0022574, SEQ ID NO: 328) having a hairpin-like structure is known as a precursor of “hsa-miR-6729-5p”.

(225) The term “hsa-miR-3196 gene” or “hsa-miR-3196” used herein includes the hsa-miR-3196 gene (miRBase Accession No. MIMAT0015080) described in SEQ ID NO: 149, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3196 gene can be obtained by a method described in Stark M S et al., 2010, PLoS One, Vol. 5, e9685. Also, “hsa-mir-3196” (miRBase Accession No. MI0014241, SEQ ID NO: 329) having a hairpin-like structure is known as a precursor of “hsa-miR-3196”.

(226) The term “hsa-miR-8069 gene” or “hsa-miR-8069” used herein includes the hsa-miR-8069 gene (miRBase Accession No. MIMAT0030996) described in SEQ ID NO: 150, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-8069 gene can be obtained by a method described in Wang H J et al., 2013, Shock, Vol. 39, p. 480-487. Also, “hsa-mir-8069”

(miRBase Accession No. MI0025905, SEQ ID NO: 330) having a hairpin-like structure is known as a precursor of “hsa-miR-8069”.

(227) The term “hsa-miR-1268a gene” or “hsa-miR-1268a” used herein includes the hsa-miR-1268a gene (miRBase Accession No. MIMAT0005922) described in SEQ ID NO: 151, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1268a gene can be obtained by a method described in Morin R D et al., 2008, *Genome Res*, Vol. 18, p. 610-621. Also, “hsa-mir-1268a” (miRBase Accession No. MI0006405, SEQ ID NO: 331) having a hairpin-like structure is known as a precursor of “hsa-miR-1268a”.

(228) The term “hsa-miR-4739 gene” or “hsa-miR-4739” used herein includes the hsa-miR-4739 gene (miRBase Accession No. MIMAT0019868) described in SEQ ID NO: 152, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4739 gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, “hsa-mir-4739” (miRBase Accession No. MI0017377, SEQ ID NO: 332) having a hairpin-like structure is known as a precursor of “hsa-miR-4739”.

(229) The term “hsa-miR-1268b gene” or “hsa-miR-1268b” used herein includes the hsa-miR-1268b gene (miRBase Accession No. MIMAT0018925) described in SEQ ID NO: 153, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1268b gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-1268b” (miRBase Accession No. MI0016748, SEQ ID NO: 333) having a hairpin-like structure is known as a precursor of “hsa-miR-1268b”.

(230) The term “hsa-miR-5698 gene” or “hsa-miR-5698” used herein includes the hsa-miR-5698 gene (miRBase Accession No. MIMAT0022491) described in SEQ ID NO: 154, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-5698 gene can be obtained by a method described in Watahiki A et al., 2011, *PLoS One*, Vol. 6, e24950. Also, “hsa-mir-5698” (miRBase Accession No. MI0019305, SEQ ID NO: 334) having a hairpin-like structure is known as a precursor of “hsa-miR-5698”.

(231) The term “hsa-miR-6752-5p gene” or “hsa-miR-6752-5p” used herein includes the hsa-miR-6752-5p gene (miRBase Accession No. MIMAT0027404) described in SEQ ID NO: 155, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6752-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6752” (miRBase Accession No. MI0022597, SEQ ID NO: 335) having a hairpin-like structure is known as a precursor of “hsa-miR-6752-5p”.

(232) The term “hsa-miR-4507 gene” or “hsa-miR-4507” used herein includes the hsa-miR-4507 gene (miRBase Accession No. MIMAT0019044) described in SEQ ID NO: 156, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4507 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4507” (miRBase Accession No. MI0016871, SEQ ID NO: 336) having a hairpin-like structure is known as a precursor of “hsa-miR-4507”.

(233) The term “hsa-miR-564 gene” or “hsa-miR-564” used herein includes the hsa-miR-564 gene (miRBase Accession No. MIMAT0003228) described in SEQ ID NO: 157, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-564 gene can be obtained by a method described in Cummins J M et al., 2006, *Proc Natl Acad Sci USA*, Vol. 103, p. 3687-3692. Also, “hsa-mir-564” (miRBase Accession No. MI0003570, SEQ ID NO: 337) having a hairpin-like structure is known as a precursor of “hsa-miR-564”.

(234) The term “hsa-miR-4497 gene” or “hsa-miR-4497” used herein includes the hsa-miR-4497 gene (miRBase Accession No. MIMAT0019032) described in SEQ ID NO: 158, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4497 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4497” (miRBase Accession No. MI0016859, SEQ ID NO: 338) having a hairpin-like structure is known as a precursor of “hsa-miR-4497”.

(235) The term “hsa-miR-6877-5p gene” or “hsa-miR-6877-5p” used herein includes the hsa-miR-6877-5p gene (miRBase Accession No. MIMAT0027654) described in SEQ ID NO: 159, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6877-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6877” (miRBase Accession No. MI0022724, SEQ ID NO: 339) having a hairpin-like structure is known as a precursor of “hsa-miR-6877-5p”.

(236) The term “hsa-miR-6087 gene” or “hsa-miR-6087” used herein includes the hsa-miR-6087 gene (miRBase Accession No. MIMAT0023712) described in SEQ ID NO: 160, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6087 gene can be obtained by a method described in Yoo J K et al., 2012, *Stem Cells Dev*, Vol. 21, p. 2049-2057. Also, “hsa-mir-6087” (miRBase Accession No. MI0020364, SEQ ID NO: 340) having a hairpin-like structure is known as a precursor of “hsa-miR-6087”.

(237) The term “hsa-miR-4731-5p gene” or “hsa-miR-4731-5p” used herein includes the hsa-miR-4731-5p gene (miRBase Accession No. MIMAT0019853) described in SEQ ID NO: 161, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4731-5p gene can be obtained by a method described in Persson H et al., 2011, *Cancer Res*, Vol. 71, p. 78-86. Also, “hsa-mir-4731” (miRBase Accession No. MI0017368, SEQ ID NO: 341) having a hairpin-like structure is known as a precursor of “hsa-miR-4731-5p”.

(238) The term “hsa-miR-615-5p gene” or “hsa-miR-615-5p” used herein includes the hsa-miR-615-5p gene (miRBase Accession No. MIMAT0004804) described in SEQ ID NO: 162, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-615-5p gene can be obtained by a method described in Cummins J M et al., 2006, *Proc Natl Acad Sci USA*, Vol. 103, p. 3687-3692. Also, “hsa-mir-615” (miRBase Accession No. MI0003628, SEQ ID NO: 342) having a hairpin-like structure is known as a precursor of “hsa-miR-615-5p”.

(239) The term “hsa-miR-760 gene” or “hsa-miR-760” used herein includes the hsa-miR-760 gene (miRBase Accession No. MIMAT0004957) described in SEQ ID NO: 163, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-760 gene can be obtained by a method described in Berezikov E et al., 2006, *Genome Res*, Vol. 16, p. 1289-1298. Also, “hsa-mir-760” (miRBase Accession No. MI0005567, SEQ ID NO: 343) having a hairpin-like structure is known as a precursor of “hsa-miR-760”.

(240) The term “hsa-miR-6891-5p gene” or “hsa-miR-6891-5p” used herein includes the hsa-miR-6891-5p gene (miRBase Accession No. MIMAT0027682) described in SEQ ID NO: 164, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6891-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6891” (miRBase Accession No. MI0022738, SEQ ID NO: 344) having a hairpin-like structure is known as a precursor of “hsa-miR-6891-5p”.

(241) The term “hsa-miR-6887-5p gene” or “hsa-miR-6887-5p” used herein includes the hsa-miR-6887-5p gene (miRBase Accession No. MIMAT0027674) described in SEQ ID NO: 165, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6887-5p gene can be obtained by a method described in Ladewig E et al., 2012, *Genome Res*, Vol. 22, p. 1634-1645. Also, “hsa-mir-6887” (miRBase Accession No. MI0022734, SEQ ID NO: 345) having a hairpin-like structure is known as a precursor of “hsa-miR-6887-5p”.

(242) The term “hsa-miR-4525 gene” or “hsa-miR-4525” used herein includes the hsa-miR-4525 gene (miRBase Accession No. MIMAT0019064) described in SEQ ID NO: 166, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4525 gene can be obtained by a method described in Jima D D et al., 2010, *Blood*, Vol. 116, e118-e127. Also, “hsa-mir-4525” (miRBase Accession No. MI0016892, SEQ ID NO: 346) having a hairpin-like structure is known as a precursor of “hsa-miR-4525”.

(243) The term “hsa-miR-1914-3p gene” or “hsa-miR-1914-3p” used herein includes the hsa-miR-1914-3p gene (miRBase Accession No. MIMAT0007890) described in SEQ ID NO: 167, a

homolog or an ortholog of a different organism species, and the like. The hsa-miR-1914-3p gene can be obtained by a method described in Bar M et al., 2008, Stem Cells, Vol. 26, p. 2496-2505. Also, “hsa-mir-1914” (miRBase Accession No. MI0008335, SEQ ID NO: 347) having a hairpin-like structure is known as a precursor of “hsa-miR-1914-3p”.

(244) The term “hsa-miR-619-5p gene” or “hsa-miR-619-5p” used herein includes the hsa-miR-619-5p gene (miRBase Accession No. MIMAT0026622) described in SEQ ID NO: 168, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-619-5p gene can be obtained by a method described in Cummins J M et al., 2006, Proc Natl Acad Sci USA, Vol. 103, p. 3687-3692. Also, “hsa-mir-619” (miRBase Accession No. MI0003633, SEQ ID NO: 348) having a hairpin-like structure is known as a precursor of “hsa-miR-619-5p”.

(245) The term “hsa-miR-5001-5p gene” or “hsa-miR-5001-5p” used herein includes the hsa-miR-5001-5p gene (miRBase Accession No. MIMAT0021021) described in SEQ ID NO: 169, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-5001-5p gene can be obtained by a method described in Hansen T B et al., 2011, RNA Biol, Vol. 8, p. 378-383. Also, “hsa-mir-5001” (miRBase Accession No. MI0017867, SEQ ID NO: 349) having a hairpin-like structure is known as a precursor of “hsa-miR-5001-5p”.

(246) The term “hsa-miR-6722-3p gene” or “hsa-miR-6722-3p” used herein includes the hsa-miR-6722-3p gene (miRBase Accession No. MIMAT0025854) described in SEQ ID NO: 170, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6722-3p gene can be obtained by a method described in Li Y et al., 2012, Gene, Vol. 497, p. 330-335. Also, “hsa-mir-6722” (miRBase Accession No. MI0022557, SEQ ID NO: 350) having a hairpin-like structure is known as a precursor of “hsa-miR-6722-3p”.

(247) The term “hsa-miR-3621 gene” or “hsa-miR-3621” used herein includes the hsa-miR-3621 gene (miRBase Accession No. MIMAT0018002) described in SEQ ID NO: 171, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-3621 gene can be obtained by a method described in Witten D et al., 2010, BMC Biol, Vol. 8, p. 58. Also, “hsa-mir-3621” (miRBase Accession No. MI0016012, SEQ ID NO: 351) having a hairpin-like structure is known as a precursor of “hsa-miR-3621”.

(248) The term “hsa-miR-4298 gene” or “hsa-miR-4298” used herein includes the hsa-miR-4298 gene (miRBase Accession No. MIMAT0016852) described in SEQ ID NO: 172, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4298 gene can be obtained by a method described in Goff L A et al., 2009, PLoS One, Vol. 4, e7192. Also, “hsa-mir-4298” (miRBase Accession No. MI0015830, SEQ ID NO: 352) having a hairpin-like structure is known as a precursor of “hsa-miR-4298”.

(249) The term “hsa-miR-675-5p gene” or “hsa-miR-675-5p” used herein includes the hsa-miR-675-5p gene (miRBase Accession No. MIMAT0004284) described in SEQ ID NO: 173, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-675-5p gene can be obtained by a method described in Cai X et al., 2007, RNA, Vol. 13, p. 313-316. Also, “hsa-mir-675” (miRBase Accession No. MI0005416, SEQ ID NO: 353) having a hairpin-like structure is known as a precursor of “hsa-miR-675-5p”.

(250) The term “hsa-miR-4655-5p gene” or “hsa-miR-4655-5p” used herein includes the hsa-miR-4655-5p gene (miRBase Accession No. MIMAT0019721) described in SEQ ID NO: 174, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4655-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4655” (miRBase Accession No. MI0017283, SEQ ID NO: 354) having a hairpin-like structure is known as a precursor of “hsa-miR-4655-5p”.

(251) The term “hsa-miR-6073 gene” or “hsa-miR-6073” used herein includes the hsa-miR-6073 gene (miRBase Accession No. MIMAT0023698) described in SEQ ID NO: 561, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6073 gene can be obtained by a method described in Voellenkle C et al., 2012, RNA, Vol. 18, p. 472-484. Also, “hsa-mir-6073”

(miRBase Accession No. MI0020350, SEQ ID NO: 580) having a hairpin-like structure is known as a precursor of “hsa-miR-6073”.

(252) The term “hsa-miR-6845-5p gene” or “hsa-miR-6845-5p” used herein includes the hsa-miR-6845-5p gene (miRBase Accession No. MIMAT0027590) described in SEQ ID NO: 562, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6845-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6845” (miRBase Accession No. MI0022691, SEQ ID NO: 581) having a hairpin-like structure is known as a precursor of “hsa-miR-6845-5p”.

(253) The term “hsa-miR-6769b-5p gene” or “hsa-miR-6769b-5p” used herein includes the hsa-miR-6769b-5p gene (miRBase Accession No. MIMAT0027620) described in SEQ ID NO: 563, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-6769b-5p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-6769b” (miRBase Accession No. MI0022706, SEQ ID NO: 582) having a hairpin-like structure is known as a precursor of “hsa-miR-6769b-5p”.

(254) The term “hsa-miR-4665-3p gene” or “hsa-miR-4665-3p” used herein includes the hsa-miR-4665-3p gene (miRBase Accession No. MIMAT0019740) described in SEQ ID NO: 564, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4665-3p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4665” (miRBase Accession No. MI0017295, SEQ ID NO: 583) having a hairpin-like structure is known as a precursor of “hsa-miR-4665-3p”.

(255) The term “hsa-miR-1913 gene” or “hsa-miR-1913” used herein includes the hsa-miR-1913 gene (miRBase Accession No. MIMAT0007888) described in SEQ ID NO: 565, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1913 gene can be obtained by a method described in Bar M et al., 2008, Stem Cells, Vol. 26, p. 2496-2505. Also, “hsa-mir-1913” (miRBase Accession No. MI0008334, SEQ ID NO: 584) having a hairpin-like structure is known as a precursor of “hsa-miR-1913”.

(256) The term “hsa-miR-1228-3p gene” or “hsa-miR-1228-3p” used herein includes the hsa-miR-1228-3p gene (miRBase Accession No. MIMAT0005583) described in SEQ ID NO: 566, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1228-3p gene can be obtained by a method described in Berezikov E et al., 2007, Mol Cell, Vol. 28, p. 328-336. Also, “hsa-mir-1228” (miRBase Accession No. MI0006318, SEQ ID NO: 311) having a hairpin-like structure is known as a precursor of “hsa-miR-1228-3p”.

(257) The term “hsa-miR-940 gene” or “hsa-miR-940” used herein includes the hsa-miR-940 gene (miRBase Accession No. MIMAT0004983) described in SEQ ID NO: 567, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-940 gene can be obtained by a method described in Lui W O et al., 2007, Cancer Res, Vol. 67, p. 6031-6043. Also, “hsa-mir-940” (miRBase Accession No. MI0005762, SEQ ID NO: 585) having a hairpin-like structure is known as a precursor of “hsa-miR-940”.

(258) The term “hsa-miR-296-3p gene” or “hsa-miR-296-3p” used herein includes the hsa-miR-296-3p gene (miRBase Accession No. MIMAT0004679) described in SEQ ID NO: 568, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-296-3p gene can be obtained by a method described in Houbaviy H B et al., 2003, Dev Cell, Vol. 5, p. 351-358. Also, “hsa-mir-296” (miRBase Accession No. MI0000747, SEQ ID NO: 586) having a hairpin-like structure is known as a precursor of “hsa-miR-296-3p”.

(259) The term “hsa-miR-4690-5p gene” or “hsa-miR-4690-5p” used herein includes the hsa-miR-4690-5p gene (miRBase Accession No. MIMAT0019779) described in SEQ ID NO: 569, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4690-5p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4690” (miRBase Accession No. MI0017323, SEQ ID NO: 587) having a hairpin-like structure is known as a precursor of “hsa-miR-4690-5p”.

(260) The term “hsa-miR-548q gene” or “hsa-miR-548q” used herein includes the hsa-miR-548q gene (miRBase Accession No. MIMAT0011163) described in SEQ ID NO: 570, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-548q gene can be obtained by a method described in Wyman S K et al., 2009, PLoS One, Vol. 4, e5311. Also, “hsa-mir-548q” (miRBase Accession No. MI0010637, SEQ ID NO: 588) having a hairpin-like structure is known as a precursor of “hsa-miR-548q”.

(261) The term “hsa-miR-663a gene” or “hsa-miR-663a” used herein includes the hsa-miR-663a gene (miRBase Accession No. MIMAT0003326) described in SEQ ID NO: 571, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-663a gene can be obtained by a method described in Cummins J M et al., 2006, Proc Natl Acad Sci USA, Vol. 103, p. 3687-3692. Also, “hsa-mir-663a” (miRBase Accession No. MI0003672, SEQ ID NO: 589) having a hairpin-like structure is known as a precursor of “hsa-miR-663a”.

(262) The term “hsa-miR-1249 gene” or “hsa-miR-1249” used herein includes the hsa-miR-1249 gene (miRBase Accession No. MIMAT0005901) described in SEQ ID NO: 572, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1249 gene can be obtained by a method described in Morin R D et al., 2008, Genome Res, Vol. 18, p. 610-621. Also, “hsa-mir-1249” (miRBase Accession No. MI0006384, SEQ ID NO: 590) having a hairpin-like structure is known as a precursor of “hsa-miR-1249”.

(263) The term “hsa-miR-1202 gene” or “hsa-miR-1202” used herein includes the hsa-miR-1202 gene (miRBase Accession No. MIMAT0005865) described in SEQ ID NO: 573, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1202 gene can be obtained by a method described in Marton S et al., 2008, Leukemia, Vol. 22, p. 330-338. Also, “hsa-mir-1202” (miRBase Accession No. MI0006334, SEQ ID NO: 591) having a hairpin-like structure is known as a precursor of “hsa-miR-1202”.

(264) The term “hsa-miR-7113-3p gene” or “hsa-miR-7113-3p” used herein includes the hsa-miR-7113-3p gene (miRBase Accession No. MIMAT0028124) described in SEQ ID NO: 574, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-7113-3p gene can be obtained by a method described in Ladewig E et al., 2012, Genome Res, Vol. 22, p. 1634-1645. Also, “hsa-mir-7113” (miRBase Accession No. MI0022964, SEQ ID NO: 592) having a hairpin-like structure is known as a precursor of “hsa-miR-7113-3p”.

(265) The term “hsa-miR-1225-3p gene” or “hsa-miR-1225-3p” used herein includes the hsa-miR-1225-3p gene (miRBase Accession No. MIMAT0005573) described in SEQ ID NO: 575, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1225-3p gene can be obtained by a method described in Berezikov E et al., 2007, Mol Cell, Vol. 28, p. 328-336. Also, “hsa-mir-1225” (miRBase Accession No. M10006311, SEQ ID NO: 307) having a hairpin-like structure is known as a precursor of “hsa-miR-1225-3p”.

(266) The term “hsa-miR-4783-3p gene” or “hsa-miR-4783-3p” used herein includes the hsa-miR-4783-3p gene (miRBase Accession No. MIMAT0019947) described in SEQ ID NO: 576, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4783-3p gene can be obtained by a method described in Persson H et al., 2011, Cancer Res, Vol. 71, p. 78-86. Also, “hsa-mir-4783” (miRBase Accession No. MI0017428, SEQ ID NO: 593) having a hairpin-like structure is known as a precursor of “hsa-miR-4783-3p”.

(267) The term “hsa-miR-4448 gene” or “hsa-miR-4448” used herein includes the hsa-miR-4448 gene (miRBase Accession No. MIMAT0018967) described in SEQ ID NO: 577, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-4448 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4448” (miRBase Accession No. MI0016791, SEQ ID NO: 594) having a hairpin-like structure is known as a precursor of “hsa-miR-4448”.

(268) The term “hsa-miR-4534 gene” or “hsa-miR-4534” used herein includes the hsa-miR-4534 gene (miRBase Accession No. MIMAT0019073) described in SEQ ID NO: 578, a homolog or an

ortholog of a different organism species, and the like. The hsa-miR-4534 gene can be obtained by a method described in Jima D D et al., 2010, Blood, Vol. 116, e118-e127. Also, “hsa-mir-4534” (miRBase Accession No. MI0016901, SEQ ID NO: 595) having a hairpin-like structure is known as a precursor of “hsa-miR-4534”.

(269) The term “hsa-miR-1307-3p gene” or “hsa-miR-1307-3p” used herein includes the hsa-miR-1307-3p gene (miRBase Accession No. MIMAT0005951) described in SEQ ID NO: 579, a homolog or an ortholog of a different organism species, and the like. The hsa-miR-1307-3p gene can be obtained by a method described in Morin R D et al., 2008, Genome Res, Vol. 18, p. 610-621. Also, “hsa-mir-1307” (miRBase Accession No. MI0006444, SEQ ID NO: 596) having a hairpin-like structure is known as a precursor of “hsa-miR-1307-3p”.

(270) A mature miRNA may become a variant due to the sequence cleaved shorter or longer by one to several upstream or downstream nucleotides or nucleotide substitution when cut out as the mature miRNA from its RNA precursor having a hairpin-like structure. This variant is called isomiR (Morin R D. et al., 2008, Genome Res., Vol. 18, p. 610-621). The miRBase Release 20 shows the nucleotide sequences represented by SEQ ID NOs: 1 to 174 and 561 to 579 as well as a large number of the nucleotide sequence variants and fragments represented by SEQ ID NOs: 355 to 560 and 597 to 618, called isomiRs. These variants can also be obtained as miRNAs having a nucleotide sequence represented by any of SEQ ID NOs: 1 to 174 and 561 to 579. Specifically, among the variants of polynucleotides consisting of the nucleotide sequence represented by any of SEQ ID NOs: 5, 8, 9, 11, 18, 20, 22, 23, 24, 28, 29, 30, 32, 34, 37, 40, 41, 47, 48, 49, 51, 52, 53, 56, 58, 59, 60, 61, 63, 64, 65, 66, 67, 69, 72, 73, 75, 78, 79, 80, 81, 82, 85, 88, 89, 91, 92, 95, 96, 103, 104, 105, 106, 107, 108, 109, 110, 112, 113, 114, 117, 118, 120, 123, 124, 125, 126, 128, 129, 130, 131, 132, 133, 134, 135, 136, 138, 139, 142, 143, 144, 145, 146, 147, 149, 151, 152, 153, 154, 156, 157, 158, 160, 161, 162, 163, 166, 167, 168, 169, 172, 173, 174, 565, 566, 567, 568, 569, 571, 572, 573, 576, 577, 579, or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t in the nucleotide sequence, examples of the longest variants registered in the miRBase Release 20 include polynucleotides represented by SEQ ID NOs: 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, 385, 387, 389, 391, 393, 395, 397, 399, 401, 403, 405, 407, 409, 411, 413, 415, 417, 419, 421, 423, 425, 427, 429, 431, 433, 435, 437, 439, 441, 443, 445, 447, 449, 451, 453, 455, 457, 459, 461, 463, 465, 467, 469, 471, 473, 475, 477, 479, 481, 483, 485, 487, 489, 491, 493, 495, 497, 499, 501, 503, 505, 507, 509, 511, 513, 515, 517, 519, 521, 523, 525, 527, 529, 531, 533, 535, 537, 539, 541, 543, 545, 547, 549, 551, 553, 555, 557, 559, 597, 599, 601, 603, 605, 607, 609, 611, 613, 615 and 617, respectively.

(271) Also, among the variants of polynucleotides consisting of a nucleotide sequence represented by any of SEQ ID NOs: 5, 8, 9, 11, 18, 20, 22, 23, 24, 28, 29, 30, 32, 34, 37, 40, 41, 47, 48, 49, 51, 52, 53, 56, 58, 59, 60, 61, 63, 64, 65, 66, 67, 69, 72, 73, 75, 78, 79, 80, 81, 82, 85, 88, 89, 91, 92, 95, 96, 103, 104, 105, 106, 107, 108, 109, 110, 112, 113, 114, 117, 118, 120, 123, 124, 125, 126, 128, 129, 130, 131, 132, 133, 134, 135, 136, 138, 139, 142, 143, 144, 145, 146, 147, 149, 151, 152, 153, 154, 156, 157, 158, 160, 161, 162, 163, 166, 167, 168, 169, 172, 173, 174, 565, 566, 567, 568, 569, 571, 572, 573, 576, 577, 579 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t in the nucleotide sequence, examples of the shortest variants registered in the miRBase Release 20 include polynucleotides having sequences represented by SEQ ID NOs: 356, 358, 360, 362, 364, 366, 368, 370, 372, 374, 376, 378, 380, 382, 384, 386, 388, 390, 392, 394, 396, 398, 400, 402, 404, 406, 408, 410, 412, 414, 416, 418, 420, 422, 424, 426, 428, 430, 432, 434, 436, 438, 440, 442, 444, 446, 448, 450, 452, 454, 456, 458, 460, 462, 464, 466, 468, 470, 472, 474, 476, 478, 480, 482, 484, 486, 488, 490, 492, 494, 496, 498, 500, 502, 504, 506, 508, 510, 512, 514, 516, 518, 520, 522, 524, 526, 528, 530, 532, 534, 536, 538, 540, 542, 544, 546, 548, 550, 552, 554, 556, 558, 560, 598, 600, 602, 604, 606, 608, 610, 612, 614, 616 and 618, respectively.

(272) In addition to these variants and fragments, examples thereof include a large number of

isomiR polynucleotides of SEQ ID NOs: 1 to 174 and 561 to 579 registered in miRBase.

(273) Examples of the polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 1 to 174 and 561 to 579 include a polynucleotide represented by any of SEQ ID NOs: 175 to 354 and 579 to 596, which are their respective precursors.

(274) The names and miRBase Accession Nos. (registration numbers) of the genes represented by SEQ ID NOs: 1 to 618 are shown in Table 1.

(275) As used herein, the term “capable of specifically binding” means that the nucleic acid probe or the primer used in the present invention binds to a particular target nucleic acid and cannot substantially bind to other nucleic acids.

(276) TABLE-US-00001 TABLE 1 SEQ miRBase ID NO: Gene name registration No. 1 hsa-miR-6768-5p MIMAT0027436 2 hsa-miR-6836-3p MIMAT0027575 3 hsa-miR-6782-5p MIMAT0027464 4 hsa-miR-3663-3p MIMAT0018085 5 hsa-miR-1908-3p MIMAT0026916 6 hsa-miR-6726-5p MIMAT0027353 7 hsa-miR-4258 MIMAT0016879 8 hsa-miR-1343-3p MIMAT0019776 9 hsa-miR-4516 MIMAT0019053 10 hsa-miR-6875-5p MIMAT0027650 11 hsa-miR-4651 MIMAT0019715 12 hsa-miR-6825-5p MIMAT0027550 13 hsa-miR-6840-3p MIMAT0027583 14 hsa-miR-6780b-5p MIMAT0027572 15 hsa-miR-6749-5p MIMAT0027398 16 hsa-miR-8063 MIMAT0030990 17 hsa-miR-6784-5p MIMAT0027468 18 hsa-miR-3679-5p MIMAT0018104 19 hsa-miR-3184-5p MIMAT0015064 20 hsa-miR-663b MIMAT0005867 21 hsa-miR-6880-5p MIMAT0027660 22 hsa-miR-1908-5p MIMAT0007881 23 hsa-miR-92a-2-5p MIMAT0004508 24 hsa-miR-7975 MIMAT0031178 25 hsa-miR-7110-5p MIMAT0028117 26 hsa-miR-6842-5p MIMAT0027586 27 hsa-miR-6857-5p MIMAT0027614 28 hsa-miR-5572 MIMAT0022260 29 hsa-miR-3197 MIMAT0015082 30 hsa-miR-6131 MIMAT0024615 31 hsa-miR-6889-5p MIMAT0027678 32 hsa-miR-4454 MIMAT0018976 33 hsa-miR-1199-5p MIMAT0031119 34 hsa-miR-1247-3p MIMAT0022721 35 hsa-miR-6800-5p MIMAT0027500 36 hsa-miR-6872-3p MIMAT0027645 37 hsa-miR-4649-5p MIMAT0019711 38 hsa-miR-6791-5p MIMAT0027482 39 hsa-miR-4433b-3p MIMAT0030414 40 hsa-miR-3135b MIMAT0018985 41 hsa-miR-128-2-5p MIMAT0031095 42 hsa-miR-4675 MIMAT0019757 43 hsa-miR-4472 MIMAT0018999 44 hsa-miR-6785-5p MIMAT0027470 45 hsa-miR-6741-5p MIMAT0027383 46 hsa-miR-7977 MIMAT0031180 47 hsa-miR-3665 MIMAT0018087 48 hsa-miR-128-1-5p MIMAT0026477 49 hsa-miR-4286 MIMAT0016916 50 hsa-miR-6765-3p MIMAT0027431 51 hsa-miR-4632-5p MIMAT0022977 52 hsa-miR-365a-5p MIMAT0009199 53 hsa-miR-6088 MIMAT0023713 54 hsa-miR-6816-5p MIMAT0027532 55 hsa-miR-6885-5p MIMAT0027670 56 hsa-miR-711 MIMAT0012734 57 hsa-miR-6765-5p MIMAT0027430 58 hsa-miR-3180 MIMAT0018178 59 hsa-miR-4442 MIMAT0018960 60 hsa-miR-4792 MIMAT0019964 61 hsa-miR-6721-5p MIMAT0025852 62 hsa-miR-6798-5p MIMAT0027496 63 hsa-miR-3162-5p MIMAT0015036 64 hsa-miR-6126 MIMAT0024599 65 hsa-miR-4758-5p MIMAT0019903 66 hsa-miR-2392 MIMAT0019043 67 hsa-miR-486-3p MIMAT0004762 68 hsa-miR-6727-5p MIMAT0027355 69 hsa-miR-4728-5p MIMAT0019849 70 hsa-miR-6746-5p MIMAT0027392 71 hsa-miR-4270 MIMAT0016900 72 hsa-miR-3940-5p MIMAT0019229 73 hsa-miR-4725-3p MIMAT0019844 74 hsa-miR-7108-5p MIMAT0028113 75 hsa-miR-3656 MIMAT0018076 76 hsa-miR-6879-5p MIMAT0027658 77 hsa-miR-6738-5p MIMAT0027377 78 hsa-miR-1260a MIMAT0005911 79 hsa-miR-4446-3p MIMAT0018965 80 hsa-miR-3131 MIMAT0014996 81 hsa-miR-4463 MIMAT0018987 82 hsa-miR-3185 MIMAT0015065 83 hsa-miR-6870-5p MIMAT0027640 84 hsa-miR-6779-5p MIMAT0027458 85 hsa-miR-1273g-3p MIMAT0022742 86 hsa-miR-8059 MIMAT0030986 87 hsa-miR-4697-5p MIMAT0019791 88 hsa-miR-4674 MIMAT0019756 89 hsa-miR-4433-3p MIMAT0018949 90 hsa-miR-4257 MIMAT0016878 91 hsa-miR-1915-5p MIMAT0007891 92 hsa-miR-4417 MIMAT0018929 93 hsa-miR-1343-5p MIMAT0027038 94 hsa-miR-6781-5p MIMAT0027462 95 hsa-miR-4695-5p MIMAT0019788 96 hsa-miR-1237-5p MIMAT0022946 97 hsa-miR-6775-5p MIMAT0027450 98 hsa-miR-7845-5p MIMAT0030420 99 hsa-miR-4746-3p MIMAT0019881 100 hsa-miR-7641 MIMAT0029782 101

hsa-miR-7847-3p MIMAT0030422 102 hsa-miR-5806-5p MIMAT0027512 103 hsa-miR-4467
MIMAT0018994 104 hsa-miR-4726-5p MIMAT0019845 105 hsa-miR-4648 MIMAT0019710 106
hsa-miR-6089 MIMAT0023714 107 hsa-miR-1260b MIMAT0015041 108 hsa-miR-4532
MIMAT0019071 109 hsa-miR-5195-3p MIMAT0021127 110 hsa-miR-3188 MIMAT0015070 111
hsa-miR-6848-5p MIMAT0027596 112 hsa-miR-1233-5p MIMAT0022943 113 hsa-miR-6717-5p
MIMAT0025846 114 hsa-miR-3195 MIMAT0015079 115 hsa-miR-6757-5p MIMAT0027414 116
hsa-miR-8072 MIMAT0030999 117 hsa-miR-4745-5p MIMAT0019878 118 hsa-miR-6511a-5p
MIMAT0025478 119 hsa-miR-6776-5p MIMAT0027452 120 hsa-miR-371a-5p MIMAT0004687
121 hsa-miR-1227-5p MIMAT0022941 122 hsa-miR-7150 MIMAT0028211 123 hsa-miR-1915-3p
MIMAT0007892 124 hsa-miR-187-5p MIMAT0004561 125 hsa-miR-614 MIMAT0003282 126
hsa-miR-19b-3p MIMAT0000074 127 hsa-miR-1225-5p MIMAT0005572 128 hsa-miR-451a
MIMAT0001631 129 hsa-miR-939-5p MIMAT0004982 130 hsa-miR-223-3p MIMAT0000280 131
hsa-miR-1228-5p MIMAT0005582 132 hsa-miR-125a-3p MIMAT0004602 133 hsa-miR-92b-5p
MIMAT0004792 134 hsa-miR-22-3p MIMAT0000077 135 hsa-miR-4271 MIMAT0016901 136
hsa-miR-642b-3p MIMAT0018444 137 hsa-miR-6075 MIMAT0023700 138 hsa-miR-6125
MIMAT0024598 139 hsa-miR-887-3p MIMAT0004951 140 hsa-miR-6851-5p MIMAT0027602
141 hsa-miR-6763-5p MIMAT0027426 142 hsa-miR-3928-3p MIMAT0018205 143 hsa-miR-4443
MIMAT0018961 144 hsa-miR-3648 MIMAT0018068 145 hsa-miR-149-3p MIMAT0004609 146
hsa-miR-4689 MIMAT0019778 147 hsa-miR-4763-3p MIMAT0019913 148 hsa-miR-6729-5p
MIMAT0027359 149 hsa-miR-3196 MIMAT0015080 150 hsa-miR-8069 MIMAT0030996 151
hsa-miR-1268a MIMAT0005922 152 hsa-miR-4739 MIMAT0019868 153 hsa-miR-1268b
MIMAT0018925 154 hsa-miR-5698 MIMAT0022491 155 hsa-miR-6752-5p MIMAT0027404 156
hsa-miR-4507 MIMAT0019044 157 hsa-miR-564 MIMAT0003228 158 hsa-miR-4497
MIMAT0019032 159 hsa-miR-6877-5p MIMAT0027654 160 hsa-miR-6087 MIMAT0023712 161
hsa-miR-4731-5p MIMAT0019853 162 hsa-miR-615-5p MIMAT0004804 163 hsa-miR-760
MIMAT0004957 164 hsa-miR-6891-5p MIMAT0027682 165 hsa-miR-6887-5p MIMAT0027674
166 hsa-miR-4525 MIMAT0019064 167 hsa-miR-1914-3p MIMAT0007890 168 hsa-miR-619-5p
MIMAT0026622 169 hsa-miR-5001-5p MIMAT0021021 170 hsa-miR-6722-3p MIMAT0025854
171 hsa-miR-3621 MIMAT0018002 172 hsa-miR-4298 MIMAT0016852 173 hsa-miR-675-5p
MIMAT0004284 174 hsa-miR-4655-5p MIMAT0019721 175 hsa-mir-6768 MI0022613 176 hsa-
mir-6836 MI0022682 177 hsa-mir-6782 MI0022627 178 hsa-mir-3663 MI0016064 179 hsa-mir-
1908 MI0008329 180 hsa-mir-6726 MI0022571 181 hsa-mir-4258 MI0015857 182 hsa-mir-1343
MI0017320 183 hsa-mir-4516 MI0016882 184 hsa-mir-6875 MI0022722 185 hsa-mir-4651
MI0017279 186 hsa-mir-6825 MI0022670 187 hsa-mir-6840 MI0022686 188 hsa-mir-6780b
MI0022681 189 hsa-mir-6749 MI0022594 190 hsa-mir-8063 MI0025899 191 hsa-mir-6784
MI0022629 192 hsa-mir-3679 MI0016080 193 hsa-mir-3184 MI0014226 194 hsa-mir-663b
MI0006336 195 hsa-mir-6880 MI0022727 196 hsa-mir-92a-2 MI0000094 197 hsa-mir-7975
MI0025751 198 hsa-mir-7110 MI0022961 199 hsa-mir-6842 MI0022688 200 hsa-mir-6857
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example 1 of SEQ ID NO: 22 — 368 isomiR example 2 of SEQ ID NO: 22 — 369 isomiR
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example 1 of SEQ ID NO: 24 — 372 isomiR example 2 of SEQ ID NO: 24 — 373 isomiR
example 1 of SEQ ID NO: 28 — 374 isomiR example 2 of SEQ ID NO: 28 — 375 isomiR
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example 2 of SEQ ID NO: 173 — 559 isomiR example 1 of SEQ ID NO: 174 — 560 isomiR
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hsa-miR-1225-3p MIMAT0005573 576 hsa-miR-4783-3p MIMAT0019947 577 hsa-miR-4448
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(277) The present specification encompasses the contents described in the specification and/or drawings of Japanese Patent Application No. 2014-125561 on which the priority of the present application is based.

Advantageous Effects of Invention

(278) According to the present invention, lung cancer can be detected easily and in high accuracy.

(279) For example, the presence or absence of lung cancer in a patient can be easily detected by using, as an index, the expression level measurement values of several miRNAs in blood, serum, and/or plasma of the patient, which can be collected with limited invasiveness.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 This figure shows the relationship between the nucleotide sequences of hsa-miR-1908-5p represented by SEQ ID NO: 22 and hsa-miR-1908-3p represented by SEQ ID NO: 5, which are produced from a precursor hsa-mir-1908 represented by SEQ ID NO: 179.

(2) FIGS. 2A and 2B FIG. 2A: the expression level measurement values of hsa-miR-6768-5p (SEQ ID NO: 1) in healthy subjects (100 persons) and lung cancer patients (17 persons) selected as a training cohort were each plotted on the ordinate. The horizontal line in the diagram depicts a threshold (10.08) that was optimized by Fisher's discriminant analysis and discriminated between the two groups. FIG. 2B: the expression level measurement values of hsa-miR-6768-5p (SEQ ID NO: 1) in healthy subjects (50 persons) and lung cancer patients (8 persons) selected as a validation cohort were each plotted on the ordinate. The horizontal line in the diagram depicts the threshold (10.08) that was set in the training cohort and discriminated between the two groups.

(3) FIGS. 3A and 3B FIG. 3A: the expression level measurement values of hsa-miR-6768-5p (SEQ ID NO: 1) in healthy subjects (100 persons, circles) and lung cancer patients (17 persons, triangles) selected as a training cohort were each plotted on the abscissa against their expression level measurement values of hsa-miR-6836-3p (SEQ ID NO: 2) on the ordinate. The line in the diagram depicts a discriminant function ($0 = -1.42x + y + 4.7$) that was optimized by Fisher's discriminant analysis and discriminated between the two groups. FIG. 3B: the expression level measurement values of hsa-miR-6768-5p (SEQ ID NO: 1) in healthy subjects (50 persons, circles) and lung cancer patients (8 persons, triangles) selected as a validation cohort were each plotted on the abscissa against their expression level measurement values of hsa-miR-6836-3p (SEQ ID NO: 2) on the ordinate. The line in the diagram depicts the threshold ($0 = -1.42x + y + 4.7$) that was set in the

training cohort and discriminated between the two groups.

(4) FIGS. 4A and 4B FIG. 4A: a discriminant ($-1.86 \times \text{hsa-miR-6768-5p} - 0.68 \times \text{hsa-miR-19b-3p} + 0.43 \times \text{hsa-miR-6073} - 0.87 \times \text{hsa-miR-6717-5p} + 25.68$) was prepared by use of Fisher's discriminant analysis from the expression level measurement values of hsa-miR-6768-5p (SEQ ID NO: 1), hsa-miR-6717-5p (SEQ ID NO: 113), hsa-miR-19b-3p (SEQ ID NO: 126), and hsa-miR-6073 (SEQ ID NO: 561) in 17 lung cancer patients, 99 healthy subjects, 75 pancreatic cancer patients, 62 biliary tract cancer patients, 32 colorectal cancer patients, 35 stomach cancer patients, 32 esophageal cancer patients, 33 liver cancer patients, and 13 benign pancreaticobiliary disease patients selected as a training cohort, and discriminant scores obtained from the discriminant were plotted on the ordinate against the sample groups on the abscissa. The dotted line in the diagram depicts a discriminant boundary that offered a discriminant score of 0 and discriminated between the two groups. FIG. 4B: discriminant scores obtained from the discriminant prepared in the training cohort as to the expression level measurement values of hsa-miR-6768-5p (SEQ ID NO: 1), hsa-miR-6717-5p (SEQ ID NO: 113), hsa-miR-19b-3p (SEQ ID NO: 126), and hsa-miR-6073 (SEQ ID NO: 561) in 8 lung cancer patients, 51 healthy subjects, 23 pancreatic cancer patients, 38 biliary tract cancer patients, 18 colorectal cancer patients, 15 stomach cancer patients, 18 esophageal cancer patients, 19 liver cancer patients, and 8 benign pancreaticobiliary disease patients selected as a validation cohort were plotted on the ordinate against the sample groups on the abscissa. The dotted line in the diagram depicts the discriminant boundary that offered a discriminant score of 0 and discriminated between the two groups.

DESCRIPTION OF EMBODIMENTS

(5) Hereinafter, the present invention will be described further specifically.

(6) 1. Target Nucleic Acid for Lung Cancer

(7) A primary target nucleic acid used as a lung cancer marker for detecting the presence and/or absence of lung cancer or lung cancer cells using the nucleic acid probe or the primer for the detection of lung cancer defined above according to the present invention can be at least one or more miRNA(s) selected from the group consisting of hsa-miR-6768-5p, hsa-miR-6836-3p, hsa-miR-6782-5p, hsa-miR-3663-3p, hsa-miR-1908-3p, hsa-miR-6726-5p, hsa-miR-4258, hsa-miR-1343-3p, hsa-miR-4516, hsa-miR-6875-5p, hsa-miR-4651, hsa-miR-6825-5p, hsa-miR-6840-3p, hsa-miR-6780b-5p, hsa-miR-6749-5p, hsa-miR-8063, hsa-miR-6784-5p, hsa-miR-3679-5p, hsa-miR-3184-5p, hsa-miR-663b, hsa-miR-6880-5p, hsa-miR-1908-5p, hsa-miR-92a-2-5p, hsa-miR-7975, hsa-miR-7110-5p, hsa-miR-6842-5p, hsa-miR-6857-5p, hsa-miR-5572, hsa-miR-3197, hsa-miR-6131, hsa-miR-6889-5p, hsa-miR-4454, hsa-miR-1199-5p, hsa-miR-1247-3p, hsa-miR-6800-5p, hsa-miR-6872-3p, hsa-miR-4649-5p, hsa-miR-6791-5p, hsa-miR-4433b-3p, hsa-miR-3135b, hsa-miR-128-2-5p, hsa-miR-4675, hsa-miR-4472, hsa-miR-6785-5p, hsa-miR-6741-5p, hsa-miR-7977, hsa-miR-3665, hsa-miR-128-1-5p, hsa-miR-4286, hsa-miR-6765-3p, hsa-miR-4632-5p, hsa-miR-365a-5p, hsa-miR-6088, hsa-miR-6816-5p, hsa-miR-6885-5p, hsa-miR-711, hsa-miR-6765-5p, hsa-miR-3180, hsa-miR-4442, hsa-miR-4792, hsa-miR-6721-5p, hsa-miR-6798-5p, hsa-miR-3162-5p, hsa-miR-6126, hsa-miR-4758-5p, hsa-miR-2392, hsa-miR-486-3p, hsa-miR-6727-5p, hsa-miR-4728-5p, hsa-miR-6746-5p, hsa-miR-4270, hsa-miR-3940-5p, hsa-miR-4725-3p, hsa-miR-7108-5p, hsa-miR-3656, hsa-miR-6879-5p, hsa-miR-6738-5p, hsa-miR-1260a, hsa-miR-4446-3p, hsa-miR-3131, hsa-miR-4463, hsa-miR-3185, hsa-miR-6870-5p, hsa-miR-6779-5p, hsa-miR-1273g-3p, hsa-miR-8059, hsa-miR-4697-5p, hsa-miR-4674, hsa-miR-4433-3p, hsa-miR-4257, hsa-miR-1915-5p, hsa-miR-4417, hsa-miR-1343-5p, hsa-miR-6781-5p, hsa-miR-4695-5p, hsa-miR-1237-5p, hsa-miR-6775-5p, hsa-miR-7845-5p, hsa-miR-4746-3p, hsa-miR-7641, hsa-miR-7847-3p, hsa-miR-6806-5p, hsa-miR-4467, hsa-miR-4726-5p, hsa-miR-4648, hsa-miR-6089, hsa-miR-1260b, hsa-miR-4532, hsa-miR-5195-3p, hsa-miR-3188, hsa-miR-6848-5p, hsa-miR-1233-5p, hsa-miR-6717-5p, hsa-miR-3195, hsa-miR-6757-5p, hsa-miR-8072, hsa-miR-4745-5p, hsa-miR-6511a-5p, hsa-miR-6776-5p, hsa-miR-371a-5p, hsa-miR-1227-5p, hsa-miR-7150, hsa-miR-1915-3p, hsa-miR-187-5p, hsa-miR-614, hsa-miR-1225-5p, hsa-miR-451a, hsa-miR-939-5p, hsa-miR-

223-3p, hsa-miR-125a-3p, hsa-miR-92b-5p, hsa-miR-22-3p, hsa-miR-6073, hsa-miR-6845-5p, hsa-miR-6769b-5p, hsa-miR-4665-3p, hsa-miR-1913, hsa-miR-1228-3p, hsa-miR-940, hsa-miR-296-3p, hsa-miR-4690-5p, hsa-miR-548q, hsa-miR-663a, hsa-miR-1249, hsa-miR-1202, hsa-miR-7113-3p, hsa-miR-1225-3p, hsa-miR-4783-3p, hsa-miR-4448 and hsa-miR-4534. Furthermore, at least one or more miRNA(s) selected from the group consisting of other lung cancer markers that can be combined with these miRNAs, i.e., hsa-miR-19b-3p, hsa-miR-1228-5p, and hsa-miR-1307-3p, can also be preferably used as a target nucleic acid. Moreover, at least one or more miRNA(s) selected from the group consisting of other lung cancer markers that can be combined with these miRNAs, i.e., hsa-miR-4271, hsa-miR-642b-3p, hsa-miR-6075, hsa-miR-6125, hsa-miR-887-3p, hsa-miR-6851-5p, hsa-miR-6763-5p, hsa-miR-3928-3p, hsa-miR-4443, hsa-miR-3648, hsa-miR-149-3p, hsa-miR-4689, hsa-miR-4763-3p, hsa-miR-6729-5p, hsa-miR-3196, hsa-miR-8069, hsa-miR-1268a, hsa-miR-4739, hsa-miR-1268b, hsa-miR-5698, hsa-miR-6752-5p, hsa-miR-4507, hsa-miR-564, hsa-miR-4497, hsa-miR-6877-5p, hsa-miR-6087, hsa-miR-4731-5p, hsa-miR-615-5p, hsa-miR-760, hsa-miR-6891-5p, hsa-miR-6887-5p, hsa-miR-4525, hsa-miR-1914-3p, hsa-miR-619-5p, hsa-miR-5001-5p, hsa-miR-6722-3p, hsa-miR-3621, hsa-miR-4298, hsa-miR-675-5p and hsa-miR-4655-5p can also be preferably used as a target nucleic acid.

(8) These miRNAs include, for example, a human gene comprising a nucleotide sequence represented by any of SEQ ID NOs: 1 to 174 and 561 to 579 (i.e., hsa-miR-6768-5p, hsa-miR-6836-3p, hsa-miR-6782-5p, hsa-miR-3663-3p, hsa-miR-1908-3p, hsa-miR-6726-5p, hsa-miR-4258, hsa-miR-1343-3p, hsa-miR-4516, hsa-miR-6875-5p, hsa-miR-4651, hsa-miR-6825-5p, hsa-miR-6840-3p, hsa-miR-6780b-5p, hsa-miR-6749-5p, hsa-miR-8063, hsa-miR-6784-5p, hsa-miR-3679-5p, hsa-miR-3184-5p, hsa-miR-663b, hsa-miR-6880-5p, hsa-miR-1908-5p, hsa-miR-92a-2-5p, hsa-miR-7975, hsa-miR-7110-5p, hsa-miR-6842-5p, hsa-miR-6857-5p, hsa-miR-5572, hsa-miR-3197, hsa-miR-6131, hsa-miR-6889-5p, hsa-miR-4454, hsa-miR-1199-5p, hsa-miR-1247-3p, hsa-miR-6800-5p, hsa-miR-6872-3p, hsa-miR-4649-5p, hsa-miR-6791-5p, hsa-miR-4433b-3p, hsa-miR-3135b, hsa-miR-128-2-5p, hsa-miR-4675, hsa-miR-4472, hsa-miR-6785-5p, hsa-miR-6741-5p, hsa-miR-7977, hsa-miR-3665, hsa-miR-128-1-5p, hsa-miR-4286, hsa-miR-6765-3p, hsa-miR-4632-5p, hsa-miR-365a-5p, hsa-miR-6088, hsa-miR-6816-5p, hsa-miR-6885-5p, hsa-miR-711, hsa-miR-6765-5p, hsa-miR-3180, hsa-miR-4442, hsa-miR-4792, hsa-miR-6721-5p, hsa-miR-6798-5p, hsa-miR-3162-5p, hsa-miR-6126, hsa-miR-4758-5p, hsa-miR-2392, hsa-miR-486-3p, hsa-miR-6727-5p, hsa-miR-4728-5p, hsa-miR-6746-5p, hsa-miR-4270, hsa-miR-3940-5p, hsa-miR-4725-3p, hsa-miR-7108-5p, hsa-miR-3656, hsa-miR-6879-5p, hsa-miR-6738-5p, hsa-miR-1260a, hsa-miR-4446-3p, hsa-miR-3131, hsa-miR-4463, hsa-miR-3185, hsa-miR-6870-5p, hsa-miR-6779-5p, hsa-miR-1273g-3p, hsa-miR-8059, hsa-miR-4697-5p, hsa-miR-4674, hsa-miR-4433-3p, hsa-miR-4257, hsa-miR-1915-5p, hsa-miR-4417, hsa-miR-1343-5p, hsa-miR-6781-5p, hsa-miR-4695-5p, hsa-miR-1237-5p, hsa-miR-6775-5p, hsa-miR-7845-5p, hsa-miR-4746-3p, hsa-miR-7641, hsa-miR-7847-3p, hsa-miR-6806-5p, hsa-miR-4467, hsa-miR-4726-5p, hsa-miR-4648, hsa-miR-6089, hsa-miR-1260b, hsa-miR-4532, hsa-miR-5195-3p, hsa-miR-3188, hsa-miR-6848-5p, hsa-miR-1233-5p, hsa-miR-6717-5p, hsa-miR-3195, hsa-miR-6757-5p, hsa-miR-8072, hsa-miR-4745-5p, hsa-miR-6511a-5p, hsa-miR-6776-5p, hsa-miR-371a-5p, hsa-miR-1227-5p, hsa-miR-7150, hsa-miR-1915-3p, hsa-miR-187-5p, hsa-miR-614, hsa-miR-19b-3p, hsa-miR-1225-5p, hsa-miR-451a, hsa-miR-939-5p, hsa-miR-223-3p, hsa-miR-1228-5p, hsa-miR-125a-3p, hsa-miR-92b-5p, hsa-miR-22-3p, hsa-miR-6073, hsa-miR-6845-5p, hsa-miR-6769b-5p, hsa-miR-4665-3p, hsa-miR-1913, hsa-miR-1228-3p, hsa-miR-940, hsa-miR-296-3p, hsa-miR-4690-5p, hsa-miR-548q, hsa-miR-663a, hsa-miR-1249, hsa-miR-1202, hsa-miR-7113-3p, hsa-miR-1225-3p, hsa-miR-4783-3p, hsa-miR-4448 and hsa-miR-4534, hsa-miR-1307-3p, hsa-miR-4271, hsa-miR-642b-3p, hsa-miR-6075, hsa-miR-6125, hsa-miR-887-3p, hsa-miR-6851-5p, hsa-miR-6763-5p, hsa-miR-3928-3p, hsa-miR-4443, hsa-miR-3648, hsa-miR-149-3p, hsa-miR-4689, hsa-miR-4763-3p, hsa-miR-6729-5p, hsa-miR-3196, hsa-miR-8069, hsa-miR-1268a, hsa-miR-4739, hsa-miR-1268b, hsa-miR-5698, hsa-miR-6752-5p, hsa-miR-4507, hsa-miR-564, hsa-miR-4497, hsa-miR-6877-5p, hsa-

miR-6087, hsa-miR-4731-5p, hsa-miR-615-5p, hsa-miR-760, hsa-miR-6891-5p, hsa-miR-6887-5p, hsa-miR-4525, hsa-miR-1914-3p, hsa-miR-619-5p, hsa-miR-5001-5p, hsa-miR-6722-3p, hsa-miR-3621, hsa-miR-4298, hsa-miR-675-5p and hsa-miR-4655-5p, respectively), a congener thereof, a transcript thereof, and a variant or a derivative thereof. In this context, the gene, the congener, the transcript, the variant, and the derivative are as defined above.

(9) The target nucleic acid is preferably a human gene comprising a nucleotide sequence represented by any of SEQ ID NOs: 1 to 618 or a transcript thereof, more preferably the transcript, i.e., a miRNA or its precursor RNA (pri-miRNA or pre-miRNA).

(10) The first target gene is the hsa-miR-6768-5p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(11) The second target gene is the hsa-miR-6836-3p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(12) The third target gene is the hsa-miR-6782-5p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(13) The fourth target gene is the hsa-miR-3663-3p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(14) The fifth target gene is the hsa-miR-1908-3p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(15) The sixth target gene is the hsa-miR-6726-5p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(16) The seventh target gene is the hsa-miR-4258 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(17) The eighth target gene is the hsa-miR-1343-3p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(18) The ninth target gene is the hsa-miR-4516 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(19) The 10th target gene is the hsa-miR-6875-5p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(20) The 11th target gene is the hsa-miR-4651 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(21) The 12th target gene is the hsa-miR-6825-5p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(22) The 13th target gene is the hsa-miR-6840-3p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(23) The 14th target gene is the hsa-miR-6780b-5p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(190) The 181st target gene is the hsa-miR-940 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(191) The 182nd target gene is the hsa-miR-296-3p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(192) The 183rd target gene is the hsa-miR-4690-5p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(193) The 184th target gene is the hsa-miR-548q gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(194) The 185th target gene is the hsa-miR-663a gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(195) The 186th target gene is the hsa-miR-1249 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(196) The 187th target gene is the hsa-miR-1202 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(197) The 188th target gene is the hsa-miR-7113-3p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(198) The 189th target gene is the hsa-miR-1225-3p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(199) The 190th target gene is the hsa-miR-4783-3p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(200) The 191st target gene is the hsa-miR-4448 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(201) The 192nd target gene is the hsa-miR-4534 gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. None of the previously known reports show that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer.

(202) The 193rd target gene is the hsa-miR-1307-3p gene, a congener thereof, a transcript thereof, or a variant or a derivative thereof. The previously known report shows that change in the expression of the gene or the transcript thereof can serve as a marker for lung cancer (Patent Literature 3).

(203) 2. Nucleic Acid Probe or Primer for Detection of Lung Cancer

(204) In the present invention, a nucleic acid capable of specifically binding to any of the target nucleic acids as the lung cancer markers described above can be used as a nucleic acid, for example, a nucleic acid probe or a primer, for the detection or diagnosis of lung cancer.

(205) In the present invention, the nucleic acid probe or the primer that can be used for detecting lung cancer or for diagnosing lung cancer permits qualitative and/or quantitative measurement of the presence, expression level, or abundance of any of the target nucleic acids as the lung cancer markers described above, for example, human-derived hsa-miR-6768-5p, hsa-miR-6836-3p, hsa-miR-6782-5p, hsa-miR-3663-3p, hsa-miR-1908-3p, hsa-miR-6726-5p, hsa-miR-4258, hsa-miR-1343-3p, hsa-miR-4516, hsa-miR-6875-5p, hsa-miR-4651, hsa-miR-6825-5p, hsa-miR-6840-3p,

hsa-miR-6780b-5p, hsa-miR-6749-5p, hsa-miR-8063, hsa-miR-6784-5p, hsa-miR-3679-5p, hsa-miR-3184-5p, hsa-miR-663b, hsa-miR-6880-5p, hsa-miR-1908-5p, hsa-miR-92a-2-5p, hsa-miR-7975, hsa-miR-7110-5p, hsa-miR-6842-5p, hsa-miR-6857-5p, hsa-miR-5572, hsa-miR-3197, hsa-miR-6131, hsa-miR-6889-5p, hsa-miR-4454, hsa-miR-1199-5p, hsa-miR-1247-3p, hsa-miR-6800-5p, hsa-miR-6872-3p, hsa-miR-4649-5p, hsa-miR-6791-5p, hsa-miR-4433b-3p, hsa-miR-3135b, hsa-miR-128-2-5p, hsa-miR-4675, hsa-miR-4472, hsa-miR-6785-5p, hsa-miR-6741-5p, hsa-miR-7977, hsa-miR-3665, hsa-miR-128-1-5p, hsa-miR-4286, hsa-miR-6765-3p, hsa-miR-4632-5p, hsa-miR-365a-5p, hsa-miR-6088, hsa-miR-6816-5p, hsa-miR-6885-5p, hsa-miR-711, hsa-miR-6765-5p, hsa-miR-3180, hsa-miR-4442, hsa-miR-4792, hsa-miR-6721-5p, hsa-miR-6798-5p, hsa-miR-3162-5p, hsa-miR-6126, hsa-miR-4758-5p, hsa-miR-2392, hsa-miR-486-3p, hsa-miR-6727-5p, hsa-miR-4728-5p, hsa-miR-6746-5p, hsa-miR-4270, hsa-miR-3940-5p, hsa-miR-4725-3p, hsa-miR-7108-5p, hsa-miR-3656, hsa-miR-6879-5p, hsa-miR-6738-5p, hsa-miR-1260a, hsa-miR-4446-3p, hsa-miR-3131, hsa-miR-4463, hsa-miR-3185, hsa-miR-6870-5p, hsa-miR-6779-5p, hsa-miR-1273g-3p, hsa-miR-8059, hsa-miR-4697-5p, hsa-miR-4674, hsa-miR-4433-3p, hsa-miR-4257, hsa-miR-1915-5p, hsa-miR-4417, hsa-miR-1343-5p, hsa-miR-6781-5p, hsa-miR-4695-5p, hsa-miR-1237-5p, hsa-miR-6775-5p, hsa-miR-7845-5p, hsa-miR-4746-3p, hsa-miR-7641, hsa-miR-7847-3p, hsa-miR-6806-5p, hsa-miR-4467, hsa-miR-4726-5p, hsa-miR-4648, hsa-miR-6089, hsa-miR-1260b, hsa-miR-4532, hsa-miR-5195-3p, hsa-miR-3188, hsa-miR-6848-5p, hsa-miR-1233-5p, hsa-miR-6717-5p, hsa-miR-3195, hsa-miR-6757-5p, hsa-miR-8072, hsa-miR-4745-5p, hsa-miR-6511a-5p, hsa-miR-6776-5p, hsa-miR-371a-5p, hsa-miR-1227-5p, hsa-miR-7150, hsa-miR-1915-3p, hsa-miR-187-5p, hsa-miR-614, hsa-miR-1225-5p, hsa-miR-451a, hsa-miR-939-5p, hsa-miR-223-3p, hsa-miR-125a-3p, hsa-miR-92b-5p, hsa-miR-22-3p, hsa-miR-6073, hsa-miR-6845-5p, hsa-miR-6769b-5p, hsa-miR-4665-3p, hsa-miR-1913, hsa-miR-1228-3p, hsa-miR-940, hsa-miR-296-3p, hsa-miR-4690-5p, hsa-miR-548q, hsa-miR-663a, hsa-miR-1249, hsa-miR-1202, hsa-miR-7113-3p, hsa-miR-1225-3p, hsa-miR-4783-3p, hsa-miR-4448, and hsa-miR-4534 or a combination thereof, congeners thereof, transcripts thereof, or variants or derivatives thereof, optionally in combination therewith, hsa-miR-19b-3p, hsa-miR-1228-5p, and hsa-miR-1307-3p or a combination thereof, congeners thereof, transcripts thereof, or variants or derivatives thereof, and, optionally in combination therewith, hsa-miR-4271, hsa-miR-642b-3p, hsa-miR-6075, hsa-miR-6125, hsa-miR-887-3p, hsa-miR-6851-5p, hsa-miR-6763-5p, hsa-miR-3928-3p, hsa-miR-4443, hsa-miR-3648, hsa-miR-149-3p, hsa-miR-4689, hsa-miR-4763-3p, hsa-miR-6729-5p, hsa-miR-3196, hsa-miR-8069, hsa-miR-1268a, hsa-miR-4739, hsa-miR-1268b, hsa-miR-5698, hsa-miR-6752-5p, hsa-miR-4507, hsa-miR-564, hsa-miR-4497, hsa-miR-6877-5p, hsa-miR-6087, hsa-miR-4731-5p, hsa-miR-615-5p, hsa-miR-760, hsa-miR-6891-5p, hsa-miR-6887-5p, hsa-miR-4525, hsa-miR-1914-3p, hsa-miR-619-5p, hsa-miR-5001-5p, hsa-miR-6722-3p, hsa-miR-3621, hsa-miR-4298, hsa-miR-675-5p and hsa-miR-4655-5p or a combination thereof, congeners thereof, transcripts thereof, or variants or derivatives thereof.

(206) The expression level of each target nucleic acid described above is increased or decreased (hereinafter, referred to as “increased/decreased”) according to the type of the target nucleic acid in a subject who has lung cancer as compared with a healthy subject. Hence, the nucleic acid of the present invention can be effectively used for measuring the expression level of the target nucleic acid in a body fluid derived from a subject (e.g., a human) who is suspected of having lung cancer and a body fluid derived from a healthy subject, and detecting lung cancer by the comparison thereof.

(207) The nucleic acid probe or the primer that can be used in the present invention is a nucleic acid probe capable of specifically binding to a polynucleotide consisting of a nucleotide sequence represented by at least one of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578, or a primer for amplifying a polynucleotide consisting of a nucleotide sequence represented by at least one of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578.

(208) The nucleic acid probe or the primer that can be further used in the present invention can

comprise a nucleic acid probe capable of specifically binding to a polynucleotide consisting of a nucleotide sequence represented by at least one of SEQ ID NOs: 126, 131, and 579, or a primer for amplifying a polynucleotide consisting of a nucleotide sequence represented by at least one of SEQ ID NOs: 126, 131, and 579.

(209) The nucleic acid probe or the primer that can be further used in the present invention can comprise a nucleic acid probe capable of specifically binding to a polynucleotide consisting of a nucleotide sequence represented by at least one of SEQ ID NOs: 135 to 174, or a primer for amplifying a polynucleotide consisting of a nucleotide sequence represented by at least one of SEQ ID NOs: 135 to 174.

(210) Specifically, these nucleic acid probes or primers comprise a combination of one or more polynucleotides selected from a polynucleotide group comprising nucleotide sequences represented by any of SEQ ID NOs: 1 to 618, or nucleotide sequences derived from the nucleotide sequences by the replacement of u with t, and a complementary polynucleotide group thereof, a polynucleotide group respectively hybridizing under stringent conditions (mentioned later) to DNAs consisting of nucleotide sequences complementary to these nucleotide sequences, and a complementary polynucleotide group thereof, and a polynucleotide group comprising 15 or more, preferably 17 or more consecutive nucleotides in the nucleotide sequences of these polynucleotide groups. These polynucleotides can be used as nucleic acid probes and primers for detecting the lung cancer markers as target nucleic acids.

(211) More specifically, examples of the nucleic acid probe or the primer that can be used in the present invention include one or more polynucleotide(s) selected from the group consisting of the following polynucleotides (a) to (e): (a) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (b) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578, (c) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (d) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and (e) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (a) to (d).

(212) In addition to at least one or more polynucleotide(s) selected from the polynucleotides (a) to (e), the nucleic acid probe or the primer that can be further used in the present invention can comprise the polynucleotides selected from the group consisting of the following polynucleotides (f) to (j): (f) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (g) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579, (h) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (i) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and (j) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (f) to (i).

(213) In addition to at least one or more polynucleotide(s) selected from the polynucleotides (a) to (j), the nucleic acid probe or the primer that can be further used in the present invention can comprise the polynucleotides selected from the group consisting of the following polynucleotides (k) to (o): (k) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (l) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174, (m) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (n) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and (o) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (k) to (n).

(214) For these polynucleotides, the “fragment thereof comprising 15 or more consecutive nucleotides” can contain the number of nucleotides in the range from, for example, 15 consecutive nucleotides to less than the total number of nucleotides of the sequence, from 17 consecutive nucleotides to less than the total number of nucleotides of the sequence, or from 19 consecutive nucleotides to less than the total number of nucleotides of the sequence, in the nucleotide sequence of each polynucleotide, though the fragment is not limited thereto.

(215) These polynucleotides or the fragments thereof used in the present invention may each be DNA or may each be RNA.

(216) The polynucleotides that can be used in the present invention can each be prepared by use of a general technique such as a DNA recombination technique, PCR, or a method using an automatic DNA/RNA synthesizer.

(217) The DNA recombination technique and the PCR can employ a technique described in, for example, Ausubel et al., *Current Protocols in Molecular Biology*, John Wiley & Sons, US (1993); and Sambrook et al., *Molecular Cloning—A Laboratory Manual*, Cold Spring Harbor Laboratory Press, US (1989).

(218) The human-derived hsa-miR-6768-5p, hsa-miR-6836-3p, hsa-miR-6782-5p, hsa-miR-3663-3p, hsa-miR-1908-3p, hsa-miR-6726-5p, hsa-miR-4258, hsa-miR-1343-3p, hsa-miR-4516, hsa-miR-6875-5p, hsa-miR-4651, hsa-miR-6825-5p, hsa-miR-6840-3p, hsa-miR-6780b-5p, hsa-miR-6749-5p, hsa-miR-8063, hsa-miR-6784-5p, hsa-miR-3679-5p, hsa-miR-3184-5p, hsa-miR-663b, hsa-miR-6880-5p, hsa-miR-1908-5p, hsa-miR-92a-2-5p, hsa-miR-7975, hsa-miR-7110-5p, hsa-miR-6842-5p, hsa-miR-6857-5p, hsa-miR-5572, hsa-miR-3197, hsa-miR-6131, hsa-miR-6889-5p, hsa-miR-4454, hsa-miR-1199-5p, hsa-miR-1247-3p, hsa-miR-6800-5p, hsa-miR-6872-3p, hsa-miR-4649-5p, hsa-miR-6791-5p, hsa-miR-4433b-3p, hsa-miR-3135b, hsa-miR-128-2-5p, hsa-miR-4675, hsa-miR-4472, hsa-miR-6785-5p, hsa-miR-6741-5p, hsa-miR-7977, hsa-miR-3665, hsa-miR-128-1-5p, hsa-miR-4286, hsa-miR-6765-3p, hsa-miR-4632-5p, hsa-miR-365a-5p, hsa-miR-6088, hsa-miR-6816-5p, hsa-miR-6885-5p, hsa-miR-711, hsa-miR-6765-5p, hsa-miR-3180, hsa-miR-4442, hsa-miR-4792, hsa-miR-6721-5p, hsa-miR-6798-5p, hsa-miR-3162-5p, hsa-miR-6126, hsa-miR-4758-5p, hsa-miR-2392, hsa-miR-486-3p, hsa-miR-6727-5p, hsa-miR-4728-5p, hsa-miR-6746-5p, hsa-miR-4270, hsa-miR-3940-5p, hsa-miR-4725-3p, hsa-miR-7108-5p, hsa-miR-3656, hsa-miR-6879-5p, hsa-miR-6738-5p, hsa-miR-1260a, hsa-miR-4446-3p, hsa-miR-3131, hsa-miR-4463, hsa-miR-3185, hsa-miR-6870-5p, hsa-miR-6779-5p, hsa-miR-1273g-3p, hsa-miR-8059, hsa-miR-4697-5p, hsa-miR-4674, hsa-miR-4433-3p, hsa-miR-4257, hsa-miR-1915-5p, hsa-miR-4417, hsa-miR-1343-5p, hsa-miR-6781-5p, hsa-miR-4695-5p, hsa-miR-1237-5p, hsa-miR-6775-5p, hsa-miR-7845-5p, hsa-miR-4746-3p, hsa-miR-7641, hsa-miR-7847-3p, hsa-miR-6806-5p, hsa-miR-4467, hsa-miR-4726-5p, hsa-miR-4648, hsa-miR-6089, hsa-miR-1260b, hsa-miR-4532, hsa-miR-

5195-3p, hsa-miR-3188, hsa-miR-6848-5p, hsa-miR-1233-5p, hsa-miR-6717-5p, hsa-miR-3195, hsa-miR-6757-5p, hsa-miR-8072, hsa-miR-4745-5p, hsa-miR-6511a-5p, hsa-miR-6776-5p, hsa-miR-371a-5p, hsa-miR-1227-5p, hsa-miR-7150, hsa-miR-1915-3p, hsa-miR-187-5p, hsa-miR-614, hsa-miR-19b-3p, hsa-miR-1225-5p, hsa-miR-451a, hsa-miR-939-5p, hsa-miR-223-3p, hsa-miR-1228-5p, hsa-miR-125a-3p, hsa-miR-92b-5p, hsa-miR-22-3p, hsa-miR-6073, hsa-miR-6845-5p, hsa-miR-6769b-5p, hsa-miR-4665-3p, hsa-miR-1913, hsa-miR-1228-3p, hsa-miR-940, hsa-miR-296-3p, hsa-miR-4690-5p, hsa-miR-548q, hsa-miR-663a, hsa-miR-1249, hsa-miR-1202, hsa-miR-7113-3p, hsa-miR-1225-3p, hsa-miR-4783-3p, hsa-miR-4448 and hsa-miR-4534, hsa-miR-1307-3p, hsa-miR-4271, hsa-miR-642b-3p, hsa-miR-6075, hsa-miR-6125, hsa-miR-887-3p, hsa-miR-6851-5p, hsa-miR-6763-5p, hsa-miR-3928-3p, hsa-miR-4443, hsa-miR-3648, hsa-miR-149-3p, hsa-miR-4689, hsa-miR-4763-3p, hsa-miR-6729-5p, hsa-miR-3196, hsa-miR-8069, hsa-miR-1268a, hsa-miR-4739, hsa-miR-1268b, hsa-miR-5698, hsa-miR-6752-5p, hsa-miR-4507, hsa-miR-564, hsa-miR-4497, hsa-miR-6877-5p, hsa-miR-6087, hsa-miR-4731-5p, hsa-miR-615-5p, hsa-miR-760, hsa-miR-6891-5p, hsa-miR-6887-5p, hsa-miR-4525, hsa-miR-1914-3p, hsa-miR-619-5p, hsa-miR-5001-5p, hsa-miR-6722-3p, hsa-miR-3621, hsa-miR-4298, hsa-miR-675-5p and hsa-miR-4655-5p represented by SEQ ID NOs: 1 to 174, and 561 to 579 are known in the art, and their acquisition methods are also known as mentioned above. Therefore, each polynucleotide that can be used as a nucleic acid probe or a primer in the present invention can be prepared by cloning the gene.

(219) Such a nucleic acid probe or a primer can be chemically synthesized using an automatic DNA synthesis apparatus. In general, a phosphoramidite method is used in this synthesis, and single-stranded DNA up to approximately 100 nucleotides can be automatically synthesized by this method. The automatic DNA synthesis apparatus is commercially available from, for example, Polygen GmbH, ABI, or Applied Biosystems, Inc.

(220) Alternatively, the polynucleotide of the present invention can also be prepared by a cDNA cloning method. The cDNA cloning technique can employ, for example, microRNA Cloning Kit Wako.

(221) In this context, the sequences of the nucleic acid probe and the primer for detecting the polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 174, and 561 to 579 do not exist as miRNAs or precursors thereof in vivo. For example, the nucleotide sequences represented by SEQ ID NO: 5 and SEQ ID NO: 22 are produced from the precursor represented by SEQ ID NO: 179. This precursor has a hairpin-like structure as shown in FIG. 1, and the nucleotide sequences represented by SEQ ID NO: 5 and SEQ ID NO: 22 have mismatch sequences with each other. Likewise, a nucleotide sequence completely complementary to the nucleotide sequence represented by SEQ ID NO: 5 or SEQ ID NO: 22 is not naturally produced in vivo. Therefore, the nucleic acid probe and the primer for detecting the nucleotide sequence represented by any of SEQ ID NOs: 1 to 174, and 561 to 579 each have an artificial nucleotide sequence that does not exist in vivo.

(222) 3. Kit or Device for Detection of Lung Cancer

(223) The present invention also provides a kit or a device for the detection of lung cancer, comprising one or more polynucleotide(s) (which can include a variant, a fragment, and a derivative; hereinafter, also referred to as a polynucleotide for detection) that can be used as a nucleic acid probe or a primer in the present invention for measuring a target nucleic acid as a lung cancer marker.

(224) The target nucleic acid as a lung cancer marker according to the present invention is preferably selected from the following group 1: hsa-miR-6768-5p, hsa-miR-6836-3p, hsa-miR-6782-5p, hsa-miR-3663-3p, hsa-miR-1908-3p, hsa-miR-6726-5p, hsa-miR-4258, hsa-miR-1343-3p, hsa-miR-4516, hsa-miR-6875-5p, hsa-miR-4651, hsa-miR-6825-5p, hsa-miR-6840-3p, hsa-miR-6780b-5p, hsa-miR-6749-5p, hsa-miR-8063, hsa-miR-6784-5p, hsa-miR-3679-5p, hsa-miR-3184-5p, hsa-miR-663b, hsa-miR-6880-5p, hsa-miR-1908-5p, hsa-miR-92a-2-5p, hsa-miR-7975,

hsa-miR-7110-5p, hsa-miR-6842-5p, hsa-miR-6857-5p, hsa-miR-5572, hsa-miR-3197, hsa-miR-6131, hsa-miR-6889-5p, hsa-miR-4454, hsa-miR-1199-5p, hsa-miR-1247-3p, hsa-miR-6800-5p, hsa-miR-6872-3p, hsa-miR-4649-5p, hsa-miR-6791-5p, hsa-miR-4433b-3p, hsa-miR-3135b, hsa-miR-128-2-5p, hsa-miR-4675, hsa-miR-4472, hsa-miR-6785-5p, hsa-miR-6741-5p, hsa-miR-7977, hsa-miR-3665, hsa-miR-128-1-5p, hsa-miR-4286, hsa-miR-6765-3p, hsa-miR-4632-5p, hsa-miR-365a-5p, hsa-miR-6088, hsa-miR-6816-5p, hsa-miR-6885-5p, hsa-miR-711, hsa-miR-6765-5p, hsa-miR-3180, hsa-miR-4442, hsa-miR-4792, hsa-miR-6721-5p, hsa-miR-6798-5p, hsa-miR-3162-5p, hsa-miR-6126, hsa-miR-4758-5p, hsa-miR-2392, hsa-miR-486-3p, hsa-miR-6727-5p, hsa-miR-4728-5p, hsa-miR-6746-5p, hsa-miR-4270, hsa-miR-3940-5p, hsa-miR-4725-3p, hsa-miR-7108-5p, hsa-miR-3656, hsa-miR-6879-5p, hsa-miR-6738-5p, hsa-miR-1260a, hsa-miR-4446-3p, hsa-miR-3131, hsa-miR-4463, hsa-miR-3185, hsa-miR-6870-5p, hsa-miR-6779-5p, hsa-miR-1273g-3p, hsa-miR-8059, hsa-miR-4697-5p, hsa-miR-4674, hsa-miR-4433-3p, hsa-miR-4257, hsa-miR-1915-5p, hsa-miR-4417, hsa-miR-1343-5p, hsa-miR-6781-5p, hsa-miR-4695-5p, hsa-miR-1237-5p, hsa-miR-6775-5p, hsa-miR-7845-5p, hsa-miR-4746-3p, hsa-miR-7641, hsa-miR-7847-3p, hsa-miR-6806-5p, hsa-miR-4467, hsa-miR-4726-5p, hsa-miR-4648, hsa-miR-6089, hsa-miR-1260b, hsa-miR-4532, hsa-miR-5195-3p, hsa-miR-3188, hsa-miR-6848-5p, hsa-miR-1233-5p, hsa-miR-6717-5p, hsa-miR-3195, hsa-miR-6757-5p, hsa-miR-8072, hsa-miR-4745-5p, hsa-miR-6511a-5p, hsa-miR-6776-5p, hsa-miR-371a-5p, hsa-miR-1227-5p, hsa-miR-7150, hsa-miR-1915-3p, hsa-miR-187-5p, hsa-miR-614, hsa-miR-1225-5p, hsa-miR-451a, hsa-miR-939-5p, hsa-miR-223-3p, hsa-miR-125a-3p, hsa-miR-92b-5p, hsa-miR-22-3p, hsa-miR-6073, hsa-miR-6845-5p, hsa-miR-6769b-5p, hsa-miR-4665-3p, hsa-miR-1913, hsa-miR-1228-3p, hsa-miR-940, hsa-miR-296-3p, hsa-miR-4690-5p, hsa-miR-548q, hsa-miR-663a, hsa-miR-1249, hsa-miR-1202, hsa-miR-7113-3p, hsa-miR-1225-3p, hsa-miR-4783-3p, hsa-miR-4448 and hsa-miR-4534.

(225) An additional target nucleic acid that can be optionally used in the measurement is selected from the following group 2: hsa-miR-19b-3p, hsa-miR-1228-5p and hsa-miR-1307-3p.

(226) An additional target nucleic acid that can be optionally further used in the measurement is selected from the following group 3: hsa-miR-4271, hsa-miR-642b-3p, hsa-miR-6075, hsa-miR-6125, hsa-miR-887-3p, hsa-miR-6851-5p, hsa-miR-6763-5p, hsa-miR-3928-3p, hsa-miR-4443, hsa-miR-3648, hsa-miR-149-3p, hsa-miR-4689, hsa-miR-4763-3p, hsa-miR-6729-5p, hsa-miR-3196, hsa-miR-8069, hsa-miR-1268a, hsa-miR-4739, hsa-miR-1268b, hsa-miR-5698, hsa-miR-6752-5p, hsa-miR-4507, hsa-miR-564, hsa-miR-4497, hsa-miR-6877-5p, hsa-miR-6087, hsa-miR-4731-5p, hsa-miR-615-5p, hsa-miR-760, hsa-miR-6891-5p, hsa-miR-6887-5p, hsa-miR-4525, hsa-miR-1914-3p, hsa-miR-619-5p, hsa-miR-5001-5p, hsa-miR-6722-3p, hsa-miR-3621, hsa-miR-4298, hsa-miR-675-5p and hsa-miR-4655-5p.

(227) The kit or the device of the present invention comprises a nucleic acid capable of specifically binding to any of the target nucleic acids as the lung cancer markers described above, preferably one or more polynucleotide(s) selected from the nucleic acid probes or the primers described in the preceding Section 2, specifically, the polynucleotides described in the preceding paragraph 2, or variant(s) thereof.

(228) Specifically, the kit or the device of the present invention can comprise at least one or more polynucleotide(s) comprising (or consisting of) a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, polynucleotide(s) comprising (or consisting of) a complementary sequence thereof, polynucleotide(s) hybridizing under stringent conditions to any of these polynucleotides, or variant(s) or fragment(s) comprising 15 or more consecutive nucleotides of any of these polynucleotide sequences.

(229) The kit or the device of the present invention can further comprise one or more polynucleotide(s) comprising (or consisting of) a nucleotide sequence represented by any of SEQ ID NOs: 126 and 131 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, polynucleotide(s) comprising (or consisting of) a complementary sequence

thereof, polynucleotide(s) hybridizing under stringent conditions to any of these polynucleotides, variant(s) or fragment(s) comprising 15 or more consecutive nucleotides of any of these polynucleotide sequences.

(230) The kit or the device of the present invention can further comprise one or more polynucleotide(s) comprising (or consisting of) a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, polynucleotide(s) comprising (or consisting of) a complementary sequence thereof, polynucleotide(s) hybridizing under stringent conditions to any of these polynucleotides, variant(s) or fragment(s) comprising 15 or more consecutive nucleotides of any of these polynucleotide sequences.

(231) The fragment that can be contained in the kit or the device of the present invention is, for example, one or more, preferably two or more polynucleotides selected from the group consisting of the following polynucleotides (1) to (3): (1) a polynucleotide comprising 15 or more consecutive nucleotides in a nucleotide sequence derived from a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578 by the replacement of u with t, or a complementary sequence thereof; (2) a polynucleotide comprising 15 or more consecutive nucleotides in a nucleotide sequence derived from a nucleotide sequence represented by any of SEQ ID NOs: 126, 131 and 579 by the replacement of u with t, or a complementary sequence thereof; and (3) a polynucleotide comprising 15 or more consecutive nucleotides in a nucleotide sequence derived from a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 by the replacement of u with t, or a complementary sequence thereof.

(232) In a preferred embodiment, the polynucleotide is a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578, or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a polynucleotide consisting of a complementary sequence thereof, a polynucleotide hybridizing under stringent conditions to any of these polynucleotides, or a variant thereof comprising 15 or more, preferably 17 or more, more preferably 19 or more consecutive nucleotides.

(233) In a preferred embodiment, the polynucleotide is a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 126, 134 and 579 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a polynucleotide consisting of a complementary sequence thereof, a polynucleotide hybridizing under stringent conditions to any of these polynucleotides, or a variant thereof comprising 15 or more, preferably 17 or more, more preferably 19 or more consecutive nucleotides.

(234) In a preferred embodiment, the polynucleotide is a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a polynucleotide consisting of a complementary sequence thereof, a polynucleotide hybridizing under stringent conditions to any of these polynucleotides, or a variant thereof comprising 15 or more, preferably 17 or more, more preferably 19 or more consecutive nucleotides.

(235) In a preferred embodiment, the fragment can be a polynucleotide comprising 15 or more, preferably 17 or more, more preferably 19 or more consecutive nucleotides.

(236) In the present invention, the size of the polynucleotide fragment is the number of bases in the range from, for example, 15 consecutive nucleotides to less than the total number of bases of the sequence, from 17 consecutive nucleotides to less than the total number of bases of the sequence, or from 19 consecutive nucleotides to less than the total number of nucleotides of the sequence, in the nucleotide sequence of each polynucleotide.

(237) Specific examples of the aforementioned combination of the polynucleotides constituting the kit or the device of the present invention can include the polynucleotides as to combinations of SEQ ID NOs shown in Table 1 (SEQ ID NOs: 1 to 174, and 561 to 579 corresponding to the miRNA markers in Table 1). However, these are given merely for illustrative purposes, and various

other possible combinations are included in the present invention.

(238) The aforementioned combination constituting the kit or the device for discriminating a lung cancer patient from a healthy subject according to the present invention is desirably, for example, a combination of two or more of the aforementioned polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs shown in Table 1. Usually, a combination of two of these polynucleotides can produce adequate performance.

(239) The combination of two polynucleotides consisting of the nucleotide sequences or the complementary sequences thereof for specifically discriminating a lung cancer patient from a healthy subject is preferably a combination comprising at least one or more of newly found polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 125, 127 to 130, 132 to 174, and 561 to 578, among the combinations constituted by two of the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 174, and 561 to 579.

(240) The combination of polynucleotides with cancer type specificity capable of discriminating a lung cancer patient not only from a healthy subject but also from other cancer patients is preferably, for example, a combination of multiple polynucleotides comprising at least one polynucleotide selected from the group consisting of polynucleotides of SEQ ID NOs: 1, 2, 3, 4, 5, 7, 9, 10, 11, 19, 21, 26, 29, 31, 52, 53, 63, 65, 69, 72, 87, 90, 113, 124, 125, 126, 128, 130, 143, 148, 160, 162, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578 and 579 (hereinafter, this group is referred to as “cancer type-specific polynucleotide group 1”), with any of the polynucleotides of the other SEQ ID NOs.

(241) The combination of polynucleotides with cancer type specificity capable of discriminating a lung cancer patient not only from a healthy subject but also from other cancer patients is more preferably a combination of multiple polynucleotides selected from the cancer type-specific polynucleotide group 1.

(242) The combination of polynucleotides with cancer type specificity capable of discriminating a lung cancer patient not only from a healthy subject but also from other cancer patients is more preferably a combination comprising at least one or more polynucleotide(s) selected from the group consisting of polynucleotides of SEQ ID NOs: 1, 2, 3, 10, 63, 113, 124, 125, 126, 128, 130, 143, 160, 561, 568, 573 and 578 (hereinafter, this group is referred to as “cancer type-specific polynucleotide group 2”) included in the cancer type-specific polynucleotide group 1, among the combinations of multiple polynucleotides selected from the cancer type-specific polynucleotide group 1.

(243) The number of the polynucleotides with cancer type specificity in the aforementioned combination can be 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more for the combination and is more preferably 4 or more for the combination. Usually, the combination of 4 of these polynucleotides can produce adequate performance.

(244) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 1 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be listed below. (1) a combination of SEQ ID NOs: 1, 53, 113, and 125 (markers: hsa-miR-6768-5p, hsa-miR-6088, hsa-miR-6717-5p, and hsa-miR-614); (2) a combination of SEQ ID NOs: 1, 10, 63, and 113 (markers: hsa-miR-6768-5p, hsa-miR-6875-5p, hsa-miR-3162-5p, and hsa-miR-6717-5p); (3) a combination of SEQ ID NOs: 1, 19, 113, and 143 (markers: hsa-miR-6768-5p, hsa-miR-3184-5p, hsa-miR-6717-5p, and hsa-miR-4443); (4) a combination of SEQ ID NOs: 1, 10, 113, and 126 (markers: hsa-miR-6768-5p, hsa-miR-6875-5p, hsa-miR-6717-5p, and hsa-miR-19b-3p); and (5) a combination of SEQ ID NOs: 1, 2, 10, and 113 (markers: hsa-miR-6768-5p, hsa-miR-6836-3p, hsa-miR-6875-5p, and hsa-miR-6717-5p).

(245) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide

sequence represented by SEQ ID NO: 2 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 2, 19, 53, and 113 (markers: hsa-miR-6836-3p, hsa-miR-3184-5p, hsa-miR-6088, and hsa-miR-6717-5p); (2) a combination of SEQ ID NOs: 2, 72, 113, and 125 (markers: hsa-miR-6836-3p, hsa-miR-3940-5p, hsa-miR-6717-5p, and hsa-miR-614); (3) a combination of SEQ ID NOs: 2, 19, 72, and 113 (markers: hsa-miR-6836-3p, hsa-miR-3184-5p, hsa-miR-3940-5p, and hsa-miR-6717-5p); (4) a combination of SEQ ID NOs: 2, 19, 113, and 579 (markers: hsa-miR-6836-3p, hsa-miR-3184-5p, hsa-miR-6717-5p, and hsa-miR-1307-3p); and (5) a combination of SEQ ID NOs: 1, 2, 19, and 113 (markers: hsa-miR-6768-5p, hsa-miR-6836-3p, hsa-miR-3184-5p, and hsa-miR-6717-5p).

(246) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 3 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 3, 125, 128, and 568 (markers: hsa-miR-6782-5p, hsa-miR-614, hsa-miR-451a, and hsa-miR-296-3p); (2) a combination of SEQ ID NOs: 1, 3, 10, and 113 (markers: hsa-miR-6768-5p, hsa-miR-6782-5p, hsa-miR-6875-5p, and hsa-miR-6717-5p); (3) a combination of SEQ ID NOs: 3, 113, 125, and 126 (markers: hsa-miR-6782-5p, hsa-miR-6717-5p, hsa-miR-614, and hsa-miR-19b-3p); (4) a combination of SEQ ID NOs: 1, 3, 126, and 573 (markers: hsa-miR-6768-5p, hsa-miR-6782-5p, hsa-miR-19b-3p, and hsa-miR-1202); and (5) a combination of SEQ ID NOs: 3, 126, 130, and 561 (markers: hsa-miR-6782-5p, hsa-miR-19b-3p, hsa-miR-223-3p, and hsa-miR-6073).

(247) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 10 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 1, 10, 113, and 143 (markers: hsa-miR-6768-5p, hsa-miR-6875-5p, hsa-miR-6717-5p, and hsa-miR-4443); (2) a combination of SEQ ID NOs: 1, 10, 113, and 569 (markers: hsa-miR-6768-5p, hsa-miR-6875-5p, hsa-miR-6717-5p, and hsa-miR-4690-5p); (3) a combination of SEQ ID NOs: 1, 10, 113, and 562 (markers: hsa-miR-6768-5p, hsa-miR-6875-5p, hsa-miR-6717-5p, and hsa-miR-6845-5p); (4) a combination of SEQ ID NOs: 1, 10, 113, and 578 (markers: hsa-miR-6768-5p, hsa-miR-6875-5p, hsa-miR-6717-5p, hsa-miR-4534); and (5) a combination of SEQ ID NOs: 1, 7, 10, and 113 (markers: hsa-miR-6768-5p, hsa-miR-4258, hsa-miR-6875-5p, and hsa-miR-6717-5p).

(248) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 63 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 1, 63, 567, and 578 (markers: hsa-miR-6768-5p, hsa-miR-3162-5p, hsa-miR-940, and hsa-miR-4534); (2) a combination of SEQ ID NOs: 1, 53, 63, and 578 (markers: hsa-miR-6768-5p, hsa-miR-6088, hsa-miR-3162-5p, and hsa-miR-4534); (3) a combination of SEQ ID NOs: 1, 63, 162, and 573 (markers: hsa-miR-6768-5p, hsa-miR-3162-5p, hsa-miR-615-5p, and hsa-miR-1202); (4) a combination of SEQ ID NOs: 1, 63, 162, and 578 (markers: hsa-miR-6768-5p, hsa-miR-3162-5p, hsa-miR-615-5p, and hsa-miR-4534); and (5) a combination of SEQ ID NOs: 1, 63, 576, and 578 (markers: hsa-miR-6768-5p, hsa-miR-3162-5p, hsa-miR-4783-3p, and hsa-miR-4534).

(249) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 113 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five

polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 1, 10, 113, and 567 (markers: hsa-miR-6768-5p, hsa-miR-6875-5p, hsa-miR-6717-5p, and hsa-miR-940); (2) a combination of SEQ ID NOs: 1, 53, 63, and 113 (markers: hsa-miR-6768-5p, hsa-miR-6088, hsa-miR-3162-5p, and hsa-miR-6717-5p); (3) a combination of SEQ ID NOs: 1, 53, 113, and 143 (markers: hsa-miR-6768-5p, hsa-miR-6088, hsa-miR-6717-5p, and hsa-miR-4443); (4) a combination of SEQ ID NOs: 2, 19, 113, and 125 (markers: hsa-miR-6836-3p, hsa-miR-3184-5p, hsa-miR-6717-5p, and hsa-miR-614); and (5) a combination of SEQ ID NOs: 2, 10, 113, and 130 (markers: hsa-miR-6836-3p, hsa-miR-6875-5p, hsa-miR-6717-5p, and hsa-miR-223-3p).

(250) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 124 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 113, 124, 125, and 126 (markers: hsa-miR-6717-5p, hsa-miR-187-5p, hsa-miR-614, and hsa-miR-19b-3p); (2) a combination of SEQ ID NOs: 124, 125, 128, and 568 (markers: hsa-miR-187-5p, hsa-miR-614, hsa-miR-451a, and hsa-miR-296-3p); (3) a combination of SEQ ID NOs: 113, 124, 125, and 162 (markers: hsa-miR-6717-5p, hsa-miR-187-5p, hsa-miR-614, and hsa-miR-615-5p); (4) a combination of SEQ ID NOs: 52, 124, 126, and 561 (markers: hsa-miR-365a-5p, hsa-miR-187-5p, hsa-miR-19b-3p, and hsa-miR-6073); and (5) a combination of SEQ ID NOs: 19, 113, 124, and 126 (markers: hsa-miR-3184-5p, hsa-miR-6717-5p, hsa-miR-187-5p, and hsa-miR-19b-3p).

(251) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 125 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 1, 113, 125, and 160 (markers: hsa-miR-6768-5p, hsa-miR-6717-5p, hsa-miR-614, and hsa-miR-6087); (2) a combination of SEQ ID NOs: 31, 113, 125, and 568 (markers: hsa-miR-6889-5p, hsa-miR-6717-5p, hsa-miR-614, and hsa-miR-296-3p); (3) a combination of SEQ ID NOs: 2, 53, 113, and 125 (markers: hsa-miR-6836-3p, hsa-miR-6088, hsa-miR-6717-5p, and hsa-miR-614); (4) a combination of SEQ ID NOs: 1, 10, 113, and 125 (markers: hsa-miR-6768-5p, hsa-miR-6875-5p, hsa-miR-6717-5p, and hsa-miR-614); and (5) a combination of SEQ ID NOs: 1, 113, 125, and 143 (markers: hsa-miR-6768-5p, hsa-miR-6717-5p, hsa-miR-614, and hsa-miR-4443).

(252) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 126 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 1, 126, 561, and 573 (markers: hsa-miR-6768-5p, hsa-miR-19b-3p, hsa-miR-6073, and hsa-miR-1202); (2) a combination of SEQ ID NOs: 113, 125, 126, and 568 (markers: hsa-miR-6717-5p, hsa-miR-614, hsa-miR-19b-3p, and hsa-miR-296-3p); (3) a combination of SEQ ID NOs: 113, 125, 126, and 561 (markers: hsa-miR-6717-5p, hsa-miR-614, hsa-miR-19b-3p, and hsa-miR-6073); (4) a combination of SEQ ID NOs: 1, 113, 125, and 126 (markers: hsa-miR-6768-5p, hsa-miR-6717-5p, hsa-miR-614, and hsa-miR-19b-3p); and (5) a combination of SEQ ID NOs: 1, 52, 126, and 561 (markers: hsa-miR-6768-5p, hsa-miR-365a-5p, hsa-miR-19b-3p, and hsa-miR-6073).

(253) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 128 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 26, 113, 125, and

128 (markers: hsa-miR-6842-5p, hsa-miR-6717-5p, hsa-miR-614, and hsa-miR-451a); (2) a combination of SEQ ID NOs: 1, 113, 125, and 128 (markers: hsa-miR-6768-5p, hsa-miR-6717-5p, hsa-miR-614, and hsa-miR-451a); (3) a combination of SEQ ID NOs: 1, 10, 113, and 128 (markers: hsa-miR-6768-5p, hsa-miR-6875-5p, hsa-miR-6717-5p, and hsa-miR-451a); (4) a combination of SEQ ID NOs: 31, 113, 125, and 128 (markers: hsa-miR-6889-5p, hsa-miR-6717-5p, hsa-miR-614, and hsa-miR-451a); and (5) a combination of SEQ ID NOs: 2, 19, 113, and 128 (markers: hsa-miR-6836-3p, hsa-miR-3184-5p, hsa-miR-6717-5p, and hsa-miR-451a).

(254) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 130 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 1, 3, 130, and 143 (markers: hsa-miR-6768-5p, hsa-miR-6782-5p, hsa-miR-223-3p, and hsa-miR-4443); (2) a combination of SEQ ID NOs: 1, 10, 113, and 130 (markers: hsa-miR-6768-5p, hsa-miR-6875-5p, hsa-miR-6717-5p, and hsa-miR-223-3p); (3) a combination of SEQ ID NOs: 1, 63, 130, and 578 (markers: hsa-miR-6768-5p, hsa-miR-3162-5p, hsa-miR-223-3p, and hsa-miR-4534); (4) a combination of SEQ ID NOs: 124, 125, 130, and 568 (markers: hsa-miR-187-5p, hsa-miR-614, hsa-miR-223-3p, and hsa-miR-296-3p); and (5) a combination of SEQ ID NOs: 2, 19, 113, and 130 (markers: hsa-miR-6836-3p, hsa-miR-3184-5p, hsa-miR-6717-5p, and hsa-miR-223-3p).

(255) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 143 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 1, 3, 126, and 143 (markers: hsa-miR-6768-5p, hsa-miR-6782-5p, hsa-miR-19b-3p, and hsa-miR-4443); (2) a combination of SEQ ID NOs: 1, 63, 130, and 143 (markers: hsa-miR-6768-5p, hsa-miR-3162-5p, hsa-miR-223-3p, and hsa-miR-4443); (3) a combination of SEQ ID NOs: 1, 10, 52, and 143 (markers: hsa-miR-6768-5p, hsa-miR-6875-5p, hsa-miR-365a-5p, and hsa-miR-4443); (4) a combination of SEQ ID NOs: 2, 19, 113, and 143 (markers: hsa-miR-6836-3p, hsa-miR-3184-5p, hsa-miR-6717-5p, and hsa-miR-4443); and (5) a combination of SEQ ID NOs: 63, 124, 130, and 143 (markers: hsa-miR-3162-5p, hsa-miR-187-5p, hsa-miR-223-3p, and hsa-miR-4443).

(256) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 160 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 1, 10, 113, and 160 (markers: hsa-miR-6768-5p, hsa-miR-6875-5p, hsa-miR-6717-5p, and hsa-miR-6087); (2) a combination of SEQ ID NOs: 7, 113, 125, and 160 (markers: hsa-miR-4258, hsa-miR-6717-5p, hsa-miR-614, and hsa-miR-6087); (3) a combination of SEQ ID NOs: 1, 113, 160, and 567 (markers: hsa-miR-6768-5p, hsa-miR-6717-5p, hsa-miR-6087, and hsa-miR-940); (4) a combination of SEQ ID NOs: 1, 113, 160, and 578 (markers: hsa-miR-6768-5p, hsa-miR-6717-5p, hsa-miR-6087, and hsa-miR-4534); and (5) a combination of SEQ ID NOs: 2, 19, 113, and 160 (markers: hsa-miR-6836-3p, hsa-miR-3184-5p, hsa-miR-6717-5p, and hsa-miR-6087).

(257) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 561 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 113, 125, 130, and 561 (markers: hsa-miR-6717-5p, hsa-miR-614, hsa-miR-223-3p, and hsa-miR-6073); (2) a combination of SEQ ID NOs: 7, 126, 143, and 561 (markers: hsa-miR-4258, hsa-miR-19b-3p, hsa-

miR-4443, and hsa-miR-6073); (3) a combination of SEQ ID NOs: 1, 113, and 126, 561 (markers: hsa-miR-6768-5p, hsa-miR-6717-5p, hsa-miR-19b-3p, and hsa-miR-6073); (4) a combination of SEQ ID NOs: 1, 126, 561, and 568 (markers: hsa-miR-6768-5p, hsa-miR-19b-3p, hsa-miR-6073, and hsa-miR-296-3p); and (5) a combination of SEQ ID NOs: 7, 113, 126, and 561 (markers: hsa-miR-4258, hsa-miR-6717-5p, hsa-miR-19b-3p, and hsa-miR-6073).

(258) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 568 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 7, 125, 126, and 568 (markers: hsa-miR-4258, hsa-miR-614, hsa-miR-19b-3p, and hsa-miR-296-3p); (2) a combination of SEQ ID NOs: 124, 125, 126, and 568 (markers: hsa-miR-187-5p, hsa-miR-614, hsa-miR-19b-3p, and hsa-miR-296-3p); (3) a combination of SEQ ID NOs: 7, 113, 125, and 568 (markers: hsa-miR-4258, hsa-miR-6717-5p, hsa-miR-614, and hsa-miR-296-3p); (4) a combination of SEQ ID NOs: 1, 113, 125, and 568 (markers: hsa-miR-6768-5p, hsa-miR-6717-5p, hsa-miR-614, and hsa-miR-296-3p); and (5) a combination of SEQ ID NOs: 113, 125, 128, and 568 (markers: hsa-miR-6717-5p, hsa-miR-614, hsa-miR-451a, and hsa-miR-296-3p).

(259) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 573 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 113, 125, 126, and 573 (markers: hsa-miR-6717-5p, hsa-miR-614, hsa-miR-19b-3p, and hsa-miR-1202); (2) a combination of SEQ ID NOs: 1, 113, 125, and 573 (markers: hsa-miR-6768-5p, hsa-miR-6717-5p, hsa-miR-614, and hsa-miR-1202); (3) a combination of SEQ ID NOs: 1, 53, 113, and 573 (markers: hsa-miR-6768-5p, hsa-miR-6088, hsa-miR-6717-5p, and hsa-miR-1202); (4) a combination of SEQ ID NOs: 1, 124, 126, and 573 (markers: hsa-miR-6768-5p, hsa-miR-187-5p, hsa-miR-19b-3p, and hsa-miR-1202); and (5) a combination of SEQ ID NOs: 1, 63, 130, and 573 (markers: hsa-miR-6768-5p, hsa-miR-3162-5p, hsa-miR-223-3p, and hsa-miR-1202).

(260) Non-limiting examples of the combination of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 578 or a complementary sequence thereof with polynucleotides consisting of nucleotide sequences represented by SEQ ID NOs of five polynucleotides selected from the cancer type-specific polynucleotide group 1 or complementary sequences thereof will be further listed below. (1) a combination of SEQ ID NOs: 1, 126, 567, and 578 (markers: hsa-miR-6768-5p, hsa-miR-19b-3p, hsa-miR-940, and hsa-miR-4534); (2) a combination of SEQ ID NOs: 1, 19, 113, and 578 (markers: hsa-miR-6768-5p, hsa-miR-3184-5p, hsa-miR-6717-5p, and hsa-miR-4534); (3) a combination of SEQ ID NOs: 31, 126, 561, and 578 (markers: hsa-miR-6889-5p, hsa-miR-19b-3p, hsa-miR-6073, and hsa-miR-4534); (4) a combination of SEQ ID NOs: 1, 126, 160, and 578 (markers: hsa-miR-6768-5p, hsa-miR-19b-3p, hsa-miR-6087, and hsa-miR-4534); and (5) a combination of SEQ ID NOs: 1, 113, 125, 578 (markers: hsa-miR-6768-5p, hsa-miR-6717-5p, hsa-miR-614, and hsa-miR-4534).

(261) The kit or the device of the present invention can also contain a polynucleotide that is already known or that will be found in the future, to enable detection of lung cancer, in addition to the polynucleotide(s) (which can include the variant(s), the fragment(s), and the derivative(s)) according to the present invention described above.

(262) The kit of the present invention can also contain an antibody for measuring a marker for lung cancer examination known in the art, such as CEA, or CYFRA21-1, in addition to the polynucleotide(s) according to the present invention described above.

(263) These polynucleotides contained in the kit of the present invention can be packaged in different containers either individually or in any combination.

(264) The kit of the present invention can contain a kit for extracting a nucleic acid (e.g., total RNA) from body fluids, cells, or tissues, a fluorescent material for labeling, an enzyme and a medium for nucleic acid amplification, an instruction manual, etc.

(265) The device of the present invention is a device for cancer marker measurement in which nucleic acids such as the polynucleotides according to the present invention described above are bonded or attached to, for example, a solid phase. Examples of the material for the solid phase include plastics, paper, glass, and silicon. The material for the solid phase is preferably a plastic from the viewpoint of easy processability. The solid phase has any shape and is, for example, square, round, reed-shaped, or film-shaped. The device of the present invention includes, for example, a device for measurement by a hybridization technique. Specific examples thereof include blotting devices and nucleic acid arrays (e.g., microarrays, DNA chips, and RNA chips).

(266) The nucleic acid array technique is a technique which involves bonding or attaching the nucleic acids one by one by use of a method [e.g., a method of spotting the nucleic acids using a high-density dispenser called spotter or arrayer onto the surface of the solid phase surface-treated, if necessary, by coating with L-lysine or the introduction of a functional group such as an amino group or a carboxyl group, a method of spraying the nucleic acids onto the solid phase using an inkjet which injects very small liquid droplets by a piezoelectric element or the like from a nozzle, or a method of sequentially synthesizing nucleotides on the solid phase] to prepare an array such as a chip and measuring a target nucleic acid through the use of hybridization using this array.

(267) The kit or the device of the present invention comprises nucleic acids capable of specifically binding to the polynucleotides of at least one or more, preferably at least two or more, more preferably at least three or more, most preferably at least five or more to all of the lung cancer marker miRNAs, respectively, of the group 1 described above. The kit or the device of the present invention can optionally further comprise nucleic acids capable of specifically binding to the polynucleotides of at least one or more, preferably at least two or more, more preferably at least three or more, most preferably at least five or more to all of the lung cancer marker miRNAs, respectively, of the group 2 described above. The kit or the device of the present invention can optionally further comprise nucleic acids capable of specifically binding to the polynucleotides of at least one or more, preferably at least two or more, more preferably at least three or more, most preferably at least five or more to all of the lung cancer marker miRNAs, respectively, of the group 3 described above.

(268) The kit or the device of the present invention can be used for detecting lung cancer as described in Section 4 below.

(269) 4. Method for Detecting Lung Cancer

(270) The present invention further provides a method for detecting lung cancer, comprising using the kit or the device of the present invention (including the aforementioned nucleic acid(s) that can be used in the present invention) described in Section 3 above to measure an expression level(s) of one or more lung cancer-derived gene(s) represented by an expression level(s) of lung cancer-derived gene(s) selected from the following group A: hsa-miR-6768-5p, hsa-miR-6836-3p, hsa-miR-6782-5p, hsa-miR-3663-3p, hsa-miR-1908-3p, hsa-miR-6726-5p, hsa-miR-4258, hsa-miR-1343-3p, hsa-miR-4516, hsa-miR-6875-5p, hsa-miR-4651, hsa-miR-6825-5p, hsa-miR-6840-3p, hsa-miR-6780b-5p, hsa-miR-6749-5p, hsa-miR-8063, hsa-miR-6784-5p, hsa-miR-3679-5p, hsa-miR-3184-5p, hsa-miR-663b, hsa-miR-6880-5p, hsa-miR-1908-5p, hsa-miR-92a-2-5p, hsa-miR-7975, hsa-miR-7110-5p, hsa-miR-6842-5p, hsa-miR-6857-5p, hsa-miR-5572, hsa-miR-3197, hsa-miR-6131, hsa-miR-6889-5p, hsa-miR-4454, hsa-miR-1199-5p, hsa-miR-1247-3p, hsa-miR-6800-5p, hsa-miR-6872-3p, hsa-miR-4649-5p, hsa-miR-6791-5p, hsa-miR-4433b-3p, hsa-miR-3135b, hsa-miR-128-2-5p, hsa-miR-4675, hsa-miR-4472, hsa-miR-6785-5p, hsa-miR-6741-5p, hsa-miR-7977, hsa-miR-3665, hsa-miR-128-1-5p, hsa-miR-4286, hsa-miR-6765-3p, hsa-miR-4632-5p, hsa-miR-365a-5p, hsa-miR-6088, hsa-miR-6816-5p, hsa-miR-6885-5p, hsa-miR-711, hsa-miR-6765-5p, hsa-miR-3180, hsa-miR-4442, hsa-miR-4792, hsa-miR-6721-5p, hsa-miR-6798-5p, hsa-miR-

3162-5p, hsa-miR-6126, hsa-miR-4758-5p, hsa-miR-2392, hsa-miR-486-3p, hsa-miR-6727-5p, hsa-miR-4728-5p, hsa-miR-6746-5p, hsa-miR-4270, hsa-miR-3940-5p, hsa-miR-4725-3p, hsa-miR-7108-5p, hsa-miR-3656, hsa-miR-6879-5p, hsa-miR-6738-5p, hsa-miR-1260a, hsa-miR-4446-3p, hsa-miR-3131, hsa-miR-4463, hsa-miR-3185, hsa-miR-6870-5p, hsa-miR-6779-5p, hsa-miR-1273g-3p, hsa-miR-8059, hsa-miR-4697-5p, hsa-miR-4674, hsa-miR-4433-3p, hsa-miR-4257, hsa-miR-1915-5p, hsa-miR-4417, hsa-miR-1343-5p, hsa-miR-6781-5p, hsa-miR-4695-5p, hsa-miR-1237-5p, hsa-miR-6775-5p, hsa-miR-7845-5p, hsa-miR-4746-3p, hsa-miR-7641, hsa-miR-7847-3p, hsa-miR-6806-5p, hsa-miR-4467, hsa-miR-4726-5p, hsa-miR-4648, hsa-miR-6089, hsa-miR-1260b, hsa-miR-4532, hsa-miR-5195-3p, hsa-miR-3188, hsa-miR-6848-5p, hsa-miR-1233-5p, hsa-miR-6717-5p, hsa-miR-3195, hsa-miR-6757-5p, hsa-miR-8072, hsa-miR-4745-5p, hsa-miR-6511a-5p, hsa-miR-6776-5p, hsa-miR-371a-5p, hsa-miR-1227-5p, hsa-miR-7150, hsa-miR-1915-3p, hsa-miR-187-5p, hsa-miR-614, hsa-miR-1225-5p, hsa-miR-451a, hsa-miR-939-5p, hsa-miR-223-3p, hsa-miR-125a-3p, hsa-miR-92b-5p, hsa-miR-22-3p, hsa-miR-6073, hsa-miR-6845-5p, hsa-miR-6769b-5p, hsa-miR-4665-3p, hsa-miR-1913, hsa-miR-1228-3p, hsa-miR-940, hsa-miR-296-3p, hsa-miR-4690-5p, hsa-miR-548q, hsa-miR-663a, hsa-miR-1249, hsa-miR-1202, hsa-miR-7113-3p, hsa-miR-1225-3p, hsa-miR-4783-3p, hsa-miR-4448 and hsa-miR-4534, optionally an expression level of lung cancer-derived gene(s) selected from the following group B: hsa-miR-19b-3p, hsa-miR-1228-5p, and hsa-miR-1307-3p, and optionally an expression level of lung cancer-derived gene(s) selected from the following group C: hsa-miR-4271, hsa-miR-642b-3p, hsa-miR-6075, hsa-miR-6125, hsa-miR-887-3p, hsa-miR-6851-5p, hsa-miR-6763-5p, hsa-miR-3928-3p, hsa-miR-4443, hsa-miR-3648, hsa-miR-149-3p, hsa-miR-4689, hsa-miR-4763-3p, hsa-miR-6729-5p, hsa-miR-3196, hsa-miR-8069, hsa-miR-1268a, hsa-miR-4739, hsa-miR-1268b, hsa-miR-5698, hsa-miR-6752-5p, hsa-miR-4507, hsa-miR-564, hsa-miR-4497, hsa-miR-6877-5p, hsa-miR-6087, hsa-miR-4731-5p, hsa-miR-615-5p, hsa-miR-760, hsa-miR-6891-5p, hsa-miR-6887-5p, hsa-miR-4525, hsa-miR-1914-3p, hsa-miR-619-5p, hsa-miR-5001-5p, hsa-miR-6722-3p, hsa-miR-3621, hsa-miR-4298, hsa-miR-675-5p and hsa-miR-4655-5p in a sample in vitro, further comparing, for example, the expression level(s) of the aforementioned gene(s) in the sample (e.g., blood, serum, or plasma) collected from a subject who is suspected of having lung cancer with a control expression level in the sample collected from a healthy subject (including a non-lung cancer patient), and evaluating the subject as having lung cancer when the expression level(s) of the target nucleic acid(s) is statistically significantly different between the samples.

(271) This method of the present invention permits limitedly-invasive early diagnosis of cancer with high sensitivity and specificity, and thereby brings about early treatment and improved prognosis. In addition, exacerbation of the disease or the effectiveness of surgical, radiotherapeutic, and chemotherapeutic treatments can be monitored.

(272) The method for extracting the lung cancer-derived gene from the sample such as blood, serum, or plasma according to the present invention is particularly preferably prepared by the addition of a reagent for RNA extraction in 3D-Gene™ RNA extraction reagent from liquid sample kit (Toray Industries, Inc.). A general acidic phenol method (acid guanidinium-phenol-chloroform (AGPC)) may be used, or Trizol™ (Life Technologies Corp.) may be used. The lung cancer-derived gene may be prepared by the addition of a reagent for RNA extraction containing acidic phenol, such as Trizol (Life Technologies Corp.) or Isogen (Nippon Gene Co., Ltd.). Alternatively, a kit such as miRNeasy™ Mini Kit (Qiagen N.V.) can be used, though the method is not limited thereto.

(273) The present invention also provides use of the kit or the device of the present invention for detecting in vitro an expression product of a lung cancer-derived miRNA gene in a sample derived from a subject.

(274) In the method of the present invention, a kit or a device comprising, each alone or in every possible composition, the polynucleotides that can be used in the present invention as described above is used as the kit or the device.

(275) In the detection or (genetic) diagnosis of lung cancer according to the present invention, each polynucleotide contained in the kit or the device of the present invention can be used as a probe or a primer. In the case of using the polynucleotide as a primer, TaqMan™ MicroRNA Assays from Life Technologies Corp., miScript PCR System from Qiagen N.V., or the like can be used, though the method is not limited thereto.

(276) The polynucleotide contained in the kit or the device of the present invention can be used as a primer or a probe according to a routine method in a method known in the art for specifically detecting the particular gene, for example, a hybridization technique such as Northern blot, Southern blot, in situ hybridization, Northern hybridization, or Southern hybridization, or a quantitative amplification technique such as quantitative RT-PCR. A body fluid such as blood, serum, plasma, or urine from a subject is collected as a sample to be assayed according to the type of the detection method used. Alternatively, total RNA prepared from such a body fluid by the method described above may be used, and various polynucleotides including cDNA prepared on the basis of the RNA may be used.

(277) The kit or the device of the present invention is useful for the diagnosis of lung cancer or the detection of the presence or absence of lung cancer. Specifically, the detection of lung cancer using the kit or the device can be performed by detecting in vitro an expression level of a gene using the nucleic acid probe or the primer contained in the kit or the device in a sample such as blood, serum, plasma, or urine from a subject suspected of having lung cancer. The subject suspected of having lung cancer can be evaluated as having lung cancer when the expression level of a target miRNA marker measured using polynucleotide(s) (including a variant(s), a fragment(s), and a derivative(s) thereof) consisting of a nucleotide sequence represented by at least one or more of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134 and 561 to 578 or a complementary sequence thereof, optionally a nucleotide sequence represented by one or more of SEQ ID NOs: 126 and 131 or a complementary sequence thereof, and optionally a nucleotide sequence represented by one or more of SEQ ID NOs: 135 to 174 or a complementary sequence thereof in the sample such as blood, serum, plasma, or urine of the subject is statistically significantly different from the expression level thereof in the sample such as blood, serum, or plasma, or urine of a healthy subject.

(278) The method of the present invention can be combined with chest X-ray examination as well as a diagnostic imaging method such as CT, MRI, or PET. The method of the present invention is capable of specifically detecting lung cancer and can substantially discriminate lung cancer from the other cancers.

(279) The method for detecting the absence of an expression product of a lung cancer-derived gene or the presence of the expression product of a lung cancer-derived gene in a sample using the kit or the device of the present invention comprises collecting a body fluid such as blood, serum, plasma, or urine from a subject, and measuring the expression level of the target gene contained therein using one or more polynucleotide(s) (including a variant(s), a fragment(s), and a derivative(s)) selected from the polynucleotide group of the present invention, to evaluate the presence or absence of lung cancer or to detect lung cancer. Using the method for detecting lung cancer according to the present invention, for example, the presence or absence of amelioration of the disease or the degree of amelioration thereof in a lung cancer patient given a therapeutic drug for the amelioration of the disease can be also evaluated or diagnosed.

(280) The method of the present invention can comprise, for example, the following steps (a), (b), and (c): (a) a step of contacting a sample derived from a subject with a polynucleotide in the kit or the device of the present invention in vitro; (b) a step of measuring an expression level of the target nucleic acid in the sample using the polynucleotide as a nucleic acid probe or a primer; and (c) a step of evaluating the presence or absence of lung cancer (cells) in the subject on the basis of the step (b).

(281) Specifically, the present invention provides a method for detecting lung cancer, comprising measuring an expression level of a target nucleic acid in a sample of a subject using a nucleic acid

capable of specifically binding to at least one or more (preferably at least two or more) polynucleotide(s) selected from the group consisting of miR-6768-5p, miR-6836-3p, miR-6782-5p, miR-3663-3p, miR-1908-3p, miR-6726-5p, miR-4258, miR-1343-3p, miR-4516, miR-6875-5p, miR-4651, miR-6825-5p, miR-6840-3p, miR-6780b-5p, miR-6749-5p, miR-8063, miR-6784-5p, miR-3679-5p, miR-3184-5p, miR-663b, miR-6880-5p, miR-1908-5p, miR-92a-2-5p, miR-7975, miR-7110-5p, miR-6842-5p, miR-6857-5p, miR-5572, miR-3197, miR-6131, miR-6889-5p, miR-4454, miR-1199-5p, miR-1247-3p, miR-6800-5p, miR-6872-3p, miR-4649-5p, miR-6791-5p, miR-4433b-3p, miR-3135b, miR-128-2-5p, miR-4675, miR-4472, miR-6785-5p, miR-6741-5p, miR-7977, miR-3665, miR-128-1-5p, miR-4286, miR-6765-3p, miR-4632-5p, miR-365a-5p, miR-6088, miR-6816-5p, miR-6885-5p, miR-711, miR-6765-5p, miR-3180, miR-4442, miR-4792, miR-6721-5p, miR-6798-5p, miR-3162-5p, miR-6126, miR-4758-5p, miR-2392, miR-486-3p, miR-6727-5p, miR-4728-5p, miR-6746-5p, miR-4270, miR-3940-5p, miR-4725-3p, miR-7108-5p, miR-3656, miR-6879-5p, miR-6738-5p, miR-1260a, miR-4446-3p, miR-3131, miR-4463, miR-3185, miR-6870-5p, miR-6779-5p, miR-1273g-3p, miR-8059, miR-4697-5p, miR-4674, miR-4433-3p, miR-4257, miR-1915-5p, miR-4417, miR-1343-5p, miR-6781-5p, miR-4695-5p, miR-1237-5p, miR-6775-5p, miR-7845-5p, miR-4746-3p, miR-7641, miR-7847-3p, miR-6806-5p, miR-4467, miR-4726-5p, miR-4648, miR-6089, miR-1260b, miR-4532, miR-5195-3p, miR-3188, miR-6848-5p, miR-1233-5p, miR-6717-5p, miR-3195, miR-6757-5p, miR-8072, miR-4745-5p, miR-6511a-5p, miR-6776-5p, miR-371a-5p, miR-1227-5p, miR-7150, miR-1915-3p, miR-187-5p, miR-614, miR-1225-5p, miR-451a, miR-939-5p, miR-223-3p, miR-125a-3p, miR-92b-5p, miR-22-3p, miR-6073, miR-6845-5p, miR-6769b-5p, miR-4665-3p, miR-1913, miR-1228-3p, miR-940, miR-296-3p, miR-4690-5p, miR-548q, miR-663a, miR-1249, miR-1202, miR-7113-3p, miR-1225-3p, miR-4783-3p, miR-4448 and miR-4534, and evaluating in vitro whether or not the subject has lung cancer using the measured expression level and a control expression level of a healthy subject measured in the same way as above.

(282) As used herein, the term “evaluation” is evaluation support based on results of in vitro examination, not physician's judgment.

(283) As described above, as for the target nucleic acids in a preferred embodiment of the method of the present invention, specifically, miR-6768-5p is hsa-miR-6768-5p, miR-6836-3p is hsa-miR-6836-3p, miR-6782-5p is hsa-miR-6782-5p, miR-3663-3p is hsa-miR-3663-3p, miR-1908-3p is hsa-miR-1908-3p, miR-6726-5p is hsa-miR-6726-5p, miR-4258 is hsa-miR-4258, miR-1343-3p is hsa-miR-1343-3p, miR-4516 is hsa-miR-4516, miR-6875-5p is hsa-miR-6875-5p, miR-4651 is hsa-miR-4651, miR-6825-5p is hsa-miR-6825-5p, miR-6840-3p is hsa-miR-6840-3p, miR-6780b-5p is hsa-miR-6780b-5p, miR-6749-5p is hsa-miR-6749-5p, miR-8063 is hsa-miR-8063, miR-6784-5p is hsa-miR-6784-5p, miR-3679-5p is hsa-miR-3679-5p, miR-3184-5p is hsa-miR-3184-5p, miR-663b is hsa-miR-663b, miR-6880-5p is hsa-miR-6880-5p, miR-1908-5p is hsa-miR-1908-5p, miR-92a-2-5p is hsa-miR-92a-2-5p, miR-7975 is hsa-miR-7975, miR-7110-5p is hsa-miR-7110-5p, miR-6842-5p is hsa-miR-6842-5p, miR-6857-5p is hsa-miR-6857-5p, miR-5572 is hsa-miR-5572, miR-3197 is hsa-miR-3197, miR-6131 is hsa-miR-6131, miR-6889-5p is hsa-miR-6889-5p, miR-4454 is hsa-miR-4454, miR-1199-5p is hsa-miR-1199-5p, miR-1247-3p is hsa-miR-1247-3p, miR-6800-5p is hsa-miR-6800-5p, miR-6872-3p is hsa-miR-6872-3p, miR-4649-5p is hsa-miR-4649-5p, miR-6791-5p is hsa-miR-6791-5p, miR-4433b-3p is hsa-miR-4433b-3p, miR-3135b is hsa-miR-3135b, miR-128-2-5p is hsa-miR-128-2-5p, miR-4675 is hsa-miR-4675, miR-4472 is hsa-miR-4472, miR-6785-5p is hsa-miR-6785-5p, miR-6741-5p is hsa-miR-6741-5p, miR-7977 is hsa-miR-7977, miR-3665 is hsa-miR-3665, miR-128-1-5p is hsa-miR-128-1-5p, miR-4286 is hsa-miR-4286, miR-6765-3p is hsa-miR-6765-3p, miR-4632-5p is hsa-miR-4632-5p, miR-365a-5p is hsa-miR-365a-5p, miR-6088 is hsa-miR-6088, miR-6816-5p is hsa-miR-6816-5p, miR-6885-5p is hsa-miR-6885-5p, miR-711 is hsa-miR-711, miR-6765-5p is hsa-miR-6765-5p, miR-3180 is hsa-miR-3180, miR-4442 is hsa-miR-4442, miR-4792 is hsa-miR-4792, miR-6721-5p is hsa-miR-6721-5p, miR-6798-5p is hsa-miR-6798-5p, miR-3162-5p is hsa-miR-3162-5p, miR-6126 is hsa-

miR-6126, miR-4758-5p is hsa-miR-4758-5p, miR-2392 is hsa-miR-2392, miR-486-3p is hsa-miR-486-3p, miR-6727-5p is hsa-miR-6727-5p, miR-4728-5p is hsa-miR-4728-5p, miR-6746-5p is hsa-miR-6746-5p, miR-4270 is hsa-miR-4270, miR-3940-5p is hsa-miR-3940-5p, miR-4725-3p is hsa-miR-4725-3p, miR-7108-5p is hsa-miR-7108-5p, miR-3656 is hsa-miR-3656, miR-6879-5p is hsa-miR-6879-5p, miR-6738-5p is hsa-miR-6738-5p, miR-1260a is hsa-miR-1260a, miR-4446-3p is hsa-miR-4446-3p, miR-3131 is hsa-miR-3131, miR-4463 is hsa-miR-4463, miR-3185 is hsa-miR-3185, miR-6870-5p is hsa-miR-6870-5p, miR-6779-5p is hsa-miR-6779-5p, miR-1273g-3p is hsa-miR-1273g-3p, miR-8059 is hsa-miR-8059, miR-4697-5p is hsa-miR-4697-5p, miR-4674 is hsa-miR-4674, miR-4433-3p is hsa-miR-4433-3p, miR-4257 is hsa-miR-4257, miR-1915-5p is hsa-miR-1915-5p, miR-4417 is hsa-miR-4417, miR-1343-5p is hsa-miR-1343-5p, miR-6781-5p is hsa-miR-6781-5p, miR-4695-5p is hsa-miR-4695-5p, miR-1237-5p is hsa-miR-1237-5p, miR-6775-5p is hsa-miR-6775-5p, miR-7845-5p is hsa-miR-7845-5p, miR-4746-3p is hsa-miR-4746-3p, miR-7641 is hsa-miR-7641, miR-7847-3p is hsa-miR-7847-3p, miR-6806-5p is hsa-miR-6806-5p, miR-4467 is hsa-miR-4467, miR-4726-5p is hsa-miR-4726-5p, miR-4648 is hsa-miR-4648, miR-6089 is hsa-miR-6089, miR-1260b is hsa-miR-1260b, miR-4532 is hsa-miR-4532, miR-5195-3p is hsa-miR-5195-3p, miR-3188 is hsa-miR-3188, miR-6848-5p is hsa-miR-6848-5p, miR-1233-5p is hsa-miR-1233-5p, miR-6717-5p is hsa-miR-6717-5p, miR-3195 is hsa-miR-3195, miR-6757-5p is hsa-miR-6757-5p, miR-8072 is hsa-miR-8072, miR-4745-5p is hsa-miR-4745-5p, miR-6511a-5p is hsa-miR-6511a-5p, miR-6776-5p is hsa-miR-6776-5p, miR-371a-5p is hsa-miR-371a-5p, miR-1227-5p is hsa-miR-1227-5p, miR-7150 is hsa-miR-7150, miR-1915-3p is hsa-miR-1915-3p, miR-187-5p is hsa-miR-187-5p, miR-614 is hsa-miR-614, miR-1225-5p is hsa-miR-1225-5p, miR-451a is hsa-miR-451a, miR-939-5p is hsa-miR-939-5p, miR-223-3p is hsa-miR-223-3p, miR-125a-3p is hsa-miR-125a-3p, miR-92b-5p is hsa-miR-92b-5p, miR-22-3p is hsa-miR-22-3p, miR-6073 is hsa-miR-6073, miR-6845-5p is hsa-miR-6845-5p, miR-6769b-5p is hsa-miR-6769b-5p, miR-4665-3p is hsa-miR-4665-3p, miR-1913 is hsa-miR-1913, miR-1228-3p is hsa-miR-1228-3p, miR-940 is hsa-miR-940, miR-296-3p is hsa-miR-296-3p, miR-4690-5p is hsa-miR-4690-5p, miR-548q is hsa-miR-548q, miR-663a is hsa-miR-663a, miR-1249 is hsa-miR-1249, miR-1202 is hsa-miR-1202, miR-7113-3p is hsa-miR-7113-3p, miR-1225-3p is hsa-miR-1225-3p, miR-4783-3p is hsa-miR-4783-3p, miR-4448 is hsa-miR-4448, and miR-4534 is hsa-miR-4534.

(284) In a preferred embodiment of the method of the present invention, specifically, the nucleic acid (specifically, probe or primer) is selected from the group consisting of the following polynucleotides (a) to (e): (a) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (b) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578, (c) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (d) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and (e) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (a) to (d).

(285) The method of the present invention can further employ a nucleic acid capable of specifically binding to at least one or more polynucleotide(s) selected from the group consisting of miR-19b-3p, miR-1228-5p, and miR-1307-3p.

(286) As for such a nucleic acid, specifically, miR-19b-3p is hsa-miR-19b-3p, miR-1228-5p is hsa-miR-1228-5p, and miR-1307-3p is hsa-miR-1307-3p.

(287) In a preferred embodiment, such a nucleic acid is specifically selected from the group

consisting of the following polynucleotides (f) to (j): (f) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (g) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579, (h) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (i) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 126, 131, and 579 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and (j) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (f) to (i).

(288) The nucleic acid further used in the method of the present invention can comprise a nucleic acid capable of specifically binding to at least one or more polynucleotide(s) selected from the group consisting of miR-4271, miR-642b-3p, miR-6075, miR-6125, miR-887-3p, miR-6851-5p, miR-6763-5p, miR-3928-3p, miR-4443, miR-3648, miR-149-3p, miR-4689, miR-4763-3p, miR-6729-5p, miR-3196, miR-8069, miR-1268a, miR-4739, miR-1268b, miR-5698, miR-6752-5p, miR-4507, miR-564, miR-4497, miR-6877-5p, miR-6087, miR-4731-5p, miR-615-5p, miR-760, miR-6891-5p, miR-6887-5p, miR-4525, miR-1914-3p, miR-619-5p, miR-5001-5p, miR-6722-3p, miR-3621, miR-4298, miR-675-5p and miR-4655-5p.

(289) As for such a nucleic acid, specifically, miR-4271 is hsa-miR-4271, miR-642b-3p is hsa-miR-642b-3p, miR-6075 is hsa-miR-6075, miR-6125 is hsa-miR-6125, miR-887-3p is hsa-miR-887-3p, miR-6851-5p is hsa-miR-6851-5p, miR-6763-5p is hsa-miR-6763-5p, miR-3928-3p is hsa-miR-3928-3p, miR-4443 is hsa-miR-4443, miR-3648 is hsa-miR-3648, miR-149-3p is hsa-miR-149-3p, miR-4689 is hsa-miR-4689, miR-4763-3p is hsa-miR-4763-3p, miR-6729-5p is hsa-miR-6729-5p, miR-3196 is hsa-miR-3196, miR-8069 is hsa-miR-8069, miR-1268a is hsa-miR-1268a, miR-4739 is hsa-miR-4739, miR-1268b is hsa-miR-1268b, miR-5698 is hsa-miR-5698, miR-6752-5p is hsa-miR-6752-5p, miR-4507 is hsa-miR-4507, miR-564 is hsa-miR-564, miR-4497 is hsa-miR-4497, miR-6877-5p is hsa-miR-6877-5p, miR-6087 is hsa-miR-6087, miR-4731-5p is hsa-miR-4731-5p, miR-615-5p is hsa-miR-615-5p, miR-760 is hsa-miR-760, miR-6891-5p is hsa-miR-6891-5p, miR-6887-5p is hsa-miR-6887-5p, miR-4525 is hsa-miR-4525, miR-1914-3p is hsa-miR-1914-3p, miR-619-5p is hsa-miR-619-5p, miR-5001-5p is hsa-miR-5001-5p, miR-6722-3p is hsa-miR-6722-3p, miR-3621 is hsa-miR-3621, miR-4298 is hsa-miR-4298, miR-675-5p is hsa-miR-675-5p, and miR-4655-5p is hsa-miR-4655-5p.

(290) In a preferred embodiment, such a nucleic acid is specifically a polynucleotide selected from the group consisting of the following polynucleotides (k) to (o): (k) a polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (l) a polynucleotide comprising a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174, (m) a polynucleotide consisting of a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, a variant thereof, a derivative thereof, or a fragment thereof comprising 15 or more consecutive nucleotides, (n) a polynucleotide comprising a nucleotide sequence complementary to a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a nucleotide sequence derived from the nucleotide sequence by the replacement of u with t, and (o) a polynucleotide hybridizing under stringent conditions to any of the polynucleotides (k) to (n).

(291) Examples of the sample used in the method of the present invention can include samples

prepared from a living tissue (preferably a lung tissue) or a body fluid such as blood, serum, plasma, or urine from the subject. Specifically, for example, an RNA-containing sample prepared from the tissue, a polynucleotide-containing sample further prepared therefrom, a body fluid such as blood, serum, plasma, or urine, a portion or the whole of a living tissue collected from the subject by biopsy or the like, or a living tissue excised by surgery can be used, and the sample for measurement can be prepared therefrom.

(292) As used herein, the subject refers to a mammal, for example, a human, a monkey, a mouse and a rat, without any limitation, and is preferably a human.

(293) The steps of the method of the present invention can be changed according to the type of the sample to be assayed.

(294) In the case of using RNA as an analyte, the detection of lung cancer (cells) can comprise, for example, the following steps (a), (b), and (c): (a) a step of binding RNA prepared from the sample of the subject or a complementary polynucleotide (cDNA) transcribed therefrom to a polynucleotide in the kit or the device of the present invention; (b) a step of measuring the sample-derived RNA or the cDNA synthesized from the RNA, bound with the polynucleotide by hybridization using the polynucleotide as a nucleic acid probe or by quantitative RT-PCR using the polynucleotide as a primer; and (c) a step of evaluating the presence or absence of lung cancer (or lung cancer-derived gene expression) on the basis of the measurement results of the step (b).

(295) For example, various hybridization methods can be used for detecting, examining, evaluating, or diagnosing lung cancer (or lung cancer-derived gene expression) in vitro according to the present invention. For example, Northern blot, Southern blot, RT-PCR, DNA chip analysis, in situ hybridization, Northern hybridization, or Southern hybridization can be used as such a hybridization method.

(296) In the case of using the Northern blot, the presence or absence of expression of each gene or the expression level thereof in the RNA can be detected or measured by use of the nucleic acid probe that can be used in the present invention. Specific examples thereof can include a method which involves labeling the nucleic acid probe (or its complementary strand) with a radioisotope (.sup.32P, .sup.33P, .sup.35S, etc.), a fluorescent material, or the like, hybridizing the labeled product with the living tissue-derived RNA from a subject transferred to a nylon membrane or the like according to a routine method, and then detecting and measuring a signal derived from the label (radioisotope or fluorescent material) on the formed DNA/RNA duplex using a radiation detector (examples thereof can include BAS-1800 II (Fujifilm Corp.)) or a fluorescence detector (examples thereof can include STORM 865 (GE Healthcare Japan Corp.)).

(297) In the case of using the quantitative RT-PCR, the presence or absence of expression of each gene or the expression level thereof in the RNA can be detected or measured by use of the primer that can be used in the present invention. Specific examples thereof can include a method which involves preparing cDNA from the living tissue-derived RNA of a subject according to a routine method, hybridizing a pair of primers (consisting of a plus strand and a reverse strand binding to the cDNA) of the present invention with the cDNA such that the region of each target gene can be amplified with the cDNA as a template, and performing PCR according to a routine method to detect the obtained double-stranded DNA. The method for detecting the double-stranded DNA can include a method of performing the PCR using the primers labeled in advance with a radioisotope or a fluorescent material, a method of electrophoresing the PCR product on an agarose gel and staining the double-stranded DNA with ethidium bromide or the like for detection, and a method of transferring the produced double-stranded DNA to a nylon membrane or the like according to a routine method and hybridizing the double-stranded DNA to a labeled nucleic acid probe for detection.

(298) In the case of using the nucleic acid array analysis, an RNA chip or a DNA chip in which the nucleic acid probes (single-stranded or double-stranded) of the present invention is attached to a substrate (solid phase) is used. Regions having the attached nucleic acid probes are referred to as

probe spots, and regions having no attached nucleic acid probe are referred to as blank spots. A gene group immobilized on a solid-phase substrate is generally called a nucleic acid chip, a nucleic acid array, a microarray, or the like. The DNA or RNA array includes a DNA or RNA macroarray and a DNA or RNA microarray. The term “chip” used herein includes all of them. 3D-Gene™ Human miRNA Oligo chip (Toray Industries, Inc.) can be used as the DNA chip, though the DNA chip is not limited thereto.

(299) Examples of the measurement using the DNA chip can include, but are not limited to, a method of detecting and measuring a signal derived from the label on the nucleic acid probes using an image detector (examples thereof can include Typhoon 9410 (GE Healthcare Japan Corp.) and 3D-Gene™ scanner (Toray Industries, Inc.)).

(300) The “stringent conditions” used herein are, as mentioned above, conditions under which a nucleic acid probe hybridizes to its target sequence to a larger extent (e.g., a measurement value equal to or larger than a mean of background measurement values+a standard deviation of the background measurement values×2) than that for other sequences.

(301) The stringent conditions are defined by hybridization and subsequent washing conditions. The hybridization conditions are not limited and are conditions involving, for example, 30° C. to 60° C. for 1 to 24 hours in a solution containing SSC, a surfactant, formamide, dextran sulfate, a blocking agent, etc. In this context, 1×SSC is an aqueous solution (pH 7.0) containing 150 mM sodium chloride and 15 mM sodium citrate. The surfactant includes, for example, SDS (sodium dodecyl sulfate), Triton, or Tween. The hybridization conditions more preferably involve 3 to 10×SSC and 0.1 to 1% SDS. Examples of the conditions of the washing, following the hybridization, which is another condition to define the stringent conditions, can include conditions involving continuous washing at 30° C. in a solution containing 0.5×SSC and 0.1% SDS, at 30° C. in a solution containing 0.2×SSC and 0.1% SDS, and at 30° C. in a 0.05×SSC solution. It is desirable that the complementary strand should maintain its hybridized state with a target plus strand even by washing under such conditions. Specifically, examples of such a complementary strand can include a strand consisting of a nucleotide sequence in a completely complementary relationship with the nucleotide sequence of the target plus strand, and a strand consisting of a nucleotide sequence having at least 80%, preferably at least 85%, more preferably at least 90% or at least 95%, for example, at least 98% or at least 99% identity to the strand.

(302) Other examples of the “stringent conditions” for the hybridization are described in, for example, Sambrook, J. & Russel, D., Molecular Cloning, A LABORATORY MANUAL, Cold Spring Harbor Laboratory Press, published on Jan. 15, 2001, Vol. 1, 7.42 to 7.45 and Vol. 2, 8.9 to 8.17, and can be used in the present invention.

(303) Examples of the conditions for carrying out PCR using a polynucleotide fragment in the kit of the present invention as a primer include treatment for approximately 15 seconds to 1 minute at 5 to 10° C. plus a T_m value calculated from the sequence of the primer, using a PCR buffer with composition such as 10 mM Tris-HCL (pH 8.3), 50 mM KCL, and 1 to 2 mM MgCl₂. Examples of the method for calculating such a T_m value include T_m value=2×(the number of adenine residues+the number of thymine residues)+4× (the number of guanine residues+the number of cytosine residues).

(304) In the case of using the quantitative RT-PCR, a commercially available kit for measurement specially designed for quantitatively measuring miRNA, such as TaqMan™ MicroRNA Assays (Life Technologies Corp.), LNA™-based MicroRNA PCR (Exiqon), or Ncode™ miRNA qRT-PCT kit (Invitrogen Corp.) may be used.

(305) For the calculation of gene expression levels, statistical treatment described in, for example, Statistical analysis of gene expression microarray data (Speed T., Chapman and Hall/CRC), and A beginner's guide Microarray gene expression data analysis (Causton H. C. et al., Blackwell publishing) can be used in the present invention, though the calculation method is not limited thereto. For example, twice, preferably 3 times, more preferably 6 times the standard deviation of

the measurement values of the blank spots are added to the average measurement value of the blank spots on the DNA chip, and probe spots having a signal value equal to or larger than the resulting value can be regarded as detection spots. Alternatively, the average measurement value of the blank spots is regarded as a background and can be subtracted from the measurement values of the probe spots to determine gene expression levels. A missing value for a gene expression level can be excluded from the analyte, preferably replaced with the smallest value of the gene expression level in each DNA chip, or more preferably replaced with a value obtained by subtracting 0.1 from a logarithmic value of the smallest value of the gene expression level. In order to eliminate low-signal genes, only a gene having a gene expression level of $2^{\text{sup.}6}$, preferably $2^{\text{sup.}8}$, more preferably $2^{\text{sup.}1'}$ or larger, in 20% or more, preferably 50% or more, more preferably 80% or more of the number of measurement samples can be selected as the analyte. Examples of the normalization of the gene expression level include, but are not limited to, global normalization and quantile normalization (Bolstad, B. M. et al., 2003, Bioinformatics, Vol. 19, p. 185-193).

(306) The present invention also provides a method comprising measuring a target gene or gene expression level in a sample derived from a subject using the polynucleotide, the kit, or the device (e.g., chip) for detection of the present invention, or a combination thereof, preparing a discriminant (discriminant function) with gene expression levels in a sample derived from a lung cancer patient and a sample derived from a healthy subject as supervising samples, and determining or evaluating the presence and/or absence of the lung cancer-derived gene in the sample.

(307) Specifically, the present invention further provides the method comprising: a first step of measuring in vitro an expression level of a target gene in multiple samples that were known to be able to determine or evaluate the presence and/or absence of the lung cancer-derived gene in the samples, using the polynucleotide, the kit, or the device (e.g., chip) for detection of the present invention, or a combination thereof; a second step of constructing a discriminant with the measurement values of the expression level of the target gene (target nucleic acids) that was obtained in the first step as supervising samples; a third step of measuring in vitro an expression level of the target gene in a sample derived from a subject in the same way as in the first step; and a fourth step of assigning the measurement value of the expression level of the target gene obtained in the third step into the discriminant obtained in the second step, and determining or evaluating the presence and/or absence of the lung cancer-derived gene in the sample on the basis of the results obtained from the discriminant, wherein the target gene can be detected using a polynucleotide for the detection, that was contained in the polynucleotide, the kit or the device (e.g., chip). In this context, the discriminant can be prepared by use of Fisher's discriminant analysis, nonlinear discriminant analysis based on Mahalanobis' distance, neural network, Support Vector Machine (SVM), or the like, though the method is not limited thereto.

(308) When a clustering boundary is a straight line or a hyperplane, the linear discriminant analysis is a method for determining the association of a cluster using Formula 1 as a discriminant. In Formula 1, x represents an explanatory variable, w represents a coefficient of the explanatory variable, and w_0 represents a constant term.

$$(309) \quad f(x) = w_0 + \sum_{i=1}^n w_i x_i \quad \text{Formula 1}$$

(310) Values obtained from the discriminant are referred to as discriminant scores. The measurement values of a newly offered data set can be assigned as explanatory variables to the discriminant to determine clusters by the signs of the discriminant scores.

(311) The Fisher's discriminant analysis, one type of linear discriminant analysis, is a dimensionality reduction method for selecting a dimension suitable for discriminating classes, and constructs a highly discriminating synthetic variable by focusing on the variance of synthetic variables and minimizing the variance of data having the same label (Venables, W. N. et al., Modern Applied Statistics with S. Fourth edition. Springer, 2002). In the Fisher's linear discriminant analysis, direction w of projection is determined so as to maximize Formula 2. In this

Formula, represents an average input, $\bar{x}_g$ represents the number of data associated with class g , and $\bar{y}_g$ represents an average input of the data associated with class g . The numerator and the denominator are the inter-classe variance and the intra-classe variance, respectively, when each data is projected in the direction of the vector w . Discriminant coefficient w_i is determined by maximizing this ratio (Takafumi Kanamori et al., “Pattern Recognition”, Kyoritsu Shuppan Co., Ltd. (2009); and Richard O. et al., Pattern Classification Second Edition, Wiley-Interscience, 2000).

(312)

$$J(w) = \frac{\sum_{g=1}^G n_g (w^T \bar{x}_g - \bar{y}_g) (w^T \bar{x}_g - \bar{y}_g)^T}{\sum_{g=1}^G \sum_{i: y_i = g} (w^T x_i - \bar{y}_g) (w^T x_i - \bar{y}_g)^T} \text{subject to } w = \sum_{i=1}^n \frac{x_i}{n}, \bar{y}_g = \sum_{i: y_i = g} \frac{x_i}{n_g} \quad \text{Formula2}$$

(313) The Mahalanobis' distance is calculated according to Formula 3 in consideration of data correlation and can be used as nonlinear discriminant analysis for determining an associated cluster that shows a closer Mahalanobis' distance from each cluster. In this Formula 3, $\bar{x}$ represents a central vector of each cluster, and S^{-1} represents an inverse matrix of the variance-covariance matrix of the cluster. The central vector is calculated from explanatory variable x , and an average vector, a median value vector, or the like can be used.

$$(314) \quad D(x, \bar{x}) = \{(x - \bar{x})^T S^{-1} (x - \bar{x})\}^{\frac{1}{2}} \quad \text{Formula3}$$

(315) SVM is a discriminant analysis method devised by V. Vapnik (The Nature of Statistical Learning Theory, Springer, 1995). Particular data points of a data set having known classes are defined as explanatory variables, and classes are defined as objective variables. A boundary plane called hyperplane for correctly classifying the data set into the known classes is determined, and a discriminant for data classification is determined using the boundary plane. Then, the measurement values of a newly offered data set can be assigned as explanatory variables to the discriminant to determine classes. In this respect, the results of the discriminant analysis may be classes, may be a probability of data to be classified into correct classes, or may be the distance from the hyperplane. In SVM, a method of nonlinearly converting a feature vector to a high dimension and performing linear discriminant in the space is known as a method for tackling nonlinear problems. An expression in which an inner product of two factors in a nonlinearly mapped space is expressed only by inputs in their original spaces is called kernel. Examples of the kernel can include a linear kernel, a RBF (Radial Basis Function) kernel, and a Gaussian kernel. While highly dimensional mapping is performed according to the kernel, the optimum discriminant, i.e., a discriminant, can be actually constructed by mere calculation according to the kernel, which avoids calculating features in the mapped space (e.g., Hideki Aso et al., Frontier of Statistical Science 6 “Statistics of pattern recognition and learning—New concepts and approaches”, Iwanami Shoten, Publishers (2004); Nello Cristianini et al., Introduction to SVM, Kyoritsu Shuppan Co., Ltd. (2008)).

(316) C-support vector classification (C-SVC), one type of SVM, involves preparing a hyperplane by supervising with the explanatory variables of two groups and classifying an unknown data set into either of the groups (C. Cortes et al., 1995, Machine Learning, Vol. 20, p. 273-297).

(317) Exemplary calculation of the C-SVC discriminant that can be used in the method of the present invention will be given below. First, all subjects are divided into two groups, i.e., a lung cancer patient group and a healthy subject group. For example, lung tissue examination can be used for a reference under which each subject is confirmed either as a lung cancer patient or as a healthy subject.

(318) Next, a data set consisting of comprehensive gene expression levels of serum-derived samples of the two divided groups (hereinafter, this data set is referred to as a training cohort) is prepared, and a C-SVC discriminant is determined by using genes that were found to differ clearly in their gene expression levels between the two groups as explanatory variables, and using this grouping as objective variables (e.g., -1 and +1). An optimizing objective function is represented by Formula 4 wherein e represents all input vectors, y represents an objective variable, a represents

a Lagrange's undetermined multiplier vector, Q represents a positive definite matrix, and C represents a parameter for adjusting constrained conditions.

$$(319) \min_{\frac{1}{2}a^T Q a - e^T a} \text{subject to } y^T a = 0, 0 \leq a_i \leq C, i = 1, \dots, l, \quad \text{Formula 4}$$

(320) Formula 5 is a finally obtained discriminant, and an associated group can be determined on the basis of the sign of a value obtained according to the discriminant. In this Formula, x represents a support vector, y represents a label indicating the association of a group, a represents the corresponding coefficient, b represents a constant term, and K represents a kernel function.

$$(321) f(x) = \text{sgn} \left(\sum_{i=1}^l y_i a_i K(x_i, x) + b \right) \quad \text{Formula 5}$$

(322) For example, a RBF kernel defined by Formula 6 can be used as the kernel function. In this Formula, x represents a support vector, and y represents a kernel parameter for adjusting the complexity of the hyperplane.

$$K(x.\text{sub}.i, x.\text{sub}.j) = \exp(-r \|x.\text{sub}.i - x.\text{sub}.j\|.\text{sup}.2), r < 0 \quad \text{Formula 6}$$

(323) In addition, an approach such as neural network, k-nearest neighbor algorithms, decision trees, or logistic regression analysis can be selected as a method for determining or evaluating the presence and/or absence of expression of a lung cancer-derived target gene in a sample derived from a subject, or for evaluating the expression level thereof by comparison with a control derived from a healthy subject.

(324) The method of the present invention can comprise, for example, the following steps (a), (b), and (c): (a) a step of measuring an expression level of a target gene in tissues containing lung cancer-derived genes derived from lung cancer patients and/or samples that are already known to contain no lung cancer-derived gene derived from healthy subjects, using the polynucleotide, the kit, or the device (e.g., DNA chip) for detection according to the present invention; (b) a step of preparing the discriminants of Formulas 1 to 3, 5, and 6 described above from the measurement values of the expression level measured in the step (a); and (c) a step of measuring an expression level of the target gene in a sample derived from a subject using the polynucleotide, the kit, or the device (e.g., DNA chip) for detection according to the present invention, assigning the obtained measurement value to the discriminants prepared in the step (b), and determining or evaluating the presence and/or absence of expression of the lung cancer-derived target gene in the sample, or evaluating the expression level thereof by comparison with a healthy subject-derived control, on the basis of the obtained results. In this context, in the discriminants of Formulas 1 to 3, 5, and 6, x represents an explanatory variable and includes a value obtained by measuring a polynucleotide selected from the polynucleotides described in Section 2 above, or a fragment thereof, etc. Specifically, the explanatory variable for discriminating a lung cancer patient from a healthy subject according to the present invention is a gene expression level selected from, for example, the following expression levels (1) to (3): (1) a gene expression level in the serum of a lung cancer patient or a healthy subject measured by any DNA comprising 15 or more consecutive nucleotides in a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134 and 561 to 578 or a complementary sequence thereof, (2) a gene expression level in the serum of a lung cancer patient or a healthy subject measured by any DNA comprising 15 or more consecutive nucleotides in a nucleotide sequence represented by any of SEQ ID NOs: 126, 131 and 579 or a complementary sequence thereof, and (3) a gene expression level in the serum of a lung cancer patient or a healthy subject measured by any DNA comprising 15 or more consecutive nucleotides in a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174 or a complementary sequence thereof.

(325) As described above, for the method for determining or evaluating the presence and/or absence of a lung cancer-derived gene in a sample derived from a subject, the preparation of a discriminant requires a discriminant prepared in a training cohort. For enhancing the discriminant accuracy of the discriminant, it is necessary for the discriminant to use genes that show clear

difference between two groups in the training cohort.

(326) Each gene that is used for an explanatory variable in a discriminant is preferably determined as follows. First, comprehensive gene expression levels of a lung cancer patient group and comprehensive gene expression levels of a healthy subject group in a training cohort are used as a data set, the degree of difference in the expression level of each gene between the two groups is determined through the use of, for example, the P value of t test, which is parametric analysis, or the P value of Mann-Whitney's U test or Wilcoxon test, which is nonparametric analysis.

(327) The gene can be regarded as being statistically significant when the critical rate (significance level) of the P value obtained by the test is smaller than, for example, 5%, 1%, or 0.01%.

(328) In order to correct an increased probability of type I error attributed to the repetition of a test, a method known in the art, for example, Bonferroni or Holm method, can be used for the correction (e.g., Yasushi Nagata et al., "Basics of statistical multiple comparison methods", Scientist Press Co., Ltd. (2007)). As an example of the Bonferroni correction, for example, the P value obtained by a test is multiplied by the number of repetitions of the test, i.e., the number of genes used in the analysis, and the obtained value can be compared with a desired significance level to suppress a probability of causing type I error in the whole test.

(329) Instead of the statistical test, the absolute value (fold change) of an expression ratio of a median value of each gene expression level between gene expression levels of a lung cancer patient group and gene expression levels of a healthy subject group may be calculated to select a gene that is used for an explanatory variable in a discriminant. Alternatively, ROC curves may be prepared using gene expression levels of a lung cancer patient group and a healthy subject group, and a gene that is used for an explanatory variable in a discriminant can be selected on the basis of an AUROC value.

(330) Next, a discriminant that can be calculated by various methods described above is prepared using any number of genes having large difference in their gene expression levels determined here. Examples of the method for constructing a discriminant that produces the largest discriminant accuracy include a method of constructing a discriminant in every combination of genes that satisfy the significance level of P value, and a method of repetitively evaluating a discriminant while increasing the number of genes for use one by one in a descending order of difference in gene expression level (Furey T S. et al., 2000, Bioinformatics, Vol. 16, p. 906-14). A gene expression level of another independent lung cancer patient or healthy subject is assigned as an explanatory variable to this discriminant to calculate discriminant results of the group to which this independent lung cancer patient or healthy subject associates. Specifically, the found gene set for diagnosis and the discriminant constructed using the gene set for diagnosis can be evaluated in an independent sample group to find a more universal gene set for diagnosis capable of detecting lung cancer and a more universal method for discriminating lung cancer.

(331) Split-sample method is preferably used for evaluating the discriminant performance (generality) of the discriminant. Specifically, a data set is divided into a training cohort and a validation cohort, and gene selection by a statistical test and discriminant preparation are performed in the training cohort. Accuracy, sensitivity, and specificity are calculated using results of discriminating a validation cohort according to the discriminant and a true group to which the validation cohort associates, to evaluate the discriminant performance. On the other hand, instead of dividing a data set, gene selection by a statistical test and discriminant preparation may be performed using all of samples, and accuracy, sensitivity, and specificity can be calculated by the discriminant of newly prepared samples according to the discriminant to evaluate the discriminant performance.

(332) The present invention provides a polynucleotide for detection and for disease diagnosis useful in the diagnosis and treatment of lung cancer, a method for detecting lung cancer using the polynucleotide, and a kit and a device for the detection of lung cancer, comprising the polynucleotide. Particularly, in order to select a gene for diagnosis and prepare a discriminant so as

to exhibit accuracy beyond a lung cancer diagnosis method using existing tumor markers CEA, a gene set for diagnosis and a discriminant for the method of the present invention, that exhibit accuracy beyond CEA, can be constructed, for example, by comparing genes expressed in serum derived from a patient confirmed to be negative using CEA but finally found to have lung cancer by detailed examination such as computed tomography using a contrast medium, with genes expressed in serum derived from a patient having no lung cancer.

(333) For example, the gene set for diagnosis is set to any combination selected from one or two or more of the polynucleotides based on a nucleotide sequence represented by any of SEQ ID NOs: 1 to 125, 127 to 130, 132 to 134, and 561 to 578, or a complementary sequence thereof as described above, optionally one or two or more of the polynucleotides based on a nucleotide sequence represented by any of SEQ ID NOs: 126, 131 and 579, or a complementary sequence thereof, and optionally one or two or more of the polynucleotides based on a nucleotide sequence represented by any of SEQ ID NOs: 135 to 174, or a complementary sequence thereof. Further, a discriminant is constructed using expression levels of the gene set for diagnosis in samples derived from class I lung cancer patients and samples derived from class II healthy subjects as a result of tissue diagnosis. As a result, the presence or absence of lung cancer-derived genes in an unknown sample can be determined with 100% accuracy at the maximum by measuring expression levels of the gene set for diagnosis in the unknown sample.

EXAMPLES

(334) Hereinafter, the present invention will be described further specifically with reference to Examples below. However, the scope of the present invention is not intended to be limited by these Examples.

Reference Example 1

(335) <Collection of Samples from Lung Cancer Patients and Healthy Subjects>

(336) Serum was collected using VENOJECT II vacuum blood collecting tube VP-AS109K60 (Terumo Corp.) from each of 100 healthy subjects and 17 lung cancer patients (8 lung adenocarcinoma cases involving 6 cases with T2N0M0, 1 case with T2N1M0, and 1 case with T2N2M0; and 8 squamous cell cancer cases involving 5 cases with T2N0M0, 1 case with T4NOMO, 1 case with T2N1M0, and 1 case with T4N2M0) confirmed to have no primary cancer other than lung cancer after acquisition of informed consent, and used as a training cohort. Likewise, serum was collected using VENOJECT II vacuum blood collecting tube VP-AS109K60 (Terumo Corp.) from each of 50 healthy subjects and 8 lung cancer patients (5 adenocarcinoma cases involving 3 cases with T2NOMO, 1 case with T3NOMO, and 1 case with T4N2M0; and 3 squamous cell cancer cases involving 1 case with T2NOMO, 1 case with T4NOMO, and 1 case with T2N1M0) confirmed to have no primary cancer other than lung cancer after acquisition of informed consent, and used as a validation cohort. The histological types and stages of these lung cancer samples are summarized in Tables 2-1 and 2-2.

(337) <Extraction of Total RNA>

(338) Total RNA was obtained from 300 μ L of the serum sample obtained from each of 175 persons in total of 150 healthy subjects and 25 lung cancer patients included in the training cohort and the validation cohort, using a reagent for RNA extraction in 3D-Gene™ RNA extraction reagent from liquid sample kit (Toray Industries, Inc.) according to the protocol provided by the manufacturer.

(339) <Measurement of Gene Expression Level>

(340) miRNAs in the total RNA obtained from the serum sample of each of 175 persons in total of 150 healthy subjects and 25 lung cancer patients included in the training cohort and the validation cohort were fluorescently labeled using 3D-Gene™ miRNA Labeling kit (Toray Industries, Inc.) according to the protocol (ver 2.20) provided by the manufacturer. The oligo DNA chip used was 3D-Gene™ Human miRNA Oligo chip (Toray Industries, Inc.) with attached probes having sequences complementary to 2,555 miRNAs among the miRNAs registered in miRBase Release

20. Hybridization between the miRNAs in the total RNA and the probes on the DNA chip under stringent conditions and washing following the hybridization were performed according to the protocol provided by the manufacturer. The DNA chip was scanned using 3D-Gene™ scanner (Toray Industries, Inc.) to obtain images. Fluorescence intensity was digitized using 3D-Gene™ Extraction (Toray Industries, Inc.). The digitized fluorescence intensity was converted to a logarithmic value having a base of 2 and used as a gene expression level, from which a blank value was subtracted. A missing value was replaced with a value obtained by subtracting 0.1 from a logarithmic value of the smallest value of the gene expression level in each DNA chip. As a result, the comprehensive gene expression levels of the miRNAs in the serum were obtained for the 25 lung cancer patients and the 150 healthy subjects. Calculation and statistical analysis using the digitized gene expression levels of the miRNAs were carried out using R language 3.0.2 (R Development Core Team (2013). R: A language and environment for statistical computing. R Foundation for Statistical Computing, URL <http://www.R-project.org/>) and MASS package 7.3-30 (Venables, W. N. & Ripley, B. D. (2002) Modern Applied Statistics with S. Fourth Edition. Springer, New York. ISBN 0-387-95457-0).

Reference Example 2

(341) <Collection of Samples from Patients with Cancers Other than Lung Cancer>

(342) Serum was collected using VENOJECT II vacuum blood collecting tube VP-AS109K60 (Terumo Corp.) from each of 75 pancreatic cancer patients, 62 biliary tract cancer patients, 32 colorectal cancer patients, 35 stomach cancer patients, 32 esophageal cancer patients, 33 liver cancer patients, and 13 benign pancreaticobiliary disease patients confirmed to have no cancer in other organs after acquisition of informed consent, and used as a training cohort together with the samples of 17 lung cancer patients and 99 healthy subjects of Reference Example 1. Likewise, serum was collected using VENOJECT II vacuum blood collecting tube VP-AS109K60 (Terumo Corp.) from each of 28 pancreatic cancer patients, 38 biliary tract cancer patients, 18 colorectal cancer patients, 15 stomach cancer patients, 18 esophageal cancer patients, 19 liver cancer patients, and 8 benign pancreaticobiliary disease patients confirmed to have no cancer in other organs after acquisition of informed consent, and used as a validation cohort together with the samples of 8 lung cancer patients confirmed to have no cancer in organs except for lung cancer and 51 healthy subjects of Reference Example 1. Subsequent operations were conducted in the same way as in Reference Example 1.

(343) TABLE-US-00002 TABLE 2-1 Training cohort

Sample name	Cancer	stage
Lung adenocarcinoma LC01	T2N0M0	
LC02	T2N0M0	
LC03	T2N0M0	
LC05	T2N0M0	
LC07	T2N0M0	
LC08	T2N2M0	
LC11	T2N0M0	
LC12	T2N1M0	
LC14	T2N0M0	
Squamous cell cancer LC15	T2N0M0	
LC18	T2N0M0	
LC20	T2N0M0	
LC21	T2N0M0	
LC22	T4N2M0	
LC23	T2N1M0	
LC24	T2N0M0	
LC25	T4N0M0	

(344) TABLE-US-00003 TABLE 2-2 Validation cohort

Sample name	Cancer	stage
Lung adenocarcinoma LC04	T2N0M0	
LC06	T2N0M0	
LC09	T3N0M0	
LC10	T4N2M0	
LC13	T2N0M0	
Squamous cell cancer LC16	T2N1M0	
LC17	T2N0M0	
LC19	T4N0M0	

Example 1

(345) <Selection of Gene Marker Using Samples in the Training Cohort, and Method for Evaluating Lung Cancer Discriminant Performance of Single Gene Marker Using Samples in the Validation Cohort>

(346) In this Example, a gene marker for discriminating a lung cancer patient from a healthy subject was selected from the training cohort and studied in samples of the validation cohort independent of the training cohort, for a method for evaluating the lung cancer discriminant performance of each selected gene marker alone.

(347) Specifically, first, the miRNA expression levels in the training cohort and the validation cohort obtained in the preceding Reference Examples were combined and normalized by quantile normalization.

(348) Next, genes for diagnosis were selected in the training cohort. Here, in order to acquire diagnostic markers with higher reliability, only genes having a gene expression level of 2.5 or higher in 50% or more of the samples in either of the lung cancer patient group in the training cohort or the healthy subject group in the training cohort were selected. In order to further acquire statistically significant genes for discriminating a lung cancer patient group from a healthy subject group, the P value obtained by two-sample t-test assuming equal variance as to each gene expression level was corrected by the Bonferroni method, and genes that satisfied $p < 0.01$ were acquired as gene markers for use in explanatory variables of a discriminant. The result is described in Table 3.

(349) In this way, hsa-miR-6768-5p, hsa-miR-6836-3p, hsa-miR-6782-5p, hsa-miR-3663-3p, hsa-miR-1908-3p, hsa-miR-6726-5p, hsa-miR-4258, hsa-miR-1343-3p, hsa-miR-4516, hsa-miR-6875-5p, hsa-miR-4651, hsa-miR-6825-5p, hsa-miR-6840-3p, hsa-miR-6780b-5p, hsa-miR-6749-5p, hsa-miR-8063, hsa-miR-6784-5p, hsa-miR-3679-5p, hsa-miR-3184-5p, hsa-miR-663b, hsa-miR-6880-5p, hsa-miR-1908-5p, hsa-miR-92a-2-5p, hsa-miR-7975, hsa-miR-7110-5p, hsa-miR-6842-5p, hsa-miR-6857-5p, hsa-miR-5572, hsa-miR-3197, hsa-miR-6131, hsa-miR-6889-5p, hsa-miR-4454, hsa-miR-1199-5p, hsa-miR-1247-3p, hsa-miR-6800-5p, hsa-miR-6872-3p, hsa-miR-4649-5p, hsa-miR-6791-5p, hsa-miR-4433b-3p, hsa-miR-3135b, hsa-miR-128-2-5p, hsa-miR-4675, hsa-miR-4472, hsa-miR-6785-5p, hsa-miR-6741-5p, hsa-miR-7977, hsa-miR-3665, hsa-miR-128-1-5p, hsa-miR-4286, hsa-miR-6765-3p, hsa-miR-4632-5p, hsa-miR-365a-5p, hsa-miR-6088, hsa-miR-6816-5p, hsa-miR-6885-5p, hsa-miR-711, hsa-miR-6765-5p, hsa-miR-3180, hsa-miR-4442, hsa-miR-4792, hsa-miR-6721-5p, hsa-miR-6798-5p, hsa-miR-3162-5p, hsa-miR-6126, hsa-miR-4758-5p, hsa-miR-2392, hsa-miR-486-3p, hsa-miR-6727-5p, hsa-miR-4728-5p, hsa-miR-6746-5p, hsa-miR-4270, hsa-miR-3940-5p, hsa-miR-4725-3p, hsa-miR-7108-5p, hsa-miR-3656, hsa-miR-6879-5p, hsa-miR-6738-5p, hsa-miR-1260a, hsa-miR-4446-3p, hsa-miR-3131, hsa-miR-4463, hsa-miR-3185, hsa-miR-6870-5p, hsa-miR-6779-5p, hsa-miR-1273g-3p, hsa-miR-8059, hsa-miR-4697-5p, hsa-miR-4674, hsa-miR-4433-3p, hsa-miR-4257, hsa-miR-1915-5p, hsa-miR-4417, hsa-miR-1343-5p, hsa-miR-6781-5p, hsa-miR-4695-5p, hsa-miR-1237-5p, hsa-miR-6775-5p, hsa-miR-7845-5p, hsa-miR-4746-3p, hsa-miR-7641, hsa-miR-7847-3p, hsa-miR-6806-5p, hsa-miR-4467, hsa-miR-4726-5p, hsa-miR-4648, hsa-miR-6089, hsa-miR-1260b, hsa-miR-4532, hsa-miR-5195-3p, hsa-miR-3188, hsa-miR-6848-5p, hsa-miR-1233-5p, hsa-miR-6717-5p, hsa-miR-3195, hsa-miR-6757-5p, hsa-miR-8072, hsa-miR-4745-5p, hsa-miR-6511a-5p, hsa-miR-6776-5p, hsa-miR-371a-5p, hsa-miR-1227-5p, hsa-miR-7150, hsa-miR-1915-3p, hsa-miR-187-5p, hsa-miR-614, hsa-miR-19b-3p, hsa-miR-1225-5p, hsa-miR-451a, hsa-miR-939-5p, hsa-miR-223-3p, hsa-miR-1228-5p, hsa-miR-125a-3p, hsa-miR-92b-5p, and hsa-miR-22-3p genes, and polynucleotides consisting of the nucleotide sequences of SEQ ID NOs: 1 to 134 related thereto were found.

(350) Among them, genes newly found as markers for examining the presence or absence of lung cancer are polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 125, 127 to 130, and 132 to 134.

(351) A discriminant for determining the presence or absence of lung cancer was further prepared by Fisher's discriminant analysis with the expression levels of these genes as an index. Specifically, any newly found polynucleotide consisting of a nucleotide sequence represented by any of SEQ ID NOs: 1 to 134 in the training cohort was apply for Formula 2 above to construct a discriminant. Calculated accuracy, sensitivity, and specificity are shown in Table 4. In this respect, a discriminant coefficient and a constant term are shown in Table 5.

(352) Accuracy, sensitivity, and specificity in the validation cohort were calculated using the discriminant thus prepared, and the discriminant performance of the selected polynucleotides was validated using independent samples (Table 4). For example, the expression level measurement value of the nucleotide sequence represented by SEQ ID NO: 1 was compared between the healthy subjects (100 persons) and the lung cancer patients (17 persons) in the training cohort. As a result, the gene expression level measurement values were found to be significantly lower in the lung

cancer patient group than in the healthy subject group (see FIG. 2A). These results were also reproducible for the healthy subjects (50 persons) and the lung cancer patients (8 persons) in the validation cohort (see FIG. 2B). Likewise, the results obtained about the other polynucleotides shown in SEQ ID NOs: 2 to 134 showed that the gene expression level measurement values were significantly lower (–) or higher (+) in the lung cancer patient group than in the healthy subject group (Table 3). These results were able to be validated in the validation cohort. For example, as for this nucleotide sequence represented by SEQ ID NO: 1, the number of samples that were correctly identified in the detection of lung cancer was calculated using the threshold (10.08) that was set in the training cohort and discriminated between the two groups. As a result, 7 true positives, 50 true negatives, 0 false positives, and 1 false negative were obtained. From these values, 98.3% accuracy, 87.5% sensitivity, and 100% specificity were obtained as detection performance. In this way, the detection performance was calculated as to all of the polynucleotides shown in SEQ ID NOs: 1 to 134, and described in Table 4.

(353) For example, 33 polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1, 2, 3, 5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 22, 26, 27, 28, 29, 33, 34, 38, 41, 42, 44, 65, 124, 125, and 133 exhibited sensitivity of 87.5%, 100%, 100%, 75%, 75%, 75%, 87.5%, 87.5%, 87.5%, 87.5%, 87.5%, 87.5%, 87.5%, 87.5%, 87.5%, 100%, 75%, 87.5%, 87.5%, 87.5%, 87.5%, 87.5%, 75%, 87.5%, 75%, 75%, 75%, 75%, 75%, 75% and 75% respectively, in the validation cohort (Table 4). In this context, the tumor markers CEA and CYFRA21-1 in blood for lung cancer reportedly have general lung cancer detection sensitivity of 69% and 43%, respectively (Non Patent Literature 3). These results demonstrated that the 33 polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1, 2, 3, 5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 22, 26, 27, 28, 29, 33, 34, 38, 41, 42, 44, 65, 124, 125, and 133 can discriminate, each alone, lung cancer in the validation cohort with sensitivity beyond the existing markers CEA and CYFRA21-1.

(354) For example, 10 polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 2, 3, 11, 13, 20, 21, 22, 30, 31, and 37 were able to correctly determine lung cancer as to all of 4 samples from lung adenocarcinoma or squamous cell cancer having a tumor size of less than 7 cm and having no lymph node metastasis, contained in the validation cohort. Thus, these polynucleotides can detect even relatively early lung cancer and contributes to the early diagnosis of lung cancer.

(355) TABLE-US-00004 TABLE 3 Expression level in P value after lung cancer patient Bonferroni with respect to SEQ ID NO: Gene name correction healthy subject 1 hsa-miR-6768-5p 6.71E-24 + 2 hsa-miR-6836-3p 1.44E-20 – 3 hsa-miR-6782-5p 2.89E-20 + 4 hsa-miR-3663-3p 2.77E-18 – 5 hsa-miR-1908-3p 3.58E-18 – 6 hsa-miR-6726-5p 1.02E-17 – 7 hsa-miR-4258 3.38E-17 – 8 hsa-miR-1343-3p 7.45E-17 – 9 hsa-miR-4516 7.91E-17 – 10 hsa-miR-6875-5p 3.69E-16 + 11 hsa-miR-4651 5.14E-16 – 12 hsa-miR-6825-5p 1.28E-14 + 13 hsa-miR-6840-3p 2.69E-14 – 14 hsa-miR-6780b-5p 3.47E-14 + 15 hsa-miR-6749-5p 3.82E-14 – 16 hsa-miR-8063 3.58E-13 – 17 hsa-miR-6784-5p 7.06E-13 + 18 hsa-miR-3679-5p 7.64E-13 + 19 hsa-miR-3184-5p 1.78E-12 + 20 hsa-miR-663b 5.72E-12 – 21 hsa-miR-6880-5p 9.41E-12 + 22 hsa-miR-1908-5p 1.84E-11 + 23 hsa-miR-92a-2-5p 1.85E-11 + 24 hsa-miR-7975 2.06E-11 – 25 hsa-miR-7110-5p 2.64E-11 + 26 hsa-miR-6842-5p 2.66E-11 + 27 hsa-miR-6857-5p 5.09E-11 + 28 hsa-miR-5572 7.39E-11 + 29 hsa-miR-3197 8.45E-11 + 30 hsa-miR-6131 1.51E-10 – 31 hsa-miR-6889-5p 2.73E-10 + 32 hsa-miR-4454 2.92E-10 – 33 hsa-miR-1199-5p 6.01E-10 – 34 hsa-miR-1247-3p 7.10E-10 + 35 hsa-miR-6800-5p 8.76E-10 + 36 hsa-miR-6872-3p 1.18E-09 – 37 hsa-miR-4649-5p 1.37E-09 – 38 hsa-miR-6791-5p 1.51E-09 + 39 hsa-miR-4433b-3p 1.57E-09 + 40 hsa-miR-3135b 1.78E-09 – 41 hsa-miR-128-2-5p 2.59E-09 – 42 hsa-miR-4675 2.65E-09 – 43 hsa-miR-4472 3.21E-09 + 44 hsa-miR-6785-5p 3.84E-09 – 45 hsa-miR-6741-5p 6.85E-09 – 46 hsa-miR-7977 8.90E-09 – 47 hsa-miR-3665 2.49E-08 – 48 hsa-miR-128-1-5p 3.03E-08 + 49 hsa-miR-4286 3.07E-08 – 50 hsa-miR-6765-3p 3.14E-08 – 51 hsa-miR-4632-5p 4.02E-08 + 52 hsa-miR-365a-5p 4.58E-08 +

53 hsa-miR-6088 7.80E-08 - 54 hsa-miR-6816-5p 1.19E-07 + 55 hsa-miR-6885-5p 1.59E-07 - 56 hsa-miR-711 1.93E-07 + 57 hsa-miR-6765-5p 2.99E-07 + 58 hsa-miR-3180 3.65E-07 + 59 hsa-miR-4442 3.89E-07 - 60 hsa-miR-4792 3.97E-07 + 61 hsa-miR-6721-5p 6.66E-07 + 62 hsa-miR-6798-5p 8.81E-07 + 63 hsa-miR-3162-5p 1.07E-06 + 64 hsa-miR-6126 1.26E-06 + 65 hsa-miR-4758-5p 1.35E-06 - 66 hsa-miR-2392 1.58E-06 + 67 hsa-miR-486-3p 3.01E-06 - 68 hsa-miR-6727-5p 3.06E-06 - 69 hsa-miR-4728-5p 3.61E-06 - 70 hsa-miR-6746-5p 5.00E-06 - 71 hsa-miR-4270 5.64E-06 - 72 hsa-miR-3940-5p 6.33E-06 + 73 hsa-miR-4725-3p 6.79E-06 + 74 hsa-miR-7108-5p 7.35E-06 + 75 hsa-miR-3656 1.20E-05 + 76 hsa-miR-6879-5p 1.22E-05 + 77 hsa-miR-6738-5p 1.25E-05 - 78 hsa-miR-1260a 1.51E-05 - 79 hsa-miR-4446-3p 1.67E-05 - 80 hsa-miR-3131 1.91E-05 - 81 hsa-miR-4463 2.63E-05 + 82 hsa-miR-3185 3.31E-05 + 83 hsa-miR-6870-5p 3.95E-05 + 84 hsa-miR-6779-5p 4.61E-05 - 85 hsa-miR-1273g-3p 4.73E-05 - 86 hsa-miR-8059 5.08E-05 - 87 hsa-miR-4697-5p 5.16E-05 - 88 hsa-miR-4674 7.31E-05 - 89 hsa-miR-4433-3p 8.12E-05 + 90 hsa-miR-4257 9.79E-05 - 91 hsa-miR-1915-5p 1.18E-04 - 92 hsa-miR-4417 1.36E-04 + 93 hsa-miR-1343-5p 1.45E-04 + 94 hsa-miR-6781-5p 1.54E-04 + 95 hsa-miR-4695-5p 1.57E-04 + 96 hsa-miR-1237-5p 1.80E-04 + 97 hsa-miR-6775-5p 2.34E-04 - 98 hsa-miR-7845-5p 2.40E-04 + 99 hsa-miR-4746-3p 2.62E-04 + 100 hsa-miR-7641 4.57E-04 - 101 hsa-miR-7847-3p 5.01E-04 - 102 hsa-miR-6806-5p 5.86E-04 - 103 hsa-miR-4467 6.28E-04 + 104 hsa-miR-4726-5p 6.35E-04 - 105 hsa-miR-4648 6.87E-04 + 106 hsa-miR-6089 8.08E-04 + 107 hsa-miR-1260b 8.29E-04 - 108 hsa-miR-4532 8.69E-04 - 109 hsa-miR-5195-3p 1.02E-03 - 110 hsa-miR-3188 1.12E-03 + 111 hsa-miR-6848-5p 1.36E-03 + 112 hsa-miR-1233-5p 1.41E-03 - 113 hsa-miR-6717-5p 1.63E-03 + 114 hsa-miR-3195 1.95E-03 + 115 hsa-miR-6757-5p 2.65E-03 - 116 hsa-miR-8072 3.49E-03 + 117 hsa-miR-4745-5p 4.17E-03 - 118 hsa-miR-6511a-5p 4.77E-03 - 119 hsa-miR-6776-5p 5.08E-03 + 120 hsa-miR-371a-5p 6.92E-03 - 121 hsa-miR-1227-5p 7.47E-03 + 122 hsa-miR-7150 8.50E-03 + 123 hsa-miR-1915-3p 9.50E-03 + 124 hsa-miR-187-5p 1.56E-18 - 125 hsa-miR-614 2.22E-14 - 126 hsa-miR-19b-3p 1.77E-13 + 127 hsa-miR-1225-5p 2.30E-08 + 128 hsa-miR-451a 5.96E-08 + 129 hsa-miR-939-5p 1.29E-07 + 130 hsa-miR-223-3p 4.79E-06 + 131 hsa-miR-1228-5p 5.66E-06 + 132 hsa-miR-125a-3p 1.47E-04 - 133 hsa-miR-92b-5p 2.51E-04 + 134 hsa-miR-22-3p 6.49E-04 +

Example 2

(356) <Method for Evaluating Lung Cancer Discriminant Performance by Combination of Multiple Gene Markers Using Samples in the Validation Cohort>

(357) In this Example, a method for evaluating lung cancer discriminant performance by a combination of the gene markers selected in Example 1 was studied. Specifically, Fisher's discriminant analysis was conducted as to 8,910 combinations of two polynucleotides comprising at least one or more of the expression level measurement values of any of the newly found polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 125, 127 to 130, and 132 to 134 among the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 134 selected in Example 1, to construct a discriminant for determining the presence or absence of lung cancer. Next, accuracy, sensitivity, and specificity in the validation cohort were calculated using the discriminant thus prepared, and the discriminant performance of the selected polynucleotides was validated using the independent samples.

(358) For example, the expression level measurement values of the nucleotide sequences represented by SEQ ID NO: 1 and SEQ ID NO: 2 were compared between the healthy subjects (100 persons) and the lung cancer patients (17 persons) in the training cohort. As a result, a scatter diagram that significantly separated the gene expression level measurement values of the lung cancer patient group from those of the healthy subject group was obtained (see FIG. 3A). These results were also reproducible for the healthy subjects (50 persons) and the lung cancer patients (8 persons) in the validation cohort (see FIG. 3B). Likewise, a scatter diagram that significantly separated the gene expression level measurement values of the lung cancer patient group from those of the healthy subject group was also obtained as to the other combinations of two expression

level measurement values comprising at least one or more of the expression level measurement values of any of the newly found polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 125, 127 to 130, and 132 to 134 among the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 134. These results were able to be validated in the validation cohort. For example, as for these nucleotide sequences represented by SEQ ID NO: 1 and SEQ ID NO: 2, the number of samples that correctly identified in the detection of lung cancer was calculated using the function ($0 = -1.42x + y + 4.7$) that was set in the training cohort and discriminated between the two groups. As a result, 7 true positives, 50 true negatives, 0 false positives, and 1 false negative were obtained. From these values, 98.3% accuracy, 87.5% sensitivity, and 100% specificity were obtained as detection performance. In this way, the detection performance was calculated as to all of the combinations of two expression level measurement values comprising at least one or more of the expression level measurement values of any of the newly found polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 125, 127 to 130, and 132 to 134 among the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 134. Among them, 133 combinations comprising the expression level measurement value of the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 1 and the detection performance thereof were described in Table 6 as an example. For example, all of 9 combinations of the expression level measurement values of the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 and 6, SEQ ID NOs: 1 and 11, SEQ ID NOs: 1 and 19, SEQ ID NOs: 1 and 34, SEQ ID NOs: 1 and 38, SEQ ID NOs: 1 and 52, SEQ ID NOs: 1 and 53, SEQ ID NOs: 1 and 56, and SEQ ID NOs: 1 and 113 exhibited sensitivity of 100% in the validation cohort. Likewise, all of the 133 combinations of two polynucleotides consisting of the nucleotide sequence represented by SEQ ID NO: 1 and a nucleotide sequence represented by any of SEQ ID NOs: 2 to 134 exhibited sensitivity of 75% or higher. These values of sensitivity were higher than the sensitivity of the existing tumor markers CEA (69%) and CYFRA21-1 (43%) in blood (Non Patent Literature 3). Likewise, 5,742 combinations of the measurement values of the polynucleotides having sensitivity beyond the existing markers CEA and CYFRA21-1 were obtained in the validation cohort. All of the nucleotide sequences 1 to 134 described in Table 3 obtained in Example 1 were employed at least once in these combinations. Thus, the combinations of two of the polynucleotides that consist of the nucleotide sequences represented by SEQ ID NOs: 1 to 134 also produced excellent lung cancer detection sensitivity.

(359) Markers for the detection of lung cancer with better sensitivity are obtained by further combining 3, 4, 5, 6, 7, 8, 9, 10 or more of the expression level measurement values of the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 134. For example, the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 134 selected in Example 1 were measured to obtain their expression levels between the healthy subject group and the lung cancer group in the validation cohort. All of the polynucleotides were ranked in the descending order of their P values based on the Student's t-test which indicates statistical significance of difference between groups (i.e., one having the lowest P value was ranked in the first place), and lung cancer detection sensitivity was evaluated using combinations of one or more polynucleotides to which the polynucleotides were added one by one from the top to the bottom according to the rank. In short, the order in which the polynucleotides were combined in this evaluation is in reverse in terms of SEQ ID NOs from SEQ ID NO: 134 to SEQ ID NOs: 133, 132, . . . shown in Table 3. As a result, the sensitivity in the validation cohort was 62.5% for 1 polynucleotide (SEQ ID NO: 134), 75% for 3 polynucleotides (SEQ ID NOs: 132 to 134), 87.5% for 5 polynucleotides (SEQ ID NOs: 130 to 134), 100% for 6 polynucleotides (SEQ ID NOs: 129 to 134), 100% for 10 polynucleotides (SEQ ID NOs: 125 to 134), 100% for 20 polynucleotides (SEQ ID NOs: 115 to 134), 100% for 30 polynucleotides (SEQ ID NOs: 105 to 134), 100% for 50 polynucleotides (SEQ ID NOs: 85 to 134), 100% for 80 polynucleotides (SEQ ID NOs: 55 to 134),

100% for 120 polynucleotides (SEQ ID NOs: 15 to 134), and 100% for 134 polynucleotides (SEQ ID NOs: 1 to 134).

(360) These results demonstrated that a combination of multiple polynucleotides can produce higher lung cancer discriminant performance than that of each polynucleotide alone or a combination of a fewer number of polynucleotides. In this context, the combinations of multiple polynucleotides are not limited to the combinations of the polynucleotides added in the order of statistically significant difference as described above, and any combination of multiple polynucleotides can be used in the detection of lung cancer.

(361) From these results, it can be concluded that all of the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 134 serve as excellent markers for the detection of lung cancer.

(362) TABLE-US-00005 TABLE 4 Training cohort Validation cohort SEQ Accu- Sensi- Specif-
Accu- Sensi- Specif- ID racy tivity icity racy tivity icity NO: (%) (%) (%) (%) (%) (%) 1 97.4 94.1
98 98.3 87.5 100 2 94.9 82.4 97 100 100 100 3 97.4 82.4 100 96.6 100 96 4 94 70.6 98 93.1 62.5
98 5 95.7 76.5 99 96.6 75 100 6 92.3 64.7 97 93.1 62.5 98 7 94.9 76.5 98 94.8 75 98 8 94.9 94.1 95
94.8 75 98 9 97.4 82.4 100 98.3 87.5 100 10 96.6 82.4 99 91.4 87.5 92 11 94.9 76.5 98 96.6 87.5
98 12 96.6 88.2 98 93.1 87.5 94 13 92.3 64.7 97 94.8 87.5 96 14 92.3 70.6 96 98.3 87.5 100 15
95.7 82.4 98 98.3 87.5 100 16 91.5 76.5 94 94.8 87.5 96 17 94 82.4 96 93.1 87.5 94 18 94.9 70.6
99 100 100 100 19 89.7 64.7 94 93.1 75 96 20 93.2 58.8 99 98.3 87.5 100 21 93.2 64.7 98 93.1
62.5 98 22 91.5 64.7 96 94.8 87.5 96 23 94 70.6 98 87.9 37.5 96 24 93.2 58.8 99 91.4 50 98 25
89.7 64.7 94 91.4 62.5 96 26 93.2 64.7 98 94.8 87.5 96 27 93.2 76.5 96 94.8 87.5 96 28 92.3 82.4
94 93.1 87.5 94 29 89.7 52.9 96 96.6 87.5 98 30 89.7 35.3 99 93.1 62.5 98 31 90.6 47.1 98 94.8
62.5 100 32 93.2 58.8 99 91.4 50 98 33 92.3 64.7 97 96.6 87.5 98 34 89.7 41.2 98 93.1 75 96 35
89.7 52.9 96 93.1 50 100 36 92.3 64.7 97 89.7 50 96 37 88.9 41.2 97 93.1 50 100 38 87.2 47.1 94
96.6 87.5 98 39 90.6 58.8 96 84.5 50 90 40 91.5 47.1 99 91.4 37.5 100 41 91.5 52.9 98 96.6 75 100
42 90.6 47.1 98 96.6 75 100 43 94 64.7 99 91.4 50 98 44 88 47.1 95 93.1 75 96 45 91.5 47.1 99
87.9 37.5 96 46 89.7 47.1 97 87.9 50 94 47 92.3 52.9 99 93.1 50 100 48 88 41.2 96 87.9 62.5 92 49
87.2 41.2 95 89.7 62.5 94 50 88.9 47.1 96 87.9 37.5 96 51 92.3 47.1 100 94.8 62.5 100 52 91.5
47.1 99 94.8 62.5 100 53 91.5 47.1 99 91.4 62.5 96 54 86.3 41.2 94 94.8 62.5 100 55 90.6 41.2 99
94.8 62.5 100 56 90.6 58.8 96 94.8 62.5 100 57 91.5 52.9 98 93.1 62.5 98 58 88.9 35.3 98 93.1
62.5 98 59 86.3 41.2 94 87.9 50 94 60 89.7 47.1 97 89.7 37.5 98 61 90.6 52.9 97 86.2 37.5 94 62
87.2 29.4 97 87.9 62.5 92 63 88.9 41.2 97 82.8 0 96 64 89.7 35.3 99 93.1 50 100 65 89.7 41.2 98
94.8 75 98 66 89.7 29.4 100 91.4 37.5 100 67 90.6 41.2 99 94.8 62.5 100 68 88 47.1 95 87.9 25 98
69 88 35.3 97 91.4 50 98 70 87.2 41.2 95 86.2 25 96 71 88 35.3 97 84.5 25 94 72 88 23.5 99 89.7
37.5 98 73 88 35.3 97 86.2 12.5 98 74 89.7 35.3 99 87.9 37.5 96 75 88 41.2 96 93.1 62.5 98 76
89.7 35.3 99 94.8 62.5 100 77 88.9 35.3 98 87.9 37.5 96 78 88 35.3 97 87.9 50 94 79 88.9 29.4 99
93.1 50 100 80 88.9 29.4 99 87.9 25 98 81 88 23.5 99 87.9 12.5 100 82 83.8 11.8 96 87.9 37.5 96
83 88.9 23.5 100 87.9 12.5 100 84 87.2 23.5 98 87.9 12.5 100 85 89.7 47.1 97 94.8 62.5 100 86
87.2 29.4 97 86.2 12.5 98 87 88 23.5 99 86.2 37.5 94 88 85.5 29.4 95 91.4 37.5 100 89 87.2 29.4
97 86.2 25 96 90 88.9 35.3 98 87.9 50 94 91 89.7 41.2 98 91.4 62.5 96 92 86.3 23.5 97 84.5 12.5
96 93 89.7 41.2 98 94.8 62.5 100 94 87.2 17.6 99 81 0 94 95 89.7 41.2 98 94.8 62.5 100 96 87.2
29.4 97 89.7 37.5 98 97 86.3 17.6 98 81 0 94 98 89.7 35.3 99 87.9 37.5 96 99 87.2 17.6 99 94.8
62.5 100 100 84.5 18.8 95 86.2 25 96 101 83.8 11.8 96 84.5 0 98 102 86.3 5.9 100 91.4 37.5 100
103 83.8 11.8 96 86.2 12.5 98 104 84.6 17.6 96 86.2 25 96 105 85.5 11.8 98 89.7 25 100 106 89.7
41.2 98 89.7 37.5 98 107 87.2 23.5 98 91.4 50 98 108 88 23.5 99 91.4 37.5 100 109 87.2 17.6 99
87.9 25 98 110 86.3 23.5 97 89.7 25 100 111 85.5 11.8 98 86.2 25 96 112 86.3 17.6 98 86.2 0 100
113 84.6 23.5 95 89.7 25 100 114 86.3 23.5 97 84.5 25 94 115 82.9 0 97 89.7 25 100 116 88 23.5
99 89.7 25 100 117 88 17.6 100 89.7 25 100 118 84.6 11.8 97 86.2 0 100 119 85.5 5.9 99 89.7 25
100 120 84.6 0 99 84.5 0 98 121 88.9 23.5 100 87.9 12.5 100 122 88 17.6 100 89.7 25 100 123
84.6 5.9 98 94.8 62.5 100 124 99.1 94.1 100 96.6 75 100 125 94 76.5 97 93.1 75 96 126 95.7 82.4

98 93.1 62.5 98 127 89.7 52.9 96 93.1 50 100 128 93.2 58.8 99 89.7 37.5 98 129 91.5 58.8 97 86.2
50 92 130 94 58.8 100 94.8 62.5 100 131 84.6 17.6 96 87.9 25 98 132 89.7 35.3 99 89.7 25 100
133 89.7 35.3 99 96.6 75 100 134 87.2 23.5 98 86.2 12.5 98

(363) TABLE-US-00006 TABLE 5 Discriminant Constant SEQ ID NO: coefficient term 1 3.665
36.958 2 3.482 28.279 3 3.305 21.564 4 3.967 46.907 5 2.921 18.418 6 3.258 31.351 7 2.321
19.901 8 2.482 17.979 9 5.340 69.250 10 3.780 34.781 11 6.053 65.389 12 2.169 14.787 13 3.363
28.960 14 3.278 29.867 15 4.768 47.106 16 2.668 21.511 17 3.933 49.822 18 2.781 19.688 19
2.340 19.400 20 3.173 27.138 21 2.395 19.027 22 4.481 51.987 23 1.923 18.732 24 2.221 21.483
25 1.879 15.097 26 3.449 21.201 27 1.940 10.546 28 2.467 16.896 29 3.381 32.369 30 1.883
19.278 31 2.995 22.556 32 2.257 25.609 33 2.593 16.685 34 4.054 25.898 35 4.316 37.567 36
2.347 13.660 37 2.787 28.233 38 4.929 45.747 39 3.956 32.281 40 2.822 21.631 41 2.892 30.757
42 3.016 22.359 43 2.179 11.954 44 2.956 26.296 45 4.228 28.830 46 2.347 22.562 47 7.619
102.957 48 2.849 21.598 49 2.506 18.167 50 1.885 16.130 51 4.534 36.471 52 3.307 19.440 53
3.370 33.776 54 4.473 45.416 55 3.058 33.429 56 4.044 33.691 57 4.924 52.340 58 4.740 41.821
59 3.556 33.458 60 2.051 13.913 61 4.118 31.479 62 2.848 30.006 63 2.967 23.118 64 3.094
33.898 65 6.747 57.639 66 3.115 18.546 67 2.952 23.150 68 6.267 79.386 69 5.244 36.656 70
3.634 23.502 71 5.682 45.289 72 4.756 58.458 73 3.941 38.866 74 4.639 42.673 75 4.686 54.180
76 3.379 28.223 77 3.897 27.668 78 2.497 17.033 79 2.622 18.728 80 2.639 18.344 81 4.764
52.837 82 2.582 18.301 83 3.517 26.318 84 6.525 46.333 85 2.880 21.133 86 3.254 24.541 87
4.996 39.036 88 3.508 36.118 89 3.944 29.161 90 3.193 21.619 91 1.406 8.631 92 5.754 47.280 93
3.850 40.213 94 5.850 61.192 95 4.464 33.686 96 4.601 58.630 97 6.817 56.624 98 3.273 21.990
99 2.934 19.283 100 1.405 10.220 101 3.974 25.352 102 3.294 21.365 103 2.273 22.405 104 4.014
26.327 105 1.371 8.370 106 5.947 79.958 107 2.441 20.646 108 3.287 38.733 109 3.026 20.705
110 3.417 20.796 111 5.205 38.779 112 2.897 32.216 113 2.584 17.226 114 3.934 32.685 115
3.076 22.309 116 5.228 64.304 117 2.180 25.963 118 2.566 14.847 119 3.282 19.125 120 3.663
26.980 121 6.563 62.775 122 4.018 31.312 123 4.220 46.687 124 2.174 20.711 125 1.889 11.995
126 1.102 5.734 127 3.626 27.002 128 0.979 9.798 129 2.534 19.444 130 1.051 6.668 131 3.974
47.286 132 1.456 9.155 133 3.272 26.342 134 1.514 8.925

(364) TABLE-US-00007 TABLE 6 Training set Validation set SEQ Accu- Sensi- Specif- Accu-
Sensi- Specif- ID racy tivity icity racy tivity icity NO: (%) (%) (%) (%) (%) (%) 1_2 98.3 94.1
99 98.3 87.5 100 1_3 100 100 100 98.3 87.5 100 1_4 97.4 88.2 99 98.3 87.5 100 1_5 99.1
100 99 98.3 87.5 100 1_6 99.1 100 99 100 100 100 1_7 99.1 100 99 98.3 87.5 100 1_8 98.3
94.1 99 98.3 87.5 100 1_9 99.1 100 99 98.3 87.5 100 1_10 100 100 100 98.3 87.5 100 1_11
98.3 100 98 100 100 100 1_12 98.3 100 98 98.3 87.5 100 1_13 98.3 100 98 98.3 87.5 100 1_14
100 100 100 98.3 87.5 100 1_15 98.3 100 98 98.3 87.5 100 1_16 99.1 100 99 98.3 87.5 100
1_17 98.3 100 98 98.3 87.5 100 1_18 99.1 100 99 98.3 87.5 100 1_19 99.1 100 99 100 100 100
1_20 98.3 94.1 99 98.3 87.5 100 1_21 98.3 94.1 99 96.6 87.5 98 1_22 98.3 100 98 98.3 87.5
100 1_23 97.4 94.1 98 98.3 87.5 100 1_24 97.4 94.1 98 98.3 87.5 100 1_25 98.3 100 98 98.3
87.5 100 1_26 99.1 100 99 96.6 87.5 98 1_27 98.3 100 98 98.3 87.5 100 1_28 98.3 100 98 98.3
87.5 100 1_29 98.3 94.1 99 98.3 87.5 100 1_30 97.4 94.1 98 98.3 87.5 100 1_31 98.3 100 98
98.3 87.5 100 1_32 97.4 94.1 98 96.6 75 100 1_33 98.3 94.1 99 98.3 87.5 100 1_34 99.1 100
99 100 100 100 1_35 98.3 100 98 98.3 87.5 100 1_36 97.4 94.1 98 96.6 75 100 1_37 98.3 100
98 98.3 87.5 100 1_38 98.3 100 98 100 100 100 1_39 97.4 94.1 98 98.3 87.5 100 1_40 98.3
100 98 96.6 75 100 1_41 98.3 100 98 98.3 87.5 100 1_42 98.3 100 98 98.3 87.5 100 1_43 97.4
94.1 98 98.3 87.5 100 1_44 97.4 94.1 98 98.3 87.5 100 1_45 97.4 94.1 98 98.3 87.5 100 1_46
96.6 88.2 98 96.6 75 100 1_47 98.3 100 98 98.3 87.5 100 1_48 99.1 100 99 98.3 87.5 100 1_49
97.4 94.1 98 96.6 75 100 1_50 97.4 94.1 98 96.6 75 100 1_51 99.1 100 99 98.3 87.5 100 1_52
99.1 100 99 100 100 100 1_53 99.1 100 99 100 100 100 1_54 98.3 100 98 98.3 87.5 100 1_55
98.3 100 98 98.3 87.5 100 1_56 100 100 100 100 100 100 1_57 98.3 100 98 98.3 87.5 100
1_58 98.3 100 98 98.3 87.5 100 1_59 98.3 100 98 98.3 87.5 100 1_60 97.4 94.1 98 98.3 87.5

100 1_61 98.3 100 98 98.3 87.5 100 1_62 97.4 94.1 98 98.3 87.5 100 1_63 98.3 100 98 98.3 87.5 100 1_64 98.3 100 98 98.3 87.5 100 1_65 97.4 94.1 98 98.3 87.5 100 1_66 98.3 100 98 98.3 87.5 100 1_67 98.3 100 98 98.3 87.5 100 1_68 98.3 100 98 98.3 87.5 100 1_69 97.4 94.1 98 98.3 87.5 100 1_70 97.4 94.1 98 98.3 87.5 100 1_71 97.4 94.1 98 98.3 87.5 100 1_72 97.4 94.1 98 98.3 87.5 100 1_73 98.3 94.1 99 98.3 87.5 100 1_74 98.3 100 98 98.3 87.5 100 1_75 97.4 100 97 98.3 87.5 100 1_76 99.1 94.1 100 98.3 87.5 100 1_77 97.4 94.1 98 98.3 87.5 100 1_78 97.4 94.1 98 98.3 87.5 100 1_79 97.4 94.1 98 98.3 87.5 100 1_80 98.3 94.1 99 98.3 87.5 100 1_81 97.4 94.1 98 98.3 87.5 100 1_82 99.1 100 99 98.3 87.5 100 1_83 98.3 100 98 98.3 87.5 100 1_84 98.3 100 98 98.3 87.5 100 1_85 97.4 88.2 99 98.3 87.5 100 1_86 97.4 94.1 98 98.3 87.5 100 1_87 98.3 100 98 98.3 87.5 100 1_88 98.3 100 98 98.3 87.5 100 1_89 97.4 94.1 98 98.3 87.5 100 1_90 97.4 94.1 98 98.3 87.5 100 1_91 98.3 94.1 99 98.3 87.5 100 1_92 97.4 94.1 98 98.3 87.5 100 1_93 98.3 100 98 98.3 87.5 100 1_94 98.3 100 98 98.3 87.5 100 1_95 98.3 94.1 99 98.3 87.5 100 1_96 98.3 100 98 98.3 87.5 100 1_97 97.4 94.1 98 98.3 87.5 100 1_98 98.3 94.1 99 98.3 87.5 100 1_99 98.3 100 98 98.3 87.5 100 1_100 99.1 100 99 98.3 87.5 100 1_101 97.4 94.1 98 98.3 87.5 100 1_102 97.4 94.1 98 98.3 87.5 100 1_103 99.1 100 99 98.3 87.5 100 1_104 97.4 94.1 98 98.3 87.5 100 1_105 97.4 94.1 98 98.3 87.5 100 1_106 98.3 94.1 99 98.3 87.5 100 1_107 96.6 88.2 98 96.6 75 100 1_108 98.3 100 98 98.3 87.5 100 1_109 99.1 100 99 98.3 87.5 100 1_110 97.4 94.1 98 98.3 87.5 100 1_111 97.4 94.1 98 98.3 87.5 100 1_112 97.4 94.1 98 98.3 87.5 100 1_113 99.1 100 99 100 100 100 1_114 98.3 100 98 98.3 87.5 100 1_115 97.4 94.1 98 98.3 87.5 100 1_116 98.3 100 98 98.3 87.5 100 1_117 97.4 94.1 98 98.3 87.5 100 1_118 97.4 94.1 98 98.3 87.5 100 1_119 97.4 94.1 98 98.3 87.5 100 1_120 98.3 100 98 98.3 87.5 100 1_121 98.3 94.1 99 98.3 87.5 100 1_122 98.3 100 98 98.3 87.5 100 1_123 98.3 100 98 98.3 87.5 100 1_124 98.3 100 98 98.3 87.5 100 1_125 97.4 94.1 98 98.3 87.5 100 1_126 99.1 100 99 98.3 87.5 100 1_127 98.3 100 98 98.3 87.5 100 1_128 98.3 100 98 98.3 87.5 100 1_129 98.3 100 98 98.3 87.5 100 1_130 98.3 100 98 98.3 87.5 100 1_131 97.4 94.1 98 98.3 87.5 100 1_132 98.3 88.2 100 98.3 87.5 100 1_133 97.4 94.1 98 98.3 87.5 100 1_134 98.3 100 98 98.3 87.5 100

Example 3

(365) <Selection of Gene Marker Using all Samples and Method for Evaluating Lung Cancer Discriminant Performance of Acquired Gene Marker>

(366) In this Example, the samples in the training cohort and the validation cohort used in Examples 1 and 2 were integrated, and selection of a gene marker and evaluation of its lung cancer discriminant performance were conducted using all of the samples.

(367) Specifically, the miRNA expression levels in the serum of the 25 lung cancer patients and the 150 healthy subjects obtained in the preceding Reference Examples were normalized by quantile normalization. In order to acquire diagnostic markers with higher reliability, only genes having a gene expression level of 2.sup.6 or higher in 50% or more of the samples in either of the lung cancer patient group or the healthy subject group were selected in the gene marker selection. In order to further acquire statistical significance for discriminating a lung cancer patient group from a healthy subject group, the P value obtained by two-sample t-test assuming equal variance as to each gene expression level was corrected by the Bonferroni method, and genes that satisfied $p < 0.01$ were selected as gene markers for use in explanatory variables of a discriminant. The acquired genes are described in Table 7. In this way, hsa-miR-4271, hsa-miR-642b-3p, hsa-miR-6075, hsa-miR-6125, hsa-miR-887-3p, hsa-miR-6851-5p, hsa-miR-6763-5p, hsa-miR-3928-3p, hsa-miR-4443, hsa-miR-3648, hsa-miR-149-3p, hsa-miR-4689, hsa-miR-4763-3p, hsa-miR-6729-5p, hsa-miR-3196, hsa-miR-8069, hsa-miR-1268a, hsa-miR-4739, hsa-miR-1268b, hsa-miR-5698, hsa-miR-6752-5p, hsa-miR-4507, hsa-miR-564, hsa-miR-4497, hsa-miR-6877-5p, hsa-miR-6087, hsa-miR-4731-5p, hsa-miR-615-5p, hsa-miR-760, hsa-miR-6891-5p, hsa-miR-6887-5p, hsa-miR-4525, hsa-miR-1914-3p, hsa-miR-619-5p, hsa-miR-5001-5p, hsa-miR-6722-3p, hsa-miR-3621, hsa-miR-4298, hsa-miR-675-5p, and hsa-miR-4655-5p genes, and the nucleotide sequences of SEQ ID NOs: 135 to 174 related thereto were found in addition to the genes described in Table 3. As with the

nucleotide sequences of SEQ ID NOs: 1 to 134, the results obtained about the polynucleotides shown in SEQ ID NOs: 135 to 174 also showed that the measurement values were significantly lower (-) or higher (+) in the lung cancer patient group than in the healthy subject group (Table 7). These results were able to be validated in the validation cohort. Thus, the presence or absence of lung cancer in the newly obtained samples can be determined by the methods described in Examples 1 and 2 by using the gene expression level measurement values described in Table 7 either alone or in combination with the gene expression level measurement values described in Table 3.

(368) TABLE-US-00008 TABLE 7 Expression level P value after in lung cancer Bonferroni patient with respect SEQ ID NO: Gene name correction to healthy subject 1 hsa-miR-6768-5p 6.12E-37 + 2 hsa-miR-6836-3p 4.68E-36 - 3 hsa-miR-6782-5p 7.67E-29 - 4 hsa-miR-3663-3p 4.91E-29 - 5 hsa-miR-1908-3p 2.76E-30 - 6 hsa-miR-6726-5p 1.23E-26 + 7 hsa-miR-4258 6.12E-28 - 8 hsa-miR-1343-3p 7.70E-26 - 9 hsa-miR-4516 1.71E-29 - 10 hsa-miR-6875-5p 1.59E-18 - 11 hsa-miR-4651 6.58E-26 + 12 hsa-miR-6825-5p 230E-22 - 13 hsa-miR-6840-3p 4.47E-24 + 14 hsa-miR-6780b-5p 7.12E-26 - 15 hsa-miR-6749-5p 3.83E-25 - 16 hsa-miR-8063 7.83E-21 - 17 hsa-miR-6784-5p 1.37E-17 + 18 hsa-miR-3679-5p 2.70E-25 - 19 hsa-miR-3184-5p 5.58E-19 + 20 hsa-miR-663b 2.07E-22 - 21 hsa-miR-6880-5p 4.49E-19 + 22 hsa-miR-1908-5p 7.91E-21 + 23 hsa-miR-92a-2-5p 6.69E-15 + 24 hsa-miR-7975 3.32E-17 + 25 hsa-miR-7110-5p 2.07E-16 + 26 hsa-miR-6842-5p 3.25E-19 - 27 hsa-miR-6857-5p 7.70E-16 + 28 hsa-miR-5572 1.14E-17 + 29 hsa-miR-3197 7.43E-21 + 30 hsa-miR-6131 8.81E-19 + 31 hsa-miR-6889-5p 7.76E-18 + 32 hsa-miR-4454 6.20E-15 - 33 hsa-miR-1199-5p 1.10E-16 - 34 hsa-miR-1247-3p 2.61E-15 - 35 hsa-miR-6800-5p 1.65E-14 - 36 hsa-miR-6872-3p 3.40E-13 + 37 hsa-miR-4649-5p 2.50E-16 - 38 hsa-miR-6791-5p 2.29E-18 - 39 hsa-miR-4433b-3p 1.12E-12 + 40 hsa-miR-3135b 7.14E-09 + 41 hsa-miR-128-2-5p 3.95E-17 + 42 hsa-miR-4675 3.41E-17 - 43 hsa-miR-4472 1.34E-15 - 44 hsa-miR-6785-5p 7.27E-16 + 45 hsa-miR-6741-5p 1.57E-11 + 46 hsa-miR-7977 4.98E-13 + 47 hsa-miR-3665 1.23E-11 + 48 hsa-miR-128-1-5p 6.12E-11 + 49 hsa-miR-4286 8.20E-12 + 50 hsa-miR-6765-3p 3.54E-12 + 51 hsa-miR-4632-5p 1.23E-14 - 52 hsa-miR-365a-5p 3.37E-12 - 53 hsa-miR-6088 2.65E-13 - 54 hsa-miR-6816-5p 3.35E-14 + 55 hsa-miR-6885-5p 1.83E-13 - 56 hsa-miR-711 2.81E-14 + 57 hsa-miR-6765-5p 1.37E-11 + 58 hsa-miR-3180 1.69E-14 + 59 hsa-miR-4442 2.64E-12 - 60 hsa-miR-4792 2.35E-11 + 61 hsa-miR-6721-5p 1.63E-09 + 62 hsa-miR-6798-5p 9.64E-11 - 63 hsa-miR-3162-5p 1.05E-08 - 64 hsa-miR-6126 3.64E-14 + 65 hsa-miR-4758-5p 3.51E-15 - 66 hsa-miR-2392 2.75E-12 + 67 hsa-miR-486-3p 2.02E-11 - 68 hsa-miR-6727-5p 3.30E-09 + 69 hsa-miR-4728-5p 9.06E-11 - 70 hsa-miR-6746-5p 1.45E-08 + 71 hsa-miR-4270 1.52E-08 + 72 hsa-miR-3940-5p 3.98E-09 + 73 hsa-miR-4725-3p 2.40E-08 - 74 hsa-miR-7108-5p 5.64E-10 + 75 hsa-miR-3656 6.69E-13 + 76 hsa-miR-6879-5p 3.97E-13 + 77 hsa-miR-6738-5p 1.60E-09 + 78 hsa-miR-1260a 1.22E-08 + 79 hsa-miR-4446-3p 3.23E-10 - 80 hsa-miR-3131 2.40E-09 + 81 hsa-miR-4463 1.54E-08 - 82 hsa-miR-3185 5.62E-10 - 83 hsa-miR-6870-5p 3.81E-08 + 84 hsa-miR-6779-5p 3.02E-07 + 85 hsa-miR-1273g-3p 2.06E-09 + 86 hsa-miR-8059 2.01E-06 - 87 hsa-miR-4697-5p 1.86E-08 + 88 hsa-miR-4674 4.38E-10 - 89 hsa-miR-4433-3p 2.20E-07 - 90 hsa-miR-4257 1.87E-08 + 91 hsa-miR-1915-5p 4.76E-10 - 92 hsa-miR-4417 2.14E-07 - 93 hsa-miR-1343-5p 1.06E-10 + 94 hsa-miR-6781-5p 4.10E-05 - 95 hsa-miR-4695-5p 3.31E-11 - 96 hsa-miR-1237-5p 3.95E-10 + 97 hsa-miR-6775-5p 4.09E-05 + 98 hsa-miR-7845-5p 2.84E-07 - 99 hsa-miR-4746-3p 9.11E-11 - 100 hsa-miR-7641 1.14E-06 - 101 hsa-miR-7847-3p 5.71E-05 + 102 hsa-miR-6806-5p 1.87E-09 - 103 hsa-miR-4467 2.48E-08 - 104 hsa-miR-4726-5p 8.08E-07 + 105 hsa-miR-4648 1.15E-08 + 106 hsa-miR-6089 1.19E-07 + 107 hsa-miR-1260b 1.62E-05 + 108 hsa-miR-4532 8.30E-09 + 109 hsa-miR-5195-3p 2.03E-07 + 110 hsa-miR-3188 4.84E-08 - 111 hsa-miR-6848-5p 6.01E-07 + 112 hsa-miR-1233-5p 3.76E-06 + 113 hsa-miR-6717-5p 2.38E-05 + 114 hsa-miR-3195 7.67E-06 - 115 hsa-miR-6757-5p 1.58E-06 - 116 hsa-miR-8072 1.17E-05 - 117 hsa-miR-4745-5p 5.89E-07 + 119 hsa-miR-6776-5p 1.26E-07 - 120 hsa-miR-371a-5p 9.22E-05 + 121 hsa-miR-1227-5p 9.64E-05 - 122

hsa-miR-7150 0.000252 - 123 hsa-miR-1915-3p 2.18E-09 - 124 hsa-miR-187-5p 2.81E-27 - 125 hsa-miR-614 1.65E-21 - 126 hsa-miR-19b-3p 1.33E-19 + 127 hsa-miR-1225-5p 6.67E-13 - 128 hsa-miR-451a 2.23E-10 - 129 hsa-miR-939-5p 1.89E-11 + 130 hsa-miR-223-3p 9.32E-11 - 131 hsa-miR-1228-5p 1.49E-09 + 132 hsa-miR-125a-3p 1.07E-05 + 133 hsa-miR-92b-5p 1.09E-11 + 134 hsa-miR-22-3p 9.71E-07 + 135 hsa-miR-4271 5.64E-07 + 136 hsa-miR-642b-3p 6.99E-06 - 137 hsa-miR-6075 1.17E-05 + 138 hsa-miR-6125 1.63E-05 + 139 hsa-miR-887-3p 1.68E-05 + 140 hsa-miR-6851-5p 1.97E-05 - 141 hsa-miR-6763-5p 3.54E-05 - 142 hsa-miR-3928-3p 4.67E-05 - 143 hsa-miR-4443 5.36E-05 + 144 hsa-miR-3648 6.01E-05 + 145 hsa-miR-149-3p 9.80E-05 - 146 hsa-miR-4689 1.01E-04 + 147 hsa-miR-4763-3p 1.20E-04 + 148 hsa-miR-6729-5p 1.28E-04 + 149 hsa-miR-3196 1.31E-04 + 150 hsa-miR-8069 1.84E-04 + 151 hsa-miR-1268a 2.58E-04 + 152 hsa-miR-4739 2.68E-04 + 153 hsa-miR-1268b 3.37E-04 + 154 hsa-miR-5698 4.34E-04 - 155 hsa-miR-6752-5p 5.63E-04 + 156 hsa-miR-4507 6.34E-04 + 157 hsa-miR-564 6.68E-04 - 158 hsa-miR-4497 8.11E-04 - 159 hsa-miR-6877-5p 8.21E-04 - 160 hsa-miR-6087 8.91E-04 - 161 hsa-miR-4731-5p 1.15E-03 - 162 hsa-miR-615-5p 1.25E-03 - 163 hsa-miR-760 1.42E-03 - 164 hsa-miR-6891-5p 1.71E-03 + 165 hsa-miR-6887-5p 1.82E-03 - 166 hsa-miR-4525 2.09E-03 - 167 hsa-miR-1914-3p 2.11E-03 - 168 hsa-miR-619-5p 2.61E-03 - 169 hsa-miR-5001-5p 3.01E-03 - 170 hsa-miR-6722-3p 3.88E-03 + 171 hsa-miR-3621 4.02E-03 - 172 hsa-miR-4298 7.88E-03 - 173 hsa-miR-675-5p 8.33E-03 - 174 hsa-miR-4655-5p 9.06E-03 +

Example 4

(369) <Method for Evaluating Lung Cancer-Specific Discriminant Performance by Combination of Multiple Gene Markers Using Samples in the Validation Cohort>

(370) In this Example, gene markers for diagnosis were selected by comparing gene expression levels of miRNAs in serum of lung cancer patients with that of a control group consisting of healthy subjects, pancreatic cancer patients, biliary tract cancer patients, colorectal cancer patients, stomach cancer patients, liver cancer patients, and benign pancreaticobiliary disease patients, in the same way as the method described in Example 1, using the gene markers selected in Example 1, and targeting the training cohort as the sample group described in Reference Example 2. The polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 561 to 579 thus selected were further combined therewith to study a method for evaluating lung cancer-specific discriminant performance.

(371) Specifically, first, the miRNA expression levels in the training cohort and the validation cohort obtained in Reference Example 2 mentioned above were combined and normalized by quantile normalization. Next, Fisher's discriminant analysis was conducted as to combinations of 1 to 4 expression level measurement values comprising at least one or more of the expression level measurement values of the polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1 to 174, and 561 to 579, to construct a discriminant for determining the presence or absence of lung cancer. Next, accuracy, sensitivity, and specificity in the validation cohort were calculated using the discriminant thus prepared, with the lung cancer patient group as a positive sample group and, on the other hand, the healthy subject group, the pancreatic cancer patient group, the biliary tract cancer patient group, the colorectal cancer patient group, the stomach cancer patient group, the liver cancer patient group, and the benign pancreaticobiliary disease patient group as a negative sample group. The discriminant performance of the selected polynucleotides was validated using the independent samples.

(372) Most of polynucleotides consisting of the nucleotide sequences represented by these SEQ ID NOs (SEQ ID NOs: 1 to 174, and 561 to 579 corresponding to the miRNA markers of Table 1) or complementary sequences thereof mentioned above were able to provide relatively high accuracy, sensitivity, and specificity in the determination of the presence or absence of lung cancer, and furthermore, were able to specifically discriminate lung cancer from the other cancers. For example, among the combinations of multiple polynucleotides selected from the group consisting of polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1, 2, 3, 4, 5,

7, 9, 10, 11, 19, 21, 26, 29, 31, 52, 53, 63, 65, 69, 72, 87, 90, 113, 124, 125, 126, 128, 130, 143, 148, 160, 162, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578 and 579 or complementary sequences thereof (the cancer type-specific polynucleotide group 1) as polynucleotides capable of specifically binding to target markers, combinations comprising at least one or more polynucleotide(s) selected from the group consisting of polynucleotides consisting of the nucleotide sequences represented by SEQ ID NOs: 1, 2, 3, 10, 63, 113, 124, 125, 126, 128, 130, 143, 160, 561, 568, 573 and 578 or complementary sequences thereof (the cancer type-specific polynucleotide group 2) included in the cancer type-specific polynucleotide group 1 were able to specifically discriminate lung cancer from the other cancers with high accuracy.

(373) The number of the polynucleotides with cancer type specificity in the combination can be 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more for the combination. The combinations of 4 or more of these polynucleotides were able to exhibit discriminant accuracy of 90% or higher.

(374) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 1 or a complementary sequence thereof is shown in Table 8-1. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 1 or a complementary sequence thereof exhibited accuracy of 94.2% in the training cohort and accuracy of 91.4% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 1 or a complementary sequence thereof exhibited accuracy of 98.7% in the training cohort and accuracy of 97.5% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 1 or a complementary sequence thereof exhibited accuracy of 99.2% in the training cohort and accuracy of 98.5% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 1 or a complementary sequence thereof exhibited accuracy of 99.7% in the training cohort and accuracy of 99.0% in the validation cohort.

(375) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 2 or a complementary sequence thereof is shown in Table 8-2. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 2 or a complementary sequence thereof exhibited accuracy of 94.0% in the training cohort and accuracy of 92.4% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 2 or a complementary sequence thereof exhibited accuracy of 97.2% in the training cohort and accuracy of 96.0% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 2 or a complementary sequence thereof exhibited accuracy of 99.5% in the training cohort and accuracy of 98.0% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 2 or a complementary sequence thereof exhibited accuracy of 100% in the training cohort and accuracy of 98.0% in the validation cohort.

(376) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 3 or a complementary sequence thereof is shown in Table 8-3. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence

represented by SEQ ID NO: 3 or a complementary sequence thereof exhibited accuracy of 85.7% in the training cohort and accuracy of 84.3% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 3 or a complementary sequence thereof exhibited accuracy of 97.0% in the training cohort and accuracy of 97.0% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 3 or a complementary sequence thereof exhibited accuracy of 99.0% in the training cohort and accuracy of 98.5% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 3 or a complementary sequence thereof exhibited accuracy of 99.5% in the training cohort and accuracy of 100% in the validation cohort.

(377) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 10 or a complementary sequence thereof is shown in Table 8-4. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 10 or a complementary sequence thereof exhibited accuracy of 64.0% in the training cohort and accuracy of 61.6% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 10 or a complementary sequence thereof exhibited accuracy of 94.0% in the training cohort and accuracy of 92.4% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 10 or a complementary sequence thereof exhibited accuracy of 99.2% in the training cohort and accuracy of 99.0% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 10 or a complementary sequence thereof exhibited accuracy of 99.2% in the training cohort and accuracy of 99.5% in the validation cohort.

(378) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 63 or a complementary sequence thereof is shown in Table 8-5. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 63 or a complementary sequence thereof exhibited accuracy of 79.4% in the training cohort and accuracy of 80.8% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 63 or a complementary sequence thereof exhibited accuracy of 95.7% in the training cohort and accuracy of 97.5% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 63 or a complementary sequence thereof exhibited accuracy of 98.2% in the training cohort and accuracy of 98.0% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 63 or a complementary sequence thereof exhibited accuracy of 99.5% in the training cohort and accuracy of 97.5% in the validation cohort.

(379) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 113 or a complementary sequence thereof is shown in Table 8-6. For example, the measurement using the combination of

one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 113 or a complementary sequence thereof exhibited accuracy of 67.8% in the training cohort and accuracy of 69.2% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 113 or a complementary sequence thereof exhibited accuracy of 97.7% in the training cohort and accuracy of 95.5% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 113 or a complementary sequence thereof exhibited accuracy of 99.5% in the training cohort and accuracy of 99.0% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 113 or a complementary sequence thereof exhibited accuracy of 99.5% in the training cohort and accuracy of 99.0% in the validation cohort.

(380) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 124 or a complementary sequence thereof is shown in Table 8-7. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 124 or a complementary sequence thereof exhibited accuracy of 79.6% in the training cohort and accuracy of 76.8% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 124 or a complementary sequence thereof exhibited accuracy of 95.0% in the training cohort and accuracy of 91.4% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 124 or a complementary sequence thereof exhibited accuracy of 98.5% in the training cohort and accuracy of 97.5% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 124 or a complementary sequence thereof exhibited accuracy of 99.0% in the training cohort and accuracy of 99.0% in the validation cohort.

(381) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 125 or a complementary sequence thereof is shown in Table 8-8. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 125 or a complementary sequence thereof exhibited accuracy of 77.6% in the training cohort and accuracy of 73.7% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 125 or a complementary sequence thereof exhibited accuracy of 94.7% in the training cohort and accuracy of 93.4% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 125 or a complementary sequence thereof exhibited accuracy of 99.0% in the training cohort and accuracy of 96.5% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 125 or a complementary sequence thereof exhibited accuracy of 99.5% in the training cohort and accuracy of 99.0% in the validation cohort.

(382) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 126 or a complementary

sequence thereof is shown in Table 8-9. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 126 or a complementary sequence thereof exhibited accuracy of 90.4% in the training cohort and accuracy of 92.4% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 126 or a complementary sequence thereof exhibited accuracy of 96.7% in the training cohort and accuracy of 95.5% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 126 or a complementary sequence thereof exhibited accuracy of 99.7% in the training cohort and accuracy of 98.0% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 126 or a complementary sequence thereof exhibited accuracy of 99.7% in the training cohort and accuracy of 99.0% in the validation cohort.

(383) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 128 or a complementary sequence thereof is shown in Table 8-10. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 128 or a complementary sequence thereof exhibited accuracy of 81.4% in the training cohort and accuracy of 81.3% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 128 or a complementary sequence thereof exhibited accuracy of 96.2% in the training cohort and accuracy of 94.9% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 128 or a complementary sequence thereof exhibited accuracy of 98.7% in the training cohort and accuracy of 97.5% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 128 or a complementary sequence thereof exhibited accuracy of 99.2% in the training cohort and accuracy of 99.5% in the validation cohort.

(384) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 130 or a complementary sequence thereof is shown in Table 8-11. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 130 or a complementary sequence thereof exhibited accuracy of 83.4% in the training cohort and accuracy of 87.4% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 130 or a complementary sequence thereof exhibited accuracy of 96.2% in the training cohort and accuracy of 94.4% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 130 or a complementary sequence thereof exhibited accuracy of 99.2% in the training cohort and accuracy of 98.5% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 130 or a complementary sequence thereof exhibited accuracy of 99.5% in the training cohort and accuracy of 99.5% in the validation cohort.

(385) Specifically, the discriminant accuracy of the measurement using the polynucleotide

consisting of the nucleotide sequence represented by SEQ ID NO: 143 or a complementary sequence thereof is shown in Table 8-12. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 143 or a complementary sequence thereof exhibited accuracy of 64.6% in the training cohort and accuracy of 66.2% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 143 or a complementary sequence thereof exhibited accuracy of 96.0% in the training cohort and accuracy of 93.9% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 143 or a complementary sequence thereof exhibited accuracy of 98.7% in the training cohort and accuracy of 98.0% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 143 or a complementary sequence thereof exhibited accuracy of 99.0% in the training cohort and accuracy of 98.0% in the validation cohort.

(386) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 160 or a complementary sequence thereof is shown in Table 8-13. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 160 or a complementary sequence thereof exhibited accuracy of 70.9% in the training cohort and accuracy of 67.2% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 160 or a complementary sequence thereof exhibited accuracy of 96.0% in the training cohort and accuracy of 92.4% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 160 or a complementary sequence thereof exhibited accuracy of 99.2% in the training cohort and accuracy of 98.5% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 160 or a complementary sequence thereof exhibited accuracy of 99.5% in the training cohort and accuracy of 99.0% in the validation cohort.

(387) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 561 or a complementary sequence thereof is shown in Table 8-14. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 561 or a complementary sequence thereof exhibited accuracy of 84.9% in the training cohort and accuracy of 81.8% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 561 or a complementary sequence thereof exhibited accuracy of 96.5% in the training cohort and accuracy of 97.5% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 561 or a complementary sequence thereof exhibited accuracy of 98.7% in the training cohort and accuracy of 98.0% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 561 or a complementary sequence thereof exhibited accuracy of 100% in the training cohort and accuracy of 99.0% in the validation cohort.

(388) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 568 or a complementary sequence thereof is shown in Table 8-15. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 568 or a complementary sequence thereof exhibited accuracy of 60.2% in the training cohort and accuracy of 67.2% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 568 or a complementary sequence thereof exhibited accuracy of 97.0% in the training cohort and accuracy of 96.0% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 568 or a complementary sequence thereof exhibited accuracy of 99.0% in the training cohort and accuracy of 96.0% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 568 or a complementary sequence thereof exhibited accuracy of 99.5% in the training cohort and accuracy of 98.5% in the validation cohort.

(389) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 573 or a complementary sequence thereof is shown in Table 8-16. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 573 or a complementary sequence thereof exhibited accuracy of 53.0% in the training cohort and accuracy of 53.5% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 573 or a complementary sequence thereof exhibited accuracy of 96.5% in the training cohort and accuracy of 95.5% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 573 or a complementary sequence thereof exhibited accuracy of 98.7% in the training cohort and accuracy of 98.0% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 573 or a complementary sequence thereof exhibited accuracy of 99.2% in the training cohort and accuracy of 98.5% in the validation cohort.

(390) Specifically, the discriminant accuracy of the measurement using the polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 578 or a complementary sequence thereof is shown in Table 8-17. For example, the measurement using the combination of one polynucleotide comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 578 or a complementary sequence thereof exhibited accuracy of 52.8% in the training cohort and accuracy of 53.5% in the validation cohort. Also, for example, the measurement using the combinations of two polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 578 or a complementary sequence thereof exhibited accuracy of 96.2% in the training cohort and accuracy of 94.9% in the validation cohort. Furthermore, for example, the measurement using the combinations of three polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 578 or a complementary sequence thereof exhibited accuracy of 98.5% in the training cohort and accuracy of 96.5% in the validation cohort. Furthermore, for example, the measurement using the combinations of four polynucleotides comprising at least one polynucleotide consisting of the nucleotide sequence represented by SEQ ID NO: 578 or a complementary sequence thereof exhibited accuracy of 99.2% in the training

cohort and accuracy of 99.0% in the validation cohort.

(391) The measurement values of the nucleotide sequences represented by SEQ ID NOs: 1, 113, 126, and 561 were compared among 17 lung cancer patients, 99 healthy subjects, 75 pancreatic cancer patients, 62 biliary tract cancer patients, 32 colorectal cancer patients, 35 stomach cancer patients, 32 esophageal cancer patients, 33 liver cancer patients, and 13 benign pancreaticobiliary disease patients in the training cohort. As a result, a scatter diagram that significantly separated the discriminant score of the lung cancer patient group from the discriminant scores of the other groups was obtained in the training cohort (see FIG. 4A). These results were also reproducible for the validation cohort (see FIG. 4B).

(392) TABLE-US-00009 TABLE 8-1 Training cohort Validation cohort Accu- Sensi- Specif- Accu- Sensi- Specif- racy tivity icity racy tivity icity SEQ ID NO: (%) (%) (%) (%) (%) (%) 1 94.2 100 94.0 91.4 87.5 91.6 1_113 98.7 100 98.7 97.5 100 97.4 1_52_126 99.2 100 99.2 98.5 100 98.4 1_53_113_125 99.2 100 99.2 98.5 100 98.4 1_10_63_113 99.2 100 99.2 98.5 100 98.4 1_19_113_143 99.2 100 99.2 99.0 100 98.9 1_10_113_126 99.7 100 99.7 99.0 100 98.9 1_2_10_113 99.7 100 99.7 98.5 100 98.4

(393) TABLE-US-00010 TABLE 8-2 Training cohort Validation cohort Accu- Sensi- Specif- Accu- Sensi- Specif- racy tivity icity racy tivity icity SEQ ID NO: (%) (%) (%) (%) (%) (%) 2 94.0 94.1 94.0 92.4 100 92.1 2_126 97.2 100 97.1 96.0 100 95.8 1_2_113 99.5 100 99.5 98.0 100 97.9 2_19_53_113 99.2 100 99.2 97.5 100 97.4 2_72_113_125 99.0 100 99.0 96.5 100 96.3 2_19_72_113 99.0 100 99.0 97.0 100 96.8 2_19_113_579 98.5 100 98.4 96.5 100 96.3 1_2_19_113 100 100 100 98.0 100 97.9

(394) TABLE-US-00011 TABLE 8-3 Training cohort Validation cohort Accuracy Sensitivity Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) 3 85.7 94.1 85.3 84.3 100 83.7 3_126 97.0 94.1 97.1 97.0 100 96.8 1_3_113 99.0 100 99.0 98.5 100 98.4 3_125_128_568 98.5 100 98.4 97.0 100 96.8 1_3_10_113 99.2 100 99.2 99.0 100 98.9 3_113_125_126 99.5 94.1 99.7 100 100 100 1_3_126_573 98.5 100 98.4 98.0 100 97.9 3_126_130_561 98.2 94.1 98.4 98.0 100 97.9

(395) TABLE-US-00012 TABLE 8-4 Training cohort Validation cohort Accuracy Sensitivity Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) 10 64.0 82.4 63.2 61.6 75.0 61.1 2_10 94.0 100 93.7 92.4 100 92.1 1_10_113 99.2 100 99.2 99.0 100 98.9 1_10_113_143 99.0 100 98.9 99.5 100 99.5 1_10_113_569 99.2 100 99.2 99.0 100 98.9 1_10_113_562 98.7 100 98.7 99.0 100 98.9 1_10_113_578 99.2 100 99.2 98.5 100 98.4 1_7_10_113 99.2 100 99.2 99.0 100 98.9

(396) TABLE-US-00013 TABLE 8-5 Training cohort Validation cohort Accuracy Sensitivity Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) 63 79.4 94.1 78.7 80.8 75.0 81.1 63_126 95.7 94.1 95.8 97.5 100 97.4 1_63_113 98.2 100 98.2 98.0 100 97.9 1_63_567_578 99.5 100 99.5 97.5 100 97.4 1_53_63_578 98.2 100 98.2 98.0 100 97.9 1_63_162_573 98.0 100 97.9 97.5 87.5 97.9 1_63_162_578 98.5 100 98.4 98.0 100 97.9 1_63_576_578 98.7 100 98.7 98.0 100 97.9

(397) TABLE-US-00014 TABLE 8-6 Training cohort Validation cohort Accuracy Sensitivity Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) 113 67.8 76.5 67.5 69.2 100 67.9 2_113 97.7 100 97.6 95.5 100 95.3 1_19_113 99.5 100 99.5 99.0 100 98.9 1_10_113_567 99.5 100 99.5 99.0 100 98.9 1_53_63_113 99.0 100 99.0 98.0 100 97.9 1_53_113_143 99.0 100 99.0 98.0 100 97.9 2_19_113_125 99.0 100 99.0 98.0 100 97.9 2_10_113_130 99.2 100 99.2 99.5 100 99.5

(398) TABLE-US-00015 TABLE 8-7 Training cohort Validation cohort Accuracy Sensitivity Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) 124 79.6 94.1 79.0 76.8 100 75.8 2_124 95.0 100 94.8 91.4 100 91.1 1_113_124 98.5 100 98.4 97.5 100 97.4 113_124_125_126 99.0 94.1 99.2 99.0 100 98.9 124_125_128_568 98.0 100 97.9 94.9 100 94.7 113_124_125_162 99.0 100 99.0 98.0 100 97.9 52_124_126_561 98.0 94.1 98.2 98.0 100 97.9

19_113_124_126 98.0 94.1 98.2 99.0 100 98.9
(399) TABLE-US-00016 TABLE 8-8 Training cohort Validation cohort Accuracy Sensitivity
Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) (%) 125 77.6 82.4
77.4 73.7 87.5 73.2 113_125 94.7 100 94.5 93.4 100 93.2 2_113_125 99.0 100 99.0 96.5 100 96.3
1_113_125_160 99.5 100 99.5 98.5 100 98.4 31_113_125_568 99.0 100 98.9 98.0 100 97.9
2_53_113_125 99.2 100 99.2 98.0 100 97.9 1_10_113_125 99.5 100 99.5 99.0 100 98.9
1_113_125_143 99.2 100 99.2 99.0 100 98.9
(400) TABLE-US-00017 TABLE 8-9 Training cohort Validation cohort Accuracy Sensitivity
Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) (%) 126 90.4 94.1
90.3 92.4 100 92.1 1_126 96.7 100 96.6 95.5 100 95.3 1_113_126 99.7 100 99.7 98.0 100 97.9
1_126_561_573 98.5 100 98.4 97.5 100 97.4 113_125_126_568 98.5 100 98.4 98.5 100 98.4
113_125_126_561 99.0 94.1 99.2 98.5 100 98.4 1_113_125_126 99.7 100 99.7 99.0 100 98.9
1_52_126_561 99.5 100 99.5 98.0 100 97.9
(401) TABLE-US-00018 TABLE 8-10 Training cohort Validation cohort Accuracy Sensitivity
Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) (%) 128 81.4 82.4
81.4 81.3 87.5 81.1 1_128 96.2 100 96.1 94.9 100 94.7 1_113_128 98.7 100 98.7 97.5 100 97.4
26_113_125_128 97.7 94.1 97.9 98.5 100 98.4 1_113_125_128 99.0 100 99.0 99.0 100 98.9
1_10_113_128 99.2 100 99.2 99.5 100 99.5 31_113_125_128 97.5 94.1 97.6 99.0 100 98.9
2_19_113_128 99.0 100 99.0 97.0 100 96.8
(402) TABLE-US-00019 TABLE 8-11 Training cohort Validation cohort Accuracy Sensitivity
Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) (%) 130 83.4 88.2
83.2 87.4 100 86.8 1_130 96.2 100 96.1 94.4 100 94.2 1_113_130 99.2 100 99.2 98.5 100 98.4
1_3_130_143 97.7 100 97.6 99.0 100 98.9 1_10_113_130 99.5 100 99.5 99.5 100 99.5
1_63_130_578 98.7 100 98.7 98.5 100 98.4 124_125_130_568 98.5 100 98.4 96.5 100 96.3
2_19_113_130 99.0 100 99.0 98.0 100 97.9
(403) TABLE-US-00020 TABLE 8-12 Training cohort Validation cohort Accuracy Sensitivity
Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) (%) 143 64.6 58.8
64.8 66.2 62.5 66.3 1_143 96.0 100 95.8 93.9 87.5 94.2 1_113_143 98.7 100 98.7 98.0 100 97.9
1_3_126_143 99.0 100 98.9 98.0 100 97.9 1_63_130_143 97.7 100 97.6 98.0 100 97.9
1_10_52_143 98.0 100 97.9 100 100 100 2_19_113_143 98.5 100 98.4 96.5 100 96.3
63_124_130_143 96.2 94.1 96.3 96.0 100 95.8
(404) TABLE-US-00021 TABLE 8-13 Training cohort Validation cohort Accuracy Sensitivity
Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) (%) 160 70.9 70.6
70.9 67.2 37.5 68.4 2_160 96.0 100 95.8 92.4 100 92.1 1_113_160 99.2 100 99.2 98.5 100 98.4
1_10_113_160 99.2 100 99.2 99.0 100 98.9 7_113_125_160 99.0 100 99.0 97.5 100 97.4
1_113_160_567 99.5 100 99.5 98.0 100 97.9 1_113_160_578 98.7 100 98.7 98.0 100 97.9
2_19_113_160 99.5 100 99.5 98.0 100 97.9
(405) TABLE-US-00022 TABLE 8-14 Training cohort Validation cohort Accuracy Sensitivity
Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) (%) 561 84.9 88.2
84.8 81.8 87.5 81.6 126_561 96.5 94.1 96.6 97.5 100 97.4 1_113_561 98.7 100 98.7 98.0 100 97.9
113_125_130_561 97.7 94.1 97.9 99.5 100 99.5 7_126_143_561 98.5 100 98.4 98.5 100 98.4
1_113_126_561 100 100 100 99.0 100 98.9 1_126_561_568 98.7 100 98.7 98.0 100 97.9
7_113_126_561 99.2 94.1 99.5 98.5 100 98.4
(406) TABLE-US-00023 TABLE 8-15 Training cohort Validation cohort Accuracy Sensitivity
Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) (%) 568 60.2 58.8
60.3 67.2 100 65.8 1_568 97.0 100 96.8 96.0 100 95.8 1_2_568 99.0 100 98.9 96.0 100 95.8
7_125_126_568 99.2 100 99.2 98.0 100 97.9 124_125_126_568 98.5 100 98.4 98.0 100 97.9
7_113_125_568 98.5 100 98.4 98.0 100 97.9 1_113_125_568 99.5 100 99.5 98.0 100 97.9
113_125_128_568 97.5 100 97.4 98.5 100 98.4
(407) TABLE-US-00024 TABLE 8-16 Training cohort Validation cohort Accuracy Sensitivity

Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) (%) 573 53.0 35.3
53.8 53.5 12.5 55.3 1_573 96.5 100 96.3 95.5 100 95.3 1_113_573 98.7 100 98.7 98.0 100 97.9
113_125_126_573 98.2 94.1 98.4 99.5 100 99.5 1_113_125_573 99.2 100 99.2 98.5 100 98.4
1_53_113_573 98.7 100 98.7 97.5 100 97.4 1_124_126_573 97.7 100 97.6 96.5 100 96.3
1_63_130_573 98.7 100 98.7 98.0 100 97.9
(408) TABLE-US-00025 TABLE 8-17 Training cohort Validation cohort Accuracy Sensitivity
Specificity Accuracy Sensitivity Specificity SEQ ID NO: (%) (%) (%) (%) (%) (%) (%) 578 52.8 52.9
52.8 53.5 50.0 53.7 1_578 96.2 100 96.1 94.9 100 94.7 1_113_578 98.5 100 98.4 96.5 100 96.3
1_126_567_578 98.5 100 98.4 97.5 100 97.4 1_19_113_578 99.2 100 99.2 99.0 100 98.9
31_126_561_578 97.5 94.1 97.6 97.5 100 97.4 1_126_160_578 98.7 100 98.7 97.0 100 96.8
1_113_125_578 98.7 100 98.7 98.5 100 98.4

INDUSTRIAL APPLICABILITY

(409) According to the present invention, lung cancer can be effectively detected by a simple and inexpensive method. This permits early detection, diagnosis and treatment of lung cancer. The method of the present invention can detect lung cancer with limited invasiveness using the blood of a patient and therefore allows lung cancer to be detected conveniently and rapidly.

(410) All publications, patents, and patent applications cited herein are incorporated herein by reference in their entirety.

Claims

1. A method for detecting lung cancer in a human subject, comprising measuring an expression level of hsa-miR-4258 in a blood, serum, or plasma sample from the subject; comparing the measured expression level of hsa-miR-4258 to a control expression level for a healthy subject; detecting decreased level of hsa-miR-4258 in the sample from the subject as compared to the control expression level from the sample from the healthy subject; wherein the decreased level of hsa-miR-4258 indicates that the subject has lung cancer; and wherein the method further comprises treating the subject for the lung cancer or performing a diagnostic procedure on the subject with the lung cancer; wherein the treatment comprises surgery, radiotherapy, chemotherapy or a combination thereof; and wherein the diagnostic procedure comprises chest X-ray examination or diagnostic imaging of the lung of the human subject.

2. The method according to claim 1, wherein the expression level of hsa-miR-4258 in the sample is measured by using a kit comprising a nucleic acid(s) that specifically bind to hsa-miR-4258.

3. The method according to claim 2, wherein the kit further comprises at least one nucleic acid capable of specifically binding to at least one polynucleotide selected from the group consist of other lung cancer markers, miR-19b-3p, miR-1228-5p, and miR-1307-3p, miR-4271, miR-642b-3p, miR-6075, miR-6125, miR-887-3p, miR-6851-5p, miR-6763-5p, miR-3928-3p, miR-4443, miR-3648, miR-149-3p, miR-4689, miR-4763-3p, miR-6729-5p, miR-3196, miR-8069, miR-1268a, miR-4739, miR-1268b, miR-5698, miR-6752-5p, miR-4507, miR-564, miR-4497, miR-6877-5p, miR-6087, miR-4731-5p, miR-615-5p, miR-760, miR-6891-5p, miR-6887-5p, miR-4525, miR-1914-3p, miR-619-5p, miR-5001-5p, miR-6722-3p, miR-3621, miR-4298, miR-675-5p, miR-4655-5p, miR-6836-3p, miR-6782-5p, miR-3663-3p, miR-1908-3p, miR-6726-5p, miR-1343-3p, miR-4516, miR-6875-5p, miR-4651, miR-6825-5p, miR-6840-3p, miR-6780b-5p, miR-6749-5p, miR-8063, miR-6784-5p, miR-3184-5p, miR-663b, miR-6880-5p, miR-1908-5p, miR-92a-2-5p, miR-7975, miR-7110-5p, miR-6842-5p, miR-6857-5p, miR-5572, miR-3197, miR-6131, miR-6889-5p, miR-4454, miR-1199-5p, miR-1247-3p, miR-6800-5p, miR-6872-3p, miR-4649-5p, miR-6791-5p, miR-4433b-3p, miR-3135b, miR-128-2-5p, miR-4675, miR-4472, miR-6785-5p, miR-6741-5p, miR-7977, miR-3665, miR-128-1-5p, miR-4286, miR-6765-3p, miR-4632-5p, miR-365a-5p, miR-6088, miR-6816-5p, miR-6885-5p, miR-711, miR-6765-5p, miR-3180, miR-4442, miR-4792, miR-6721-5p, miR-6798-5p, miR-6126, miR-4758-5p, miR-2392, miR-486-3p, miR-6727-

5p, miR-4728-5p, miR-6746-5p, miR-4270, miR-3940-5p, miR-4725-3p, miR-7108-5p, miR-3656, miR-6879-5p, miR-6738-5p, miR-1260a, miR-4446-3p, miR-3131, miR-4463, miR-3185, miR-6870-5p, miR-6779-5p, miR-1273g-3p, miR-8059, miR-4697-5p, miR-4674, miR-4433-3p, miR-4257, miR-1915-5p, miR-4417, miR-1343-5p, miR-6781-5p, miR-4695-5p, miR-1237-5p, miR-6775-5p, miR-7845-5p, miR-4746-3p, miR-7641, miR-7847-3p, miR-6806-5p, miR-4467, miR-4726-5p, miR-4648, miR-6089, miR-1260b, miR-4532, miR-5195-3p, miR-3188, miR-6848-5p, miR-1233-5p, miR-6717-5p, miR-3195, miR-6757-5p, miR-8072, miR-4745-5p, miR-6511a-5p, miR-6776-5p, miR-371a-5p, miR-1227-5p, miR-7150, miR-1915-3p, miR-187-5p, miR-614, miR-1225-5p, miR-451a, miR-939-5p, miR-223-3p, miR-125a-3p, miR-92b-5p, miR-22-3p, miR-6073, miR-6845-5p, miR-6769b-5p, miR-4665-3p, miR-1913, miR-1228-3p, miR-940, miR-296-3p, miR-4690-5p, miR-548q, miR-663a, miR-1249, miR-1202, miR-7113-3p, miR-1225-3p, miR-4783-3p, miR-4448 and miR-4534.

4. The method according to claim 1, comprising performing the diagnostic procedure on the subject.

5. The method according to claim 1, wherein the decreased level of hsa-miR-4258 in the sample is measured by using a device comprising a nucleic acid(s) that specifically bind to hsa-miR-4258.

6. The method according to claim 5, wherein the device further comprises at least one nucleic acid capable of specifically binding to at least one polynucleotide selected from the group consist of other lung cancer markers, miR-19b-3p, miR-1228-5p, and miR-1307-3p, miR-4271, miR-642b-3p, miR-6075, miR-6125, miR-887-3p, miR-6851-5p, miR-6763-5p, miR-3928-3p, miR-4443, miR-3648, miR-149-3p, miR-4689, miR-4763-3p, miR-6729-5p, miR-3196, miR-8069, miR-1268a, miR-4739, miR-1268b, miR-5698, miR-6752-5p, miR-4507, miR-564, miR-4497, miR-6877-5p, miR-6087, miR-4731-5p, miR-615-5p, miR-760, miR-6891-5p, miR-6887-5p, miR-4525, miR-1914-3p, miR-619-5p, miR-5001-5p, miR-6722-3p, miR-3621, miR-4298, miR-675-5p, miR-4655-5p, miR-6836-3p, miR-6782-5p, miR-3663-3p, miR-1908-3p, miR-6726-5p, miR-1343-3p, miR-4516, miR-6875-5p, miR-4651, miR-6825-5p, miR-6840-3p, miR-6780b-5p, miR-6749-5p, miR-8063, miR-6784-5p, miR-3184-5p, miR-663b, miR-6880-5p, miR-1908-5p, miR-92a-2-5p, miR-7975, miR-7110-5p, miR-6842-5p, miR-6857-5p, miR-5572, miR-3197, miR-6131, miR-6889-5p, miR-4454, miR-1199-5p, miR-1247-3p, miR-6800-5p, miR-6872-3p, miR-4649-5p, miR-6791-5p, miR-4433b-3p, miR-3135b, miR-128-2-5p, miR-4675, miR-4472, miR-6785-5p, miR-6741-5p, miR-7977, miR-3665, miR-128-1-5p, miR-4286, miR-6765-3p, miR-4632-5p, miR-365a-5p, miR-6088, miR-6816-5p, miR-6885-5p, miR-711, miR-6765-5p, miR-3180, miR-4442, miR-4792, miR-6721-5p, miR-6798-5p, miR-6126, miR-4758-5p, miR-2392, miR-486-3p, miR-6727-5p, miR-4728-5p, miR-6746-5p, miR-4270, miR-3940-5p, miR-4725-3p, miR-7108-5p, miR-3656, miR-6879-5p, miR-6738-5p, miR-1260a, miR-4446-3p, miR-3131, miR-4463, miR-3185, miR-6870-5p, miR-6779-5p, miR-1273g-3p, miR-8059, miR-4697-5p, miR-4674, miR-4433-3p, miR-4257, miR-1915-5p, miR-4417, miR-1343-5p, miR-6781-5p, miR-4695-5p, miR-1237-5p, miR-6775-5p, miR-7845-5p, miR-4746-3p, miR-7641, miR-7847-3p, miR-6806-5p, miR-4467, miR-4726-5p, miR-4648, miR-6089, miR-1260b, miR-4532, miR-5195-3p, miR-3188, miR-6848-5p, miR-1233-5p, miR-6717-5p, miR-3195, miR-6757-5p, miR-8072, miR-4745-5p, miR-6511a-5p, miR-6776-5p, miR-371a-5p, miR-1227-5p, miR-7150, miR-1915-3p, miR-187-5p, miR-614, miR-1225-5p, miR-451a, miR-939-5p, miR-223-3p, miR-125a-3p, miR-92b-5p, miR-22-3p, miR-6073, miR-6845-5p, miR-6769b-5p, miR-4665-3p, miR-1913, miR-1228-3p, miR-940, miR-296-3p, miR-4690-5p, miR-548q, miR-663a, miR-1249, miR-1202, miR-7113-3p, miR-1225-3p, miR-4783-3p, miR-4448 and miR-4534.
